

Math 101 The Fundamentals of Mathematics: Algebra

Numbered Transcript

MATH 101 · Dr. Robert Snellman

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Lecture 1: Foundations of Algebra

At a glance — Foundations of algebra — expressions and equations.

- 1.1 All right, so welcome to Foundations of Mathematics, and we're talking about algebra this course. So one of the questions you want to ask yourself, and I get asked a lot is, what is mathematics? What is algebra? What is trigonometry? What is calculus? And all these different types of things. And ultimately, I like to explain it as, mathematics is really a language to try and explain various things. And algebra plays a very integral part of that. So what we want to break down is, what is algebra? Why is it useful? Is it useful? Things like that.
- 1.2 So, algebra has a long history, as does everything in mathematics. But ultimately, people attribute algebra to a mathematician by the name of Mohamed Al-Khwarizmi, who wrote this book, Algebra, which is typically where people say the foundations of algebra were laid. Now, there's evidence to suggest that other civilizations also had this. There's tablets dating back to the Babylonians in 2000 BC, where what they were doing on those tablets could be considered algebra. There's things dating back to various people. Diophantus was one of them.
- 1.3 But typically, people attribute the foundational principles of algebra coming from Al-Khwarizmi, which is great because ultimately our number system comes from the Arabic line of mathematicians, which we should be very grateful for, because otherwise we'd be doing math with Roman numerals and things like that, and that would be a disaster, in my opinion. So we can attribute all of that back to the Arabic mathematicians.
- 1.4 Now, algebra at its base essentially says, well, what are we trying to do? We're trying to apply symbols, which we're going to call variables, to represent certain quantities of interest. So I want to motivate this just from a real-world example, because at the end of the day, I don't want to just study a topic for the sake of studying a topic. I want to know, what is the, what is the ROI for me, the return on my investment? What is it going to do for me in my life? It's the age-old question that students will ask is, "When am I ever going to use this? When is this going to be relevant?"
- 1.5 My answer to that question is, "Maybe never." But I think the principles of thinking through these problems and assigning abstract quantities to represent physical quantities of interest, that is always valuable. Being able to think through things critically and things like that. You may not ever sit down on your day-to-day and say, "I want to solve the equation $2x$ equals 7, and solve for x ." You're probably not going to do that on a daily basis. But being able to abstract certain principles from algebra and apply those in your daily life, you'll use that every single day.
- 1.6 So let's start with the following situation. So let's suppose that you're in sales. Let's suppose you're in sales and you get a base salary per month. As an example, you have a

base salary, and let's say that that's \$1,000 per month. This will be \$1,000 per month. But you also end up getting a commission on whatever you sell. So you're guaranteed \$1,000 a month, but you also are guaranteed to have a commission. Now, your commission is going to be the following. So let's say you get \$100 per whatever you're selling, whatever widget you're selling, whatever that is.

- 1.7 You're selling phones, you're selling insurance, you're selling houses, you're selling whatever. Let's just say, for the sake of argument, that you get \$100 for everything that you sell. So I sell one thing, I get \$100, I sell five things, I get \$500, things like that. Now, what we can do is, we could say, OK, well, let's say in month one, in month one, I end up selling, let's say, 25 widgets. I sell 25 widgets. Well, then how much money do I make at the end of the month?
- 1.8 And we can sit here and we can say, OK, well, if I sell 25 widgets, OK, well, I know if I'm going to take, if I sell 25 and I get 100 times that, so every one of them I get \$100, I made \$2,500. So that's how much I made, just in commission from selling those 25 widgets. But then I also have my base salary of \$1,000 a month. So my total wages, if you will, my wages are my \$1,000 that I'm guaranteed every single month, that my employer comes to me and says, "Look, I'm going to give you \$1,000 every month." OK, fine. But then I sold the 25 widgets, and therefore I also made an additional \$2,500. OK, so what do I get? I get \$3,500 at the end of the month.
- 1.9 OK, fine. If I go on to month two, I could say month two, what happens? Well, month two, let's say we sell, I don't know, 30 widgets. Same principle applies. Well, if I sell 30 widgets, 30 widgets at \$100 a piece, that's \$3,000. I'm going to have a base salary of \$1,000 on top of that, so I get my wage of \$4,000. OK, And that's all fine and good. And that's probably how we think about this on the day-to-day.
- 1.10 But algebra wants to try and extract this, or to abstract this, so that I could say, OK, in any given month, if I sell 75 widgets, if I sell 80 widgets, if I sell 50 widgets, whatever that is, that I can just have some formula that I can just go to, plug in the number of widgets that I made or that I sold, and that'll give me my corresponding wages. That's what we would like to do. So what we need to do is try and figure out, based on these, can I abstract a formula from this? So let's see what we're doing. So let's think about it.
- 1.11 My wages is always guaranteed to be 1,000 plus something. Well, what is that something? It's going to be \$100 times however many widgets that I sold. That's what we have here. I had \$100 times 25. That was the \$2,500 I made just in my commission from selling widgets. And my wages from month two for selling 30 widgets was \$3,000. So my total wages were \$4,000. My commission was \$3,000. Total wages were \$4,000. So if I look at this, I can say, OK, well, if I'm going to get \$1,000 base plus 100 times however many widgets I sell, all right, so how many widgets?
- 1.12 Well, I don't know what that is, but I can say that it's going to be times the number of widgets sold. All right? So I look at that and I'm like, OK, well, does that actually make

sense? Well, my wages, the units for the wages should be dollars. That's how much I make each month. OK, so units are dollars. 1,000, its units are in dollars. That's fine. What are the units of 100 here? The 100 is representing what? Well, that's my commission. OK? So that's saying I'm going to get \$100 per widget sold. So the units are going to be \$100 per widget. Do we agree? So I'm going to have dollars per widget.

- 1.13** Those are the units of the 100 saying, "I'm going to give you \$100 for every widget that you sell." So if the units here are dollars and the units here are dollars per widget, well, then what do I need to multiply this by in order to cancel, "cancel" the widgets in the denominator? Well, I need to multiply by widgets here, so that, units-wise, the widgets cancel and I'm left with dollars. That's going to give me my overall wage.
- 1.14** It's at least plausible that I'm taking the 100 times the number of widgets sold from a unit standpoint, actually gives me the correct amount that I'm looking for. All right? OK, but this is clunky. I don't want some formula that just has widgets sold here. I'd like something a little more concrete, and that's where variables enter the fray. So variables, what is a variable? This is just a symbol representing an abstract quantity. That's all it is. You can literally label it whatever you want.
- 1.15** Now, classically, the variable is typically denoted by x . This is where the stress and the anxiety with mathematics starts coming in. When you start throwing symbols to things, then it gets a little involved. But that's OK. But variable is literally just an assignment of whatever you want it to be. You can call it x , y , z , w , γ , α , β . It doesn't matter what you want to call it, just you give it a corresponding symbol. If I come back up to my formula up here, I can come up with a formula for my wages. Well, wages starts with w , so I'm going to give that the variable w . That seems plausible, at least.
- 1.16** I can say that my wages will be equal to 1,000 plus 100 times x , where x is going to be the number of widgets that I sell. That's going to be my formula for my wage. Now, what's nice about this is that I can then test this. I can test this against what my actual data tells me. So if I go back to month one, I say, OK, in month one, what did I have? Well, what would my corresponding variable x represent in this case? Well, x is representing the number of widgets that I sold. I would say that x is equal to 25. OK, well, if x is equal to 25, that's great.
- 1.17** What do I get for w ? All you do is, you take the x is equal to 25 and you simply substitute it into your formula. That's it. All right? So what am I going to get? I'm going to get 1,000 plus 100 times 25. And that's exactly my 2,500 plus 1,000, which is 3,500, which we saw before. Just doing the mental math calculation. All right? And I can do that for month two and month three and month four and just keep going and going and going. Whatever number of widgets I sell, I can just plug it into the formula and get my corresponding wage for that month.
- 1.18** So there's a value to this. I don't have to sit down every single time and just look at what the data is. All I can do is, what I really just need to do is say, oh, I have a formula

for this. All I need to do is plug in x every single month and it'll populate what my wage is going to be. OK, so that's the introduction of variables into algebra. OK. So again, what's the point in all of this? Well, as I said in the beginning, mathematics is a language, so we need to understand the vocabulary associated to that language, so that we can all understand each other. OK, so let's think about this.

- 1.19** What can I do with these symbols, these variables? How can I manipulate them? Can I add them? Can I subtract them? Can I divide them, multiply? Can I do the basic arithmetic that we're accustomed to doing with symbols, with variables. Am I allowed to do things like that? Well, the first thing we need to understand is, what is an expression? And this is one thing that's a little frustrating with mathematics, but I don't think it's exclusive to math.
- 1.20** I think it's universal with any discipline, is that when you're setting up language for this, there's this conflict between what we have in our vernacular, our natural language that we speak and words that have certain meanings there, versus using those words in a specific context. So for example, in statistics, you have certain distributions given your data. And one of those distributions is called the normal distribution. Well, the word normal means—I don't even know what that word means. Normal, relative to what? Normal relative, this is the normal temperature for this time of year based on historical data?
- 1.21** Well, that doesn't seem to have anything to do with some sort of distribution, maybe. Maybe it does, maybe it doesn't. Or normal compared to whose standard? It's all relative in that sense. But using the word normal in a statistical context, there's a very specific meaning for what that thing is. It is this distribution; that's what it is. An expression, mathematically or in algebra, is very different from how we would say in our natural language of an expression. "I'm expressing my feelings, I'm expressing my emotions, I'm expressing myself." It's very different from that.
- 1.22** An expression literally is something of the following form. So this is as easiest to give you an example as to what an expression is as to try, rather than giving you just an abstract notion of what an expression is. So something that looks like this. Going back to our wage example, we had something that was 1,000 plus $100x$. Now, remember, I said that that was equal to w , which was my wages. That was my formula associated to that. But if I simply just gave you this, that's an example of an expression. It's just some combination of variables and numbers, but it doesn't have any equality to it.
- 1.23** It doesn't say, this is equal to something. It's not an equation. An equation is different than an expression. This would be something that would be an expression versus—so this would be an expression—versus something that, if I said 1,000 plus $100x$ is equal to, say, 50, this would be an equation. You can kind of tell the difference. One of them, the way I very ridiculously remember this is that an equation has an equality and an expression doesn't. OK? All right.

- 1.24** So what we want to understand now is, OK, if I can take expressions like this—1,000 plus $100x$ — what arithmetic operations am I allowed to do with these things? OK, so if I took the expression, let's say, let's use smaller numbers, $2x$ plus 4, OK? So that's an expression. I say I want to add in, let's say, $3x$ minus 6, can I somehow combine things? And what am I allowed to combine? Well, I can say this is equal to what? Well, this is the notion of what we call simplifying. Simplifying expressions.
- 1.25** Because at the end of the day, if I have 50 different expressions added together, and they all have similar variables, so they all have an x and some constants and things like that in them, why am I writing out all 50? Why don't I just combine them down? Why don't I just condense them down into something that's a little more tractable, something that only has two or three terms maybe, instead of 2,000 terms. And that's the notion of simplifying expressions. So the idea is that you combine, the phrase is, you combine like terms.
- 1.26** OK, so what the heck does that mean? You combine like terms. Well, if I look at the expression up here, I say, look, I have something that involves x 's, I have something that involves x 's, and they're connected with some arithmetic operation. I have two x 's, and then I have three x 's, and I'm adding them together. OK, so if I have two of something and I have three something else and I add them, 2 plus 3 is giving me 5. So I'm going to have 5 x 's in total. So I can write this as $5x$, just combining those two terms alone.
- 1.27** I can then look at the other terms that look similar to each other. So this one is a constant. And this one is a constant. But I have to be very careful about this because the arithmetic operation that I'm using is subtraction. And I have to understand, OK, well, then how do I combine them? I've got a plus 4 here, and I've got a minus 6 here. So what am I really doing? It's like I'm saying, I have 4 of something, and then I'm going to take away 6 of that same something. So what am I left with?
- 1.28** Well, physically, it doesn't quite make sense. If I give you 4 wooden blocks and I say, "Take away 6 wooden blocks," you're like, "That doesn't make any sense. You only gave me 4 wooden blocks." Mathematically, we can say, OK, what am I going to have? If you give me 4 of something and you take away 6 of something, then there's negative 2 of that same thing. OK? 4 minus 6 is going to be my negative 2. So what am I going to get here? I'm going to get $5x$ minus 2.
- 1.29** Again, from a physical standpoint, it doesn't make sense for me to hand you 4 things and say, "Give me 6 things back." You're like, "I can't. I can only give you 4 things back." But our number system allows for us to have negative numbers in that sense, right? It's kind of, if you want to think of a real-world analog of this, let's say, going back to the wage example. OK, so let's say you made \$4,000 in a single month, but you spent \$4,500. So you spent more than you earned that month. We would say you're "in the red" that month because you overspent.

- 1.30** So what does that really mean? It's like, OK, I have \$4,000, but with all the expenses that I made throughout the month, I'm now \$500 in debt. I have negative \$500 that ended up happening that month. That's the notion. That's how I think about negative numbers. But when you simplify expressions, you combine the like terms, so things that look alike. So if I had a variable with a constant, a variable with a constant, I can combine those. If I just have numerical values, I can combine those and go from there. All right? So that's the notion of simplifying.
- 1.31** Now, in an expression like this, these things also have names. OK, so again, x , we already said, that's my variable, whatever it's representing. The number that's in front of the variable is called the coefficient of that variable. So that's part of our vocabulary. So this is the coefficient of that variable. And then this thing that's just hanging out by himself, this negative 2, is just a constant. That's just a constant. Now, with algebra, there's a lot of things that we can do, and the rules can get a little complicated. So I want to walk through some of the common rules that we have in algebra, one of which is called the distributive rule. So, distribution.
- 1.32** All right, so the distribution rule says the following thing. Let's suppose I give you a variable x . Let's suppose I give you a variable x , and to that variable x , I want you to add something to it. So let's say we add 5 things to it, whatever that is. All right. Then I want to double that quantity. I want to double that quantity. Whatever this is representing, I want to double that. I want twice as many x plus 5s as possible. Again, if you think about this from a numerical standpoint, let's suppose I give you the number x is equal to 2, and I say, I want you to take x is equal to 2, and I want you to add 5 to it. OK, 2 plus 5 is 7. Great.
- 1.33** So now I have 7, and I want you to double that. You're like, oh, OK, 14. Great. Concretely, if I give you a physical number, we can work through it mentally and it makes sense. OK, you give me 2, you add 5, that's 7, and double, it's 14. What I'm saying is, I don't need to give you the 2. I can just say it's an abstract quantity. It's x , whatever it is. If I want x to be 2, I set x equal to 2. If I want x to be 7, I set x equal to 7. If I want x to be negative 5, I set x equal to negative 5. It gives me more versatility.
- 1.34** But the question I want to understand is, how do I actually look at this quantity? Well, think about doubling. What does that really mean? If I give you one wooden block, if I give you one wooden block and I say, "Double it," well, then you have two wooden blocks. Well, what I'm really saying is, symbolically, I'm saying I have something of that form. Do we agree? So I have one wooden block. I say double it. I give you a second wooden block. That's going to give me two wooden blocks.
- 1.35** So if I now put in a variable, then it simply just tells me, if I take x and I add it to x , I get 2 times x . So what does this really mean? Let's try and reverse that. If I gave you 2 times x , then I can really decompose that as x plus x . So really what I'm saying here is 2 times this is really what? I'm taking this quantity, that's my box, so that's my x plus 5, and I'm adding it to x plus 5. I've got two copies of x plus 5. That's what we're doing. Going with

the same analogy over here with the adding the boxes, it would be like this. Here's my box, here's one box, here's two box. That's kind of the idea.

- 1.36** The nice thing about algebra is, once you get the hang of it, you realize that the symbols really are arbitrary. Whether I call them x or y or smiley face or whatever it is, a square, it doesn't matter. We're just trying to manipulate the symbols in certain ways according to mathematical rules. But now I can say, "Oh, sweet. I know how to do this," because these have like terms. And so I can just combine the like terms. I can say, OK, well, I have an x here and I have an x here and they're connected by an addition. So what is x plus x ? Well, that's 2 times x . And then I have a 5 here and a 5 here, and 5 plus 5, that's 10.
- 1.37** And so I can say, "Oh, sweet. So when I have an x plus 5, and I multiply by 2, that's equal to $2x$ plus 10." And I would say, "Well, that's great you've got the answer, but what's the underlying principle? What's the underlying principle that we're getting at here?" What I really did here is, this 2 kind of came in and it multiplied the x . It then came in and multiplied the 5. Do we agree? 2 times x gives me $2x$, and then 2 times positive 5 is positive 10. And that's what we get. And that's what the distributive rule tells us.
- 1.38** So the distributive rule says that, if you have a quantity, whatever that is, let's call it a , and you bring it in to multiply b plus c —we've been using one variable, and now I'm just throwing 3 of them at you, a , b , and c —whatever the quantities are, the rule is, this a comes in and multiplies to b to give me a times b , and then this a comes in to multiply to c , to give me a times c . That's my distributive rule. That's how we handle that.
- 1.39** Now, one thing you need to be abundantly cautious with is negative signs. Negative signs are like the bane of the existence in mathematics, which is really frustrating because minus signs are super important. But you have to be very careful with them when dealing with this. I want to illustrate this in the following way. Now, people get mad at me for saying this because they disagree with me, and that's OK. I think having disagreement is OK, some of the time. All right.
- 1.40** But addition and subtract are literally the same thing. And that seems really weird. You're like, "Wait, what are you talking about? If you give me 2 plus 2, that's 4, and you're saying that's the same as subtraction?" I'm not saying the answer is the same. I'm saying, in principle, they can be viewed as the same thing. OK, well, how do I see that? Well, if I take 2 minus 3, how do I see that as addition? How do I see that as addition? Well, I can view this as taking the quantity 2 and adding it to the quantity negative 3. Do we agree? If I take 2 minus 3, it's the same thing as just taking the concept 2 and adding it to the concept negative 3.
- 1.41** So you can view subtraction and addition very similarly. That's what I'm getting at. Again, I'm not saying that 2 plus 3 and 2 minus 3 are the same output. That's clearly false. But I'm saying that, conceptually, they're the same thing. You can view them as

one and the same. With that in mind, we can think about distribution in the following way. Let's suppose I had something where I want to combine expressions. I want to take a 3 times x minus 1, and I want to subtract out 4 times x plus 2. I want to simplify this down using our distributive rule that we just came up with. How would I approach this particular problem?

- 1.42 The tricky thing that you have to keep in mind is this negative. That negative. And I want to get you to think about this in a couple different ways. One thing that I think has to be said about mathematics in general is, and this is a question I get from students all the time— the number one question I get from students, especially when they first start learning mathematics, is they say, "So what is the answer you want? What's the correct answer?" And I think that's a very— personally, I think that's very sad. Because mathematics is really about creativity. Mathematics is about coming up, using underlying principles to solve problems.
- 1.43 So what do you mean by, "What is the answer you want me to give you?" It's like, I want you to give me your thought process and your creativity in trying to resolve the problem at hand. I don't care how you do it. As long as you use the principles correctly and get to a certain conclusion, and you can justify that conclusion, I don't care if you're doing it the way I want you to do it. Does that make sense? There's not one way to get to the answer, just like there's not one way to get to a physical destination. There's a lot of different roads or airplanes or whatever that can get you to your physical destination. As long as you get to the destination, that's the important thing.
- 1.44 How you get there doesn't necessarily matter, so long as you're using correct principles. So I want to illustrate how to simplify this in a couple different ways. One way to do it is to say the following. Well, I can distribute the 3, using my distributive law that I just came up with. So 3 times x would give me $3x$. And then it's like, OK, well, what do I do with the minus? Well, I can view this a couple different ways. I can view this as 3 times negative one, or I can view this as 3 times positive 1, but it's separated by a negative sign. However you want to think about it is completely fine. 3 times negative one is still negative 3. 3 times 1 is 3, but if I put the negative sign in it, either way, I'm going to get a 3 with a minus sign.
- 1.45 That simplifies this part. But then the question here is, well, what do I do? You say, "Well, Robbie, I don't know, what number am I looking at here? Am I looking at the number 4? And the number 4 is distributing into the x plus 2? Or am I looking at this as the number negative 4 and distributing that into the quantity?" The answer is yes. It doesn't matter how you view it, as long as you're consistent. There's a couple of ways that you can do this. You can say one way is, view this as the number negative 4 and distribute negative 4 into both quantities. What would I get? Negative 4 times x is minus $4x$, and then negative 4 times positive 2 is negative 8.
- 1.46 That's one way to think about it. That's one way to think about it. Alternatively, I could have viewed this as, what? I could have said, well, this is equal to 3 times x minus 1

plus, let's do this, minus 4 times x plus 2. But now I'm going to say, well, I'm going to leave the minus sign in there and I'm going to distribute the positive 4 through. What would this look like? This would look like $3x$ minus 3 minus, OK, if 4 comes in, I'm going to get a $4x$ and I'm going to get a plus 8, but the minus sign has to stay outside.

- 1.47** Now I'm going to bring the minus sign in to each term. If the minus sign comes in, you think of minus as minus 1. If I bring minus 1 into both terms, I get my $3x$ minus 3 minus $4x$ minus 8. Either way, I don't care how you think about it. Again, plus and minus, how you want to think about it, 2 minus 3 or 2 plus negative 3, it doesn't matter to me.
- 1.48** Now, once we have it in this form, we can combine our like terms, just like we did previously. I've got 3 x 's here, and I'm going to take away 4 of them. Again, physically, that doesn't make sense, but mathematically, we can say I'm going to have negative x . Going back to our monthly debt example, if I make \$4,000 and I spend \$4,500, then I have negative \$500. I'm \$500 in debt. OK, so if I have $3x$ dollars and I overspent by $1x$, then I'm going to have negative x . I'm going to have in debt. OK.
- 1.49** And then negative 3 minus 8, well, if I have negative 3 and I take away another 8, I'm going to get minus 11. OK, so that would be my simplified expression for this entire thing. OK. So then where do we go from there? All right, where do we go from there? Simplifying expressions is one thing, but what's the utility to it all? Why does this matter? OK, well, let's go back to our previous wage example. We were talking about wages. Remember, we had that our wages was equal to my base salary of 1,000 plus 100 for every widget that I sold. That was our wages formula.
- 1.50** What if I ask myself the question, OK, how many widgets do I need to sell so that I come out with \$5,000 at the end of the month? That's what I want to understand. That's a natural question. It's a very natural question. How many widgets do I have to sell this month in order to make \$5,000 by the end of the month? How would I couch that in a mathematical framework? The question is, so how many widgets to sell to make \$5,000 this month? Why is that a natural question? Why is that a natural question?
- 1.51** Well, think about it. In your day-to-day, you have monthly expenses. You have your rent, your mortgage, you have your car payment, you have gas, food, insurance, whatever the case may be. Whatever your monthly expenses are, you need to be able to make sure that, by the end of the month, you can cover those monthly expenses. Let's say my monthly expenses are \$5,000. Well, then it's a natural question to say, "Well, how many widgets do I need to sell so that I make \$5,000 by the end of the month, so I can at least break even?" It's a very natural thing to ask.
- 1.52** OK, well, mathematically, what am I really saying? I'm saying, mathematically, find x , such that, such that what? Such that the \$1,000 in salary that I make every single month plus the 100 times x is equal to my 5,000. That's what I'm trying to figure out. I'm trying to say my wages, that's what I want to be 5,000. So I'm making a variable

assignment of w is equal to 5,000, and I'm saying that that's equal to 1,000 plus $100x$. It begs the question, how do we actually solve this now? How do we solve this corresponding equation? All right.

- 1.53** Again, it's an equation. Why? Well, for silly reasons, there's an equality sign there. There's an equal sign, so it's going to be an equation. It's not an expression. I have an expression on the left-hand side, and I have an expression on the right-hand side, and I'm setting those two expressions equal. And by doing that, it becomes an equation. And so I want to understand, how do I actually make sense of solving this? The way I like to think about it is the following. I like to think about it like a little balance beam.
- 1.54** Or you can think of these scales, where they have the little balance where you put certain weights on one side and you put certain weights on the other, and then eventually it balances out and all of that. I like to think about it like this. I like to think about this center point as my equality sign. And whatever's on the left-hand side is the expression that I have on the left. So that's my 1,000 plus $1,000x$. I have 1,000 plus, or sorry, 1,000 plus $100x$. And then on the right-hand side, I'm going to have 5,000.
- 1.55** So I'm very physically minded when it comes to trying to understand things. So what I have in mind is, I imagine I've got, say, 1,000 pebbles on the left-hand side representing the 1,000. And then I have 100 x 's, whatever those are. So you can imagine it like this. On the left-hand side, I've got a bunch of little pebbles here and so on. I've got 1,000 of those. And then I've got 100 x 's. Let's call those boxes, whatever those are. Those are all x 's. I've got 100 of those. That's what's sitting on the left side of the balance.
- 1.56** And then on the right side of the balance, what do I have? Well, if I had 1,000 of these little pebbles over here, it's like I've got 5,000 of those pebbles over here. That's the idea. So if you imagine you have in front of you, you've got 1,000 pebbles plus 100 of these little boxes on the left-hand side, and then you have 5,000 pebbles on the right-hand side; and you want to try and say, "What am I allowed to do to make sure that they balance? What am I allowed to do to make sure they balance?"
- 1.57** So, for example, let's say I have the balance that looks like this, so your guys' left, my right. The right-hand side, I've got two black markers. On the left-hand side, I've got one black marker and one red marker. In order to keep them balanced, what I could do is, I could say, well, what do they have in common on both sides? What do they have in common on both sides? Well, they each have a black marker. So what I could do is, I could say, OK, let's take a black marker away from this one.
- 1.58** And then, in order to make sure it stays balanced, because if it was balanced before and I take something away only on one side, then it's going to tip the scales. Do we agree? It's going to tip the scales. OK. And so what do I need to do? Well, I need to remove the same quantity or the same weight or whatever it is on the right-hand side. So if I fix that, then suddenly it comes back to balance. And if I wanted to say, OK, well, then what does the red thing have to equal? Well, then the only thing on the left is the

red thing. The only thing on the right is the black thing. The red marker would have to then equal the black marker.

- 1.59** That's the idea that I'm trying to get across in terms of the balance. OK. So what could I do now? I could say, well, I have 1,000 of these little pebbles on the left, and I've got 5,000 of them on the right. So in order to balance them, I could just take away all of these 1,000 over here, but I'd have to take away 1,000 of them on this side. Does that make sense? In order to keep the balance at equilibrium. OK? So how do I do that? Algebraically, what am I really saying? I'm saying, if I have 1,000 plus $100x$ is equal to 5,000, I'm saying let's subtract 1,000 on this side, and whatever I do to one side, I have to do to the other to keep things balanced. That's really what I'm saying. OK?
- 1.60** So if I take 1,000 pebbles and I throw all of them away, I don't have any more pebbles on the left-hand side. What's the only thing that I have left? I have $100x$. If I had 5,000 pebbles on the right and I take away 1,000 of them, how many do I have now? Well, I've got 4,000 left. OK, so I've got 4,000. OK, that's good. Why is that good? Well, I've eliminated some of the terms. I've made the equation smaller, and I've given myself a better chance of being able to figure out what the corresponding quantity x is.
- 1.61** Now, once I get to this point, I can say, OK, well, if I have $100x$ boxes on the left and I've got 4,000 now, so now this is 4,000. If I've got 4,000 pebbles on the right, I need to somehow say, OK, well, how do I isolate x ? How do I isolate x ? Well, how would I only get one x left? Well, I need to divide out all of the 100 x 's in order to get one single x . What that means is, you're going to have to divide out by however many you have. That's my 100. But again, whatever you do on one side, you have to do on the other. I'm dividing that out.
- 1.62** The 100 divided by 100 cancels. That gives me my x . And 4,000 divided by 100, you just cancel the two zeros. I get 40. So that's great because that tells me what? That says, what did I just answer? What was the question that I just answered? The original question was, how many widgets do you have to sell in order to make \$5,000 by the end of the month? And we just answered the question by saying, you have to sell 40. Now we can go back and check our answer because, look, if I sell 40 widgets, how much money do I make on that? Well, I get \$100 per widget, so that gives me four grand.
- 1.63** But remember, I had my base salary of \$1,000, so my entire wage is indeed what I was looking for. I now have \$5,000. OK? Well, I mean, I don't know about you, but my monthly expenses don't typically work out to be a perfectly nice number. It's like, my car payment isn't just an even \$400. It's \$362.42 or something like that. It's not that clean. So what we want to try and understand is, like, well, OK, so what if I wasn't trying to find the number of widgets that I need in order to get a specific value? What if I have a range to some extent?
- 1.64** OK, let's say my monthly expenses are \$3,927. Well, I only get \$100 per widget and I get \$1,000 in salary. So I'm not going to be able to say, oh, sell 4.7 widgets in order to

get the right amount of money by the end of the month. That seems a little ridiculous. If I'm selling phones or laptops or whatever the case may be, if I'm selling those objects, I can't sell one-half of those objects. That's not going to work. OK, so we need a mathematical framework which allows us to handle that, and that's where inequalities come into play. We want to talk about inequalities.

- 1.65** Rather than talking about equalities, where I have an exact quantity, I want to give myself an inequality, which gives me some flexibility. It gives me a range of values that I'm allowed to talk about. OK, so we're going to talk about inequalities now. Inequalities are a little scary when you first see them, and that's OK. It's completely fine. They're scary for everybody because their notation is a little weird to first understand. And then understanding that there's flexibility with the notation is another issue. Let's talk about inequalities. Inequalities versus equalities. All right.
- 1.66** So I claim that, in our day-to-day language, we actually have equalities. They show up quite a bit. OK, so let me illustrate that to you. Let's suppose you're applying for a job and you are adamant that you are not taking that job unless the salary is at least \$75,000 a year. You're like, it's not even worth my time to work there if I don't make at least \$75,000 a year. We use this language all the time—at least, at most, things like that. What are you actually saying? OK, so you want at least \$75,000 per year. That is your condition.
- 1.67** You apply for the job, they don't tell you what the salary is, because that's like a thing. And so you apply for the job, they say \$75,000 a year once they offer you the job. What does that mean? That means, what you're claiming is that you will deny the offer if they do not come to the table with \$75,000 or more per year. That's what you're claiming, right? So you apply for the job, the employer says, "Hey, we want you. We're going to pay you \$60,000 a year." And you're like, "Thank you, but no, thank you. I won't take anything less than \$75,000 a year."
- 1.68** But if the employer comes to you and says, "Hey, we really want you. You're a great fit for the job. We're going to pay you \$80,000 a year, \$80,000," well, then what would your response be? Your response would be, "I'll take it." That's not an equality thing. You're not saying, "I am only going to take the job if and only if you give me exactly \$75,000." You're saying, "I'm going to take the job, provided you give me at least \$75,000 a year."
- 1.69** So what does this "at least" really mean? If we set a variable, we say that x is equal to the pay per year, then what you are saying is that x needs to be greater than or equal to 75,000. That's really what you're saying. So this translates mathematically to saying that x is greater than or equal to 75,000. So this is greater than or equal to. Why do I say greater than or equal to? So this little, I don't know, carrot symbol that's going to the left is the greater than part of it. And then I stick a line underneath it to say that it could be equal to. So it's what's called an "or" statement. That's why it's a greater than

or equal to. It's either greater than, specifically, so maybe \$80,000, or equal to, 75,000 on the nose.

- 1.70** Now, the confusing thing about the notation is that this is quite literally the same statement if I reverse the order, but I keep the symbol in the right direction. OK, so what if I took the 75,000 and put it on the left and I took the x and put it on the right? Well, if I put 75,000 here and x here, well, then what would the symbol need to do in order for it to be the exact same statement? It doesn't make sense to keep the symbol the same direction. It doesn't make sense to keep the symbol the same direction because I've switched their orders. All right?
- 1.71** What I need to do is say, well, look, the point of this is pointing at the 75,000. So in order to keep this the same, I need that point to still point at the 75,000 and then keep the equality there. Those two are quite literally the same thing. What's confusing is that when you deal with equalities, when you deal with equalities, this actually happens, but you don't notice it because the equality is a symmetric-looking thing. For example, to contrast this, if I said x is equal to 2 and 2 is equal to x , you would say those two quantities, you're saying the same thing.
- 1.72** What I'm really doing, though, is if I color this, if I put two little red dots on this side of the equality, by flipping it around, where do the two dots go now? Two dots are on the right. They're over here. But we don't see that because the equal sign is a symmetric-looking thing, geometrically. Whereas if I had two red dots on this side of the inequality—those are the ones that are close to x —then what needs to happen? Well, those two red dots need to remain close to x on that side. So it switches.
- 1.73** So that's partly why inequalities are a little confusing, is you're like, well, wait a second. Is this really the same quantity as this quantity? Verbally, it's yes, because the way you read this is, you say x is greater than or equal to 75,000. You read this by saying 75,000 is less than or equal to x . So if you think about that statement, you're like, oh, that's saying the same thing— x is greater than or equal to 75,000. It's the same thing as saying 75,000 is less than or equal to x .
- 1.74** Now, we can represent this on a number line. So this is our number line where we locate 75,000. And I want to say, what are the values of x that are valid here? Well, am I allowed to be 75,000? Yes, I am. So I can fill in the dot there. But I want everything that's greater than or equal to 75,000. So I'm going to shade in everything to the right on my number line. So 80,000 lives up here somewhere. If they offer \$80,000, that's up here somewhere. So that's valid.
- 1.75** But what is not represented there is the \$60,000 that they came up with with their valuation. "We're going to pay you 60 grand a year." You say, "Thanks, but no, thanks." So that's inequalities. Now, inequalities, there's four types. There's four types of inequalities. You have greater than. You have greater than. You have greater than or

equal to. So, greater than or equal to. Then you have the analogs of those, with the less thans. You have less than, and then you have less than or equal to.

1.76 Now, just like addition and subtraction, and I said they're one and the same in terms of principles, you could argue that these are one and the same as well, greater than or less than. You can say, look, if I have a quantity that x is greater than 2, right, and I want to claim that x is greater than 2, that's great. Well, you could view that as a less than, by saying that 2 is less than x . Just like if I want to say 3 minus 2, I could view that as 3 plus negative 2—depends on how you want to look at the corresponding operation.

1.77 Now, the way that these look in terms of these intervals, OK, so if I want to look at my number lines associated to that, let's suppose all of these are x is greater than 2, x is greater than or equal to 2, x is less than 2, or x is less than or equal to 2. What do these look like geometrically from an interval standpoint, so our number line standpoint? If I look at this one, I locate the number 2. So let's say here's 2, here's 3, here's 1, here's 0, and so on. And I want to say that x is strictly bigger than 2, meaning x can't equal 2. It has to be strictly bigger than 2. It could be 2.1 or 2.3 or 2.0005, whatever, just something slightly bigger than 2.

1.78 The way you denote that is, you have an open circle at 2, and then you include everything that's bigger than 2. So you would say, OK, what's bigger than 2? Well, everything this direction. Everything from there. That's how you can graphically represent these things. x is greater than or equal to 2, what would I have? OK, I've got 0, 1, 2, 3. I'm allowed to equal to, so I have a closed-in dot there, meaning that you're allowed to equal to, and then everything bigger than 2. Then I'd have everything this way.

1.79 What about x is less than 2? Same thing, except which direction am I going to shade now? 2, 3. Open circle because it's a less than, and I'm going to shade everything to the left, because everything is less than 2. Everything less than 2. That direction. Then x is less than or equal to 2, what do I get? Zero, 1, 2, 3. I'm going to have everything that direction as well. That's how we can represent our corresponding inequalities for these things. OK.

1.80 So let's look at the following thing. What if I wanted to ask myself the question, going back to the wage example that we had, let's suppose we have the following setup. Let's suppose we have the following setup. Let's try and make this as realistic as we can. We're going to say, all right, I have certain expenses each month, and I make a certain amount of money each month, and I want to try and understand the following situation. Let's say our expenses look like the following.

1.81 Our expenses are going to be, you have rent or your mortgage. You've got food, you've got a car, you've got insurance, you've got all sorts of expenses down here, each and every month. And at the end of the month, you total up your expenses. And let's say the total expense is like \$3,942, whatever that is, for that particular month. And then

you say, OK, well, I also want to put away something for savings. I'm saving for a trip. I want to put away \$500 a month in savings. So you can think of that as like an expense, to some extent. I want to be able to have an additional \$500 that I can spend on savings. So I want \$500 a month.

1.82 So what I need for my total, my total per month, I need what? 3,942 plus 500. OK, so what is that? That's 4442. That's what I need per month. That's what I need per month. OK, so going back to the sales example, I need my wages, which were 1,000 plus 100x. I need to ask myself the question, how many widgets do I have to sell in order to make \$4,442 by the end of the month? We want to know, what is x, so that 1,000 plus 100x is equal to 4,442? That's what we're trying to answer. We're trying to balance the equation. We're trying to balance both sides.

1.83 OK, so algebraically, what can I do? I can take away 1,000 over here. I can take away 1,000 on this side. What am I left with? Those cancel. I'm left with 100x. This becomes, what—3442? That's what I get. Then I need to divide out by the 100, divide out by the 100 on both sides. OK, let's cancel. The 100 just does what? Decimal-wise, it just moves the decimal over twice. So it says that you need to sell 34.42 widgets. And you're like, well, I can't do that. I can't do that. If I'm selling, say, insurance plans or something like that, I can't sell 34 insurance plans and then sell 42 hundredths of an insurance plan. It's like, you either sell the plan or you don't. It's a discrete thing. You can't sell a fractional insurance plan.

1.84 So what you really want to say is, what do I really want to do? Well, I want to sell 35. I want to sell 35 insurance plans. I want to sell 35 widgets over the course of the month. So we really need to sell 35 widgets that corresponding month. Now, how do inequalities come into this? Inequalities are really saying, we want x to satisfy what? We want it to be 35, but are you going to be sad if you sold 40? No, it's just more money in your pocket, right? Are you going to be sad if you sold 30? Well, you're going to be like, "Yeah, I'm probably not going to be able to put \$500 in the bank each month."

1.85 But you don't care if you sell 35, 36, 37, 40, 50, 100. You don't really care about that. So you want x to be greater than or equal to 35. That's what you want at the end of the day, right? You'll settle if you sold 37; you'll settle if you sold 38, 39. You'd be happy. But you're not going to be happy if you sold 34 or 33 or 32 or 31. That's where these inequalities are coming into play. So how do we represent this? Inequalities we can represent using this notation, x is greater than or equal to 35. Again, we're saying that x could be anything greater than 35.

1.86 In our example, we're selling concrete objects. And so we're saying that that's a discrete thing. It's an integer thing, 35, 36, 37, 38, 39, and so on. But this is literally saying, x is just bigger than 35. So it could be 35.2, 35.6, 35.7894, whatever it is. It's just something bigger than 35. What is our interval notation for this? Interval notation. The interval notation looks like this. You say, because it's an equal sign, because there's a

greater than or equal to, you're going to have a bracket. If it were simply just a greater than, you're going to have a parentheses instead.

- 1.87** So let me illustrate this. The interval notation is saying this: I want everything greater than or equal to 5. Remember on our number or 35, our number line said, if I have 35, 36, and so on, 34, I wanted everything from 35 and this direction. OK, going off forever. How do I notate that with intervals? I say, well, what's the furthest left point? The furthest left point is 35. What's the furthest right point? Well, there isn't one. There isn't one. It goes off to infinity on the right-hand side. OK, so I have 35 to infinity. It's just an 8 turned on its side, basically. It's a lazy 8. So you have 35 and you have infinity.
- 1.88** Now, can you ever attain infinity? Well, the answer is no. You can never do something an infinite amount of times. So you have a parentheses on the right-hand side. Parentheses just indicates you can never attain that value. You can never attain infinity. Now, on the left-hand side, can I ever attain 35? Well, yes, because it's a greater than or equal to. You can equal 35. You can sell 35 plans in a month. So you put the left bracket on there, but it's a bracket because of the equality. You can actually attain it, 35. If I had said x is greater than 35, so I can't attain 35, then what would I do? I would say I have got 35 to infinity. I can't attain infinity, and I can't attain 35. I put a parentheses on the left-hand side.
- 1.89** It's just a notation. It's just a notation. OK, that's our notation for inequalities. But just like with expressions and equations, we want to try and understand, how do we solve inequalities? How do we solve them? Let me give you a quick example. This will be our last example. Solving inequalities. So let's say I go back to the monthly expenses and wages that you have. You go through this particular example. You need to make, you need to say, find x such that my wages, 1,000 plus $100x$, is at least my expenses, is at least my expenses.
- 1.90** So let's say my expenses, let's go back to the example that we had. Our expenses here were, that we wanted with savings, was \$4,442. So I have to make at least \$4,442. I'm going to say that's greater than or equal to \$4,442. I want it to be at least that. Just like with the example with the employment, I need to at least make \$75,000. You give me \$80, I'm fine. You give me \$90, I'm fine. But you give me \$60, I'm out. OK? Makes sense? How would we go about solving this? You think about this like an equality. You think about this just like an equality, like you had over here. What would I do? OK, let's subtract \$1,000 from both sides.
- 1.91** OK, so that's your step one. Let's subtract \$1,000 on both sides. OK? That's great. Those then cancel. I end up having $100x$ is greater than or equal to—what do we get? 3442. Now I want to divide by the 100, just like I did in the example over here. I divided by 100 on both sides. What you have to be abundantly careful with, when dealing with inequalities, is that when you divide by this coefficient, the 100, if this coefficient is positive, the inequality doesn't change. The inequality stays the same. But if this

coefficient were, say, negative, the inequality will flip, and I'll illustrate that very quickly.

- 1.92** So if it's positive, the inequality stays the same, and I'm fine. It says you have to sell at least 34.42 widgets. In context, that means you need to sell 35 or more. But if I gave you an inequality that looked like this, for example, it looks like $3 - 2x$ is greater than or equal to, let's say, 6. If I gave you something that looks like this, we'd approach it the same way. Subtract 3 on both sides. I subtract 3, I subtract 3. I'm left with $-2x$ is greater than or equal to 3. And then I would just say, oh, divide by negative 2. Divide by negative 2.
- 1.93** What that does is, it flips the inequality. It flips the inequality, and I'll show you why. You can either just accept that that's one of the rules of the game, is that when you divide by negative 2, it flips the inequality. That's fine. I don't personally like things like that. I like to actually know why things are the way they are. I want to illustrate that. If I divide by minus 2, divide by minus 2, the x , the inequality flips. It goes from greater than or equal to less than or equal to, minus 3 over 2. Why is that the case? Well, addition and subtraction are the same, right?
- 1.94** OK, so this inequality is the same as this. Why don't I just add $2x$ to both sides? Let's add $2x$ on both sides. This becomes zero, and this becomes $3 + 2x$. And then let's just subtract 3 on both sides. Do we agree? I just add one thing to the other side. If I have a balance and I had the two black markers here and the red and black marker here, and I simply just replace the red and black marker here and the black markers over here, it's still balanced. It doesn't matter what side I put them on, as long as the symbol is the correct symbol.
- 1.95** But then dividing by positive coefficients doesn't change the inequalities. So I get $-\frac{3}{2}$ is greater than or equal to x . But we said as long as the arrow is pointing in the correct direction, I get the same answer. The point is pointing at the x here. The point is pointing at the x here. This one is saying, x is less than or equal to negative 3 halves. This one is saying negative 3 halves is greater than or equal to x . But this one over here, we didn't divide by a negative. We simply just moved the negative to the other side and switched them out.
- 1.96** But that's what justifies the inequality switching. When you have an equality, again, when you have an equality and you put points here, with those being red, and you divide by minus 2, those cancel and you get x is minus 3 halves. But where did the red dots go? The red dots are there. You just can't tell. That's the beginning of algebra with our expressions and equalities, and we'll delve into more of this in the coming class.

Lecture 2: Linear Equations

At a glance — Linear equations.

- 2.1** All right, so now that we have the basics of expressions, equations, inequalities, and things like that, we want to try and broaden our understanding a little bit more and include other variables rather than just a single variable. So how can I, originally, in the, in the examples that we've had, we had one variable which was controlling, in the examples that we did, the number of widgets that I was selling. OK? Now what we want to do is say, well, what if I'm selling multiple things and those things are giving me commissions at different rates?
- 2.2** OK, let's say I'm selling two things now, rather than selling simply just one widget. I'm selling, for concreteness, let's say, I'm selling insurance plans on the one hand, and there's a certain commission associated with that. And then I'm also selling, I don't know, let's say, I'm doing medical plans and I'm doing life insurance plans as well. OK? I'm doing those two different things from an insurance standpoint. All right?
- 2.3** The medical ones give me a certain commission, let's say, \$50 a plan, and then the life insurance ones give me, say, \$100 a plan. And so I sell those throughout the month, and I want to try and understand what my wages are. So I'm going to have two different variables here—the number of medical plans that I'm selling versus the number of life insurance policies I'm selling, right? I'm going to have this difference, OK?
- 2.4** So the question we want to ask is, well, can I visualize this? Can I actually write down something visually, not just the algebra, and manipulate the symbols and get to the answer that way? Can I actually graph these things and understand what they look like? And the answer is yes, but we need to understand, what is that coordinate system? So we need to get some more vocabulary out of the way. So if I have a single variable, so one variable, single variable, linear equation, linear equation. One variable, I'm going to call it x , and a linear equation.
- 2.5** What do I mean by that? These things look like, or they can be made to look like, I should say, ax is equal to b , where x is my variable. OK? And a and b are numbers, just whatever I want them to be. OK? So we saw a lot of these. OK, we saw a lot of these. We had something, for example, for example, we could have something like $2x$ is equal to negative 6, something like that. That's completely fine. But what do I mean that it can be made to look like that?
- 2.6** Well, we had other examples that were things like $2x$ plus 3 is equal to 5. And I can say, well, that doesn't look like this. It doesn't look like number variable equals number, but it can be made to, right? It's equivalent to it. Why? Because I just need to subtract 3 on both sides, and now I have $2x$ is equal to 2, and I can say, oh, OK, now that looks like

this. OK? So you might have to do some algebraic manipulation to get it to look like ax equals b , but at the end of the day, that is a linear equation. OK?

- 2.7** Now a two variable linear equation is going to look like this. Now I've got two variables associated to it. So I can have some number times one of my variables plus, say, another constant—let me call it c , because I'm going to use b later— is equal to d , OK, where now x and y are my variables, and a , c , and d are just numbers. OK, that's fine. So going back to the example that we had regarding sales, OK, we could have something that's like, for example, let's say every medical policy that I sell gives me a \$50 commission on that. So if I say I take \$50 times the number of— this will be the number of medical plans that I sell, OK?
- 2.8** So x is representing the number of medical plans that I sell. I sell two of them, I got \$100 in commission, I sell six of them, I made \$300 in commission and so forth. But then let's say, I have life insurance policies that I'm also selling. And so these would be the number of life policies, OK? And that's going to give me an additional \$100 for every policy that I sell. OK, and then what would this equal? Well, this would be equal to—let's say I want this to be \$5,000. OK, let's suppose I want it to be that. Then what am I really saying?
- 2.9** I'm saying, I want my total commission that I make per month to equal \$5,000. So I need to sell—how many medical plans do I need to sell and how many life policies do I need to sell? OK, that's kind of the question that we want to keep in mind. All right, now this introduces a level of complexity here, right? Because in the original example, when I had a single variable problem and I was selling my widgets, whatever it was, right, I could only sell that widget. So I could only ask myself the single question, how many widgets do I need to sell to make x amount of dollars at the end of the month, right? I'm only selling one product.
- 2.10** Now, this gets far more complicated because it's saying, well, how many medical plans do you need to sell and how many life policies do you need to sell in order to make your \$5,000 in commission? Well, that's very different. For example, I could say, well, in a given month, I sold zero medical plans, so I didn't get any commission for selling those. But I sold 50 life policies, and if I sold 50 life policies, then that would give me my 5,000. Or I could say, I didn't sell any life policies, and I sold—what would I need here? 100 medical plans, and I still made \$5,000.
- 2.11** But what if I sold, you know, I don't know, like, 10 medical plans, how many life policies would I need? Or if I sold 20 medical plans, how many life policies would I need? Or if I sold 20 life policies, how many medical plans would I need? Does that make sense? So they play off of each other. OK, there's a dependency here. And so we want to try and understand, can I represent this visually? Can I represent it visually? Can I draw it out and say, like, this is what my wages are going to be, given certain data. And the answer is yes.

- 2.12** And so this gets to what's called the Cartesian coordinate system. Cartesian coordinates. It's Cartesian because it's from Rene Descartes. He was the one who initially came up with this. And his kind of groundbreaking contribution to this was, he had, on the one side, algebraic things such as these, for example. I've got this algebraic expression in terms of variables. I've got an expression on the left, expression on the right, an equation associated to that, and coming up with a visual depiction of how that's being represented. That was in what's typically considered, what's called analytic geometry.
- 2.13** So he was the one who kind of married algebra with analytic geometry, basically. So that's why this is named after him. And what he observed was, he said, "Look, I could graph this in a coordinate system." So he gives himself two coordinates. You can think about these as kind of like directions, to some extent. I have a vertical direction and then I have a horizontal direction, OK? And as long as I specify some sort of base point which is considered the origin, OK, that's where the two directions intersect, OK, that's my origin. So the origin.
- 2.14** As long as I have an origin, then any location in space I can represent as a certain distance in this direction and then a certain distance in this direction, OK? And that kind of makes sense, if you think about it, right? If I'm, if I'm standing, like, if I declare this to be my origin, OK, where I'm standing to be the origin, and I allow myself to move in this direction here, and I allow myself to move in this direction here, OK, so that's like my y and my x directions, OK? Then I could say, well, where one of you is sitting, I can describe where you're sitting by saying, "I go over a certain number this way, and I go up a certain number that way." Right?
- 2.15** I would simply walk over here a couple distances and then walk over there a couple distances, and I would be in this chair, or I walk over here and up a little bit, and I'd be in that chair. So it's a way of describing locations. So how does he do it? He says, "Look, if I have a point out here, how do I represent that point?" Well, he gave this the label x and he gave this the label y, saying that this is my "x" axis, and this is my "y" axis. And it's simply saying, if I want to represent this point, I need to go over a certain distance in the x and I need to go up a certain distance in the y. OK?
- 2.16** Now, if this point is given by, let's say, just for the sake of example, let's say this is 6, comma, 6, 5, that's how that point is going to be represented—where the 6 is representing this distance from the origin out here is 6, and the distance from the origin up to here is 5. OK, now again, for vocabulary, the 6 is called the x coordinate and the 5 is called the y coordinate. OK? So every point can be represented by a pair, sometimes called a tuple, OK, of an x coordinate and a y coordinate.
- 2.17** So if I wanted to get this point, let's say down here, maybe this point is minus 3, minus 2. What does that mean? I went over 1, 2, 3 here, and I came down 1, 2, here. Right? That's how we represent that. Now, obviously, that is not the only way to represent how you get from a certain point to another point. OK. We don't typically walk this

way, right? We don't walk this way. All right, what do we do? Well, we walk from a point to another point. We don't say, if I want to walk from here to here, it's like, oh, OK, in order to get there, I'm going to walk this way and then this way. Like, that's kind of ridiculous. Right?

- 2.18** But we see this happening a lot actually in infrastructure, for example. OK? So, where I live, there's basically the grid system is how a lot of the city is laid out. OK. So you have this grid system where you have, these represent the roads and then the buildings are represented by the squares. The buildings are represented by these things in here. OK? So this type of coordinate system actually makes a lot of sense in terms of, like, where I live. OK, because if I'm at this intersection, something like this, and I want to get to this intersection, what do I need to do? Well, there's multiple ways to get there, but I could go over this way and up this way, or up this way and over this way, or there's some sort of combination, depending on traffic, that you can kind of get there.
- 2.19** But it is kind of a very boxy-like movement, right. But if there's no buildings or infrastructure actually allows me to go from point to point, then I may as well just take the straight-line distance between the two. Does that make sense? OK, so what Descartes realized is that, when looking at linear equations in two variables, so things that look like this, you can actually graph them on this by simply saying, look, just like the example down here said, there's a lot of different ways to get to 5,000. I could sell certain number of medical plans, a certain number of life policies.
- 2.20** But in particular, if you fix one of the medical, if you fix medical plans, you say, "Look, I sold 20 medical plans this month. I'm going to sell 20 medical plans," so x is going to be 20, you can say, well, how many life policies do I need to sell in order to make the 5,000? Well, that's dependent on the number of medical plans you sold, right? And vice versa, if I say I'm going to sell x number or y number of life policies, how many medical plans do I need to sell in order to make my five grand, right? They depend upon each other. OK.
- 2.21** So the game, when trying to figure this out graphically, is the following. I'm going to give you an example with numbers that are a little more tractable or a little smaller. So we can kind of graph this out. But the game is, can you solve for one of the variables in terms of the other one? OK. And this is a difference. This is kind of a different way of thinking than the previous, the previous class, where we were talking about simply just solving for a single variable in terms of that number equals, that variable equals a number. We're not saying that now. We're saying, we want to solve for a variable in terms of an expression involving another variable, OK? It's different.
- 2.22** So here's kind of the example. So given, let's say, $2x$ minus y is equal to 4, OK? Given $2x$ minus y equals 4, I want you to visualize this on the xy coordinate system, the Cartesian coordinate system. So given this, I want you to visualize this in the xy plane. Now, I'm using a lot of different language here because, just like in our natural language, there's lots of different words that will describe the same thing, for example, right? And so

when you first start learning something, right, it's kind of confusing to be, like, oh, this object is whatever, right? This object is red. It is the color red, right?

- 2.23** And then somebody comes to you later and they said, "Well, I read that book a month ago." And when you first start learning the language, you're like, "That doesn't make—wait, you color red that book a month ago? Like, what do you mean?" It's like, well, it sounds the same. The language is different, right? It means something different. OK, I'm kind of doing the opposite of that right now. I'm saying, look, the xy coordinate system that we have, we can call this multiple things. We can call it the xy coordinate system. That's one way; that's referring to this. We can call it the Cartesian coordinate system because it's named after Rene Descartes. OK, so we can call it that.
- 2.24** I'm saying the same thing. I could call this the xy plane. Same thing. They all mean the same thing. They're just saying, you have an x-axis, you have a y-axis, and you want to represent, what is the graph of this thing? What does it look like on that, on the system? OK, but we're using different language to explain what that is. OK, so here's, here's an approach. Here's an approach, OK? We can solve for y in terms of x. That's something we can do. We have two variables. We can solve for one of the variables in terms of the other one. OK, we can try that and see how that goes.
- 2.25** Well, how would I do that? What am I really saying? I'm saying to write this as y equals something. That's the goal. Let's solve for y equals something, all right? So I come up to this equation and I say, well, how would I do that? How would I do that? Well, there's a lot of ways you can do that. I could say, look, well, y lives right here. And so I could just move the $2x$ to the other side. That's completely fine. OK, and this is, that's, this is very common, actually. OK, so let's look at this. If I have $2x$ minus y is equal to 4, I could say, let's subtract $2x$ from both sides. OK?
- 2.26** Personally, this is not what I would do. OK? Negative signs terrify me, right? And that's not hyperbole. I don't like negative signs. They freak me out. OK? So personally, I wouldn't do this. I'll show you what I would do and that, but either way, I want to show you that it doesn't matter what you do. You're going to end up at the same, at the same point. OK, so if I subtract $2x$ on both sides, that's what I get. What do I have? Minus y is equal to 4 minus $2x$. But I don't want negative y. I need to get rid of the negative. So I'm going to divide by—what is this, really? It's like negative one.
- 2.27** OK, so I'm going to divide by negative one. Divide by negative one. Those cancel. I get y is equal to—OK, what do I get here? Whenever you divide by something, it's always kind of confusing what you're actually doing. OK, so if we go back to fractions and how fractions actually work, right, if I have something that looks like, let's do, I don't know, x plus y divided by 2. And I want to say, like, what does that actually mean? Well, how do you add fractions, right?
- 2.28** If I asked, how do you add the fractions $\frac{1}{2}$ plus, let's say, 5 halves, something like that, you'd say, OK, well, they have the same denominator, they have the same bottom. And

when they have the same bottom, then you can just add the tops. So this is just going to be 1 plus 5 all over 2, which we would really just say is 6 over 2, and then call that equal to 3. That's fine. What I'm asking to do now is, I'm asking you to reverse that process. Start from the right and move to the left.

- 2.29** What do I mean by that? What does x plus y over 2 mean? Well, it really means, take this one, divide it by 2, and then take that one, divide it by 2 as well. Do we agree? I could say, what does 1 plus 5 over 2 really mean? And I could say, oh, it means 1 divided by 2 plus 5 divided by 2. It's the exact same thing. So I'm trying to kind of reverse that process. OK, so I get my x over 2 plus y over 2. So what do I mean by 4 minus $2x$ over negative 1? I really mean 4 divided by negative 1 minus $2x$ over minus 1. Which is what? 4 divided by negative 1 is negative 4. And negative $2x$ over negative 1 is plus $2x$.
- 2.30** And I finally have what I wanted— y is equal to something in terms of x . OK, but just like I said, that's a lot of negatives running around. OK? That kind of freaks me out on a lot of levels. OK, so how would I have approached this personally? I would have done the following. Again, the goal is solve for y in terms of x . That's the overall goal. OK, so what if instead of subtracting $2x$, I just said, oh, well, what if I just add y to the other side? What if I did $2x$ minus y equals 4, and I just said, let's add y to both sides and see what happens?
- 2.31** OK, so what do I get? I get $2x$ is equal to 4 plus y . And then I can say, well, to solve for y , I just need to subtract 4 on both sides. OK, so minus 4, minus 4, I get $2x$ minus 4 equals y . OK? And because equalities, you can just reverse the orders—remember, like, I could just put the y on the left and the $2x$ minus 4 on the right—I still get the same, the same thing. I get y is equal to $2x$ minus 4. So that's my answer. And you look at it and you're like, "Yeah, but, Robbie, that's not what you got when you did it the first way." And I would say, "Well, why isn't it what I got?"
- 2.32** What I got the first way was negative 4 plus $2x$. Well, algebraically, what is that? I can switch their orders. $2x$ is positive plus minus 4, yes? I can just switch their orders in that case. But what is plus negative 4? Well, that's just $2x$ minus 4. That's exactly what I have. So they're the exact same things. That's kind of the annoying thing with algebra. Somebody gives you their answer and they say, "But that's not what I got. I got this." And it's like, "You're literally saying the same thing," right? You're literally saying the same thing.
- 2.33** OK, so let's look, at the end of the day, we have y is equal to $2x$ minus 4. Great. OK, so we've now solved for y in terms of x . That's my $2x$ minus 4. Well, big deal. Like, what does that, what does that tell me? OK, well, maybe it tells me nothing, but that's OK. y is equal to $2x$ minus 4. OK, this is like a philosophical thing, but it's just a total aside. But solving for y in terms of x , maybe it tells you nothing. But does that mean that there's no value in doing that? Does that mean that there's no value in just doing this? Because

you're like, oh, well, it doesn't tell me anything. It's not worth anything. Like, it's not worth my time.

- 2.34** And it's like, well, you've, you've gained something through that struggle. Does that make sense? You gained something by trying to solve for something, even if whatever you end up at the end doesn't make any difference to your day-to-day. You still learn to do something, right? You still learn to do that, right? You go to the gym, you lift weights. At the end of the day, are you bigger? Not really. Does your body feel better? Eventually.
- 2.35** Over time, these things help you to develop a resilience, to develop a greater appreciation for the thing that you're actually doing, even if at the end of the day you're like, "I don't see any difference, right? I ran five miles." You're like, "I don't see any difference in the scale, or I don't see any difference in my endurance, or I don't see any difference in muscle tone," or whatever the case may be, right? In time, those things develop and they get better. OK, all right.
- 2.36** So y is equal to $2x$ minus 4. How do I graph this? One way to do this is to say, look, each point in the xy plane has an x coordinate and a y coordinate. So what you can do is, you can create a table now. OK, you can create a table where you're going to say, I'm going to give sample x values over here. And once I give you that sample x value, my y value, which is now $2x$ minus 4, is determined for me. Does that make sense? OK, go back to the, go back to the sales example. You're selling medical plans and life policies and you want that to equal \$5,000. If I tell you I sold 20 medical plans, you can say how many life policies you have to sell in order to make five grand, right?
- 2.37** If I tell you I sold 30 medical plans, you can say, ah, this is how many life policies you're going to have to sell. What that's saying is that, if I tell you x , x determines y , OK? So the language here is to say that the variable y is dependent upon x . And we'll make this very precise when we talk about functions and all that. But we're saying y is determined by x . So what I could do is, I could say, all right, I'm just going to start plugging in some values. So let's say I look at negative 1 here, I look at negative 2, I look at 0, I look at 1, I look at 2, and I just start choosing values of x .
- 2.38** OK, so let's say x is minus 2 minus 1, 0, 1, 2. And now I want to determine what the corresponding y value is. All right, so what do I do? I take that x value and I plug it into this formula to get the y value. OK, so what do I get here? I get, y is going to be 2 times minus 2 minus 4. What is that? 2 times negative 2 is negative 4. Negative 4 minus 4 is minus 8. So when x is negative 2, y is negative 8. When x is negative 1, I get 2 times minus 1 minus 4. So that's what? Negative 2 minus 4 is negative 6. When x is 0, I get 2 times 0, minus 4, which is negative 4.
- 2.39** You can kind of see a pattern, right, with what's happening. I'm going negative 8, negative 6, negative 4. What's the next one going to be? Well, we would guess it's going to be negative 2, right? Well, let's check to see if that works. OK, if I take 2 times

1 minus 4, what is that? 2 minus 4 is negative 2. And if I plug in 2, I get 2 times 2 minus 4, which is actually 4 minus 4 is 0. OK, so that's special. And how, like, so what's, so what's actually happening here is, I'm just specifying a value of x and I'm getting the corresponding value of y . All right?

- 2.40** OK, so how would I graph this? This is saying, when x is negative 2, y is negative 8. So in terms of my points, OK, maybe I make another column here that says my points. The points look like what? Well, remember, points in the xy plane have an x coordinate and a y coordinate. The x coordinate is negative 2. The y coordinate is negative 8. So this would be negative 2, comma, minus 8. That's how I would write that. This one I would write as minus 1, minus 6. This would be 0, minus 4. This one is 1, comma, minus 2. And this one is 2, comma, 0. That's what the points are in the Cartesian plane.
- 2.41** Now, you have to be very careful here because, mathematically, this could be very confusing. You could say, well, maybe these are intervals. These are saying the values that go in between 0 and comma negative 4. Because the same notation is used if you go back to the, the previous class when we were talking about widgets and inequalities, we were saying that if I looked at x is greater than or equal to a certain thing, I wrote it as this interval, OK? So you have to be careful with the context. Are these points or are they representing intervals? OK, in our case, we're talking about points, OK?
- 2.42** But once we have those points, we can go and graph them. We can say, OK, if I go over to negative 2, I'm going to go down negative 8. So 1, 2, 3, 4, 5, 6, 7, 8. So I go down negative 2, 8. There's a point here, minus 2, minus 8. OK, I can then go to the negative 1, negative 6. So negative 1, 1, 2, 3, 4, 5, 6. So that's somewhere here. And keep going 0, negative 4, 1, 2, 3, 4, 1, negative 2, 2, 0. And so on. And I just keep drawing these dots. And what I notice is that these dots, well, what are they forming? See if I can pull this off. They're forming a line. OK, they're forming a line.
- 2.43** So what we notice—so one observation to take away from this is that the graph of a two variable linear equation is a line, OK? Is a line. The proof is in the language to some extent. All right? We have linear. So what's this graph going to look like? It's going to look like a line in this case. OK, that's a terrible way of remembering it because it's very false when you go up to higher dimensions and more variables, OK. They don't look like lines anymore, but for this example, that'll work. So they look like lines. OK, well, big deal. What can I say about them? What can I say about them?
- 2.44** Well, if I step back and I look at this, this is one thing that I think is very important from a mathematical standpoint, is again, I'm not trying to teach you what to think. I'm trying to teach you how to think and how to approach things like this, OK? So if I ask you the question, OK, well, here's the line of this thing, y equals $2x$ minus 4, here's the line, well, what can you say about it? What can you say about it? All right, and I love this, this analogy, that—and it gets to me, it gets to the statement that I made about teaching you how to think, OK, is the following.

- 2.45 You don't need to be an art expert to be able to walk into a museum and look at a painting and say, "That color is blue." You don't. You don't need to have a degree in art history or Impressionism or anything like that to be able to walk into an art gallery and say, "That color is blue." Right? Now, you know, your friends might think that that's a ridiculous statement to make. "Oh, yeah, clearly, that's blue. That's red. Like, that's great. I can do it, too, right? Like, I can do it, too." But the point I'm trying to make is the following. Sometimes the observations that seem so obvious are precisely the ones that are the most profound.
- 2.46 OK? Sometimes the observations that are the most obvious, that we're almost like, "Oh, that's too stupid. I don't want to say that. I don't want to say that that color is blue," sometimes those are the most profound. What would be an example of that? Well, let's say somebody brings a painting to you, and they want to say, like, I don't know, is this a counterfeit or is it a legitimate painting? Somebody makes the observation that color is blue, and somebody's like, "Yeah, it's blue. Big deal."
- 2.47 But if the painting is being claimed to have come from a particular era, in that era, there was no color blue that was actually available, that observation is actually brilliant, because now the observation is, well, that color's blue. That color didn't exist in the era at which that painting is claiming to have come from. Therefore, it's probably likely that it's counterfeit. But that observation seemed ridiculous at the time. Does that make sense, what I'm trying to get through?
- 2.48 Right. OK. So, for example, if I ask you about this line, what are some things that we can say about it? Well, we can say, well, it looks like this, right? It comes from down here, and it goes to up there. That seems like a ridiculous statement, but maybe that's important. OK, it is important because this thing has a particular slope to it. It has a certain tilt to it, and that slope is incredibly important. OK, I could also say, well, this thing, it seems to cross this line here and it seems to cross this line here. Is that relevant? Turns out, yes, that's incredibly relevant. Why?
- 2.49 Well, for example, let's say this line is representing— it's a bad example, but it's an example to illustrate my point—let's suppose this line is representing your revenue for your corresponding business. OK? So x is representing maybe the things that you're selling, OK, and y is representing the revenue. Why would this point be relevant in that point? Well, that's the point at which you're breaking even. You start your business. OK, maybe this is, maybe x is the number of years your business has been running. OK, let's say this, x is the number of years your business is running, and y is the revenue that your business is generating.
- 2.50 OK, well, you start your business. Things are rough for the first couple of years, OK? Maybe you're not making a lot of money and all that. You're still gaining traction, notoriety, marketing, all that kind of stuff, right? And so you're in the red, OK? You're kind of in debt; you're not making any money. But then after year two, starting at year two, suddenly you break even and now the sky's the limit. This point would be

incredibly relevant to know. When does this point happen? When does my break-even point happen? Right? That's incredibly important to know.

- 2.51** This one's also important to know because it says, look, to some extent, using that example, this is saying, what are my startup costs? What are my first costs going to be, year zero, when I start my business, what are the costs going to be? OK, so we want to characterize these things. This thing has a name, it's called the x-intercept. And this one has a name, it's called the y-intercept. OK, so just by making "silly" or ridiculous observations to this, we've already come up with three important characteristics of linear equations.
- 2.52** Number one, its level of tilt, which will be known as its slope. OK, that's important. We've already recognized the x-intercept and the y-intercept. Those are important quantities. OK, how do I characterize them? OK, how do I characterize them? Well, let's think about this for a second. How do you find the x and the y-intercepts associated to this? How do I find my x and my y-intercepts? Well, let's stare at this for a second. My x-intercept occurs here. Well, what's a defining characteristic of that point? The defining characteristic of this point is that it is 2, comma, 0.
- 2.53** Do we agree? It occurs on the x-axis where the y coordinate is 0. So a key observation here is that the y coordinate equals 0. That's a key defining characteristic of this. Similarly, for they-intercept, the y-intercept was, what? When I—let's see, what was this guy? This was negative 4? So this point was 0, comma, negative 4. The defining characteristic here is that the x coordinate is equal to zero. Yeah. There's your technique. There's your technique. Just by making simple observations about this thing and thinking deeply about it, you now have two techniques.
- 2.54** You immediately know how to find, given this linear equation, if I were to say, find the x coordinate, or the x-intercept, excuse me, find the x-intercept, I have to think, OK, x-intercept, what does that mean? X-intercept is going to look where the y coordinate is 0. So given this equation, I can say the x coordinate or the x-intercept, I set y equal to 0 and solve for x. That's my technique. So what do I get? $2x - 0 = 4$. $2x$ is equal to 4. x is equal to 2. 2, comma, 0 is my x-intercept.
- 2.55** Conversely, if I want to find the x coordinate or, sorry, the, the y-intercept, the y-intercept is setting the x coordinate equal to 0. Y-intercept. I come up here and I say, OK, set x equal to 0. So I have $2 \times 0 - y = 4$. I get $-y = 4$. I get $y = -4$, and I get the y coordinate of my, my y-intercept, 0, comma, negative 4. So now we have a technique for actually finding x and y coordinates. And even myself was kind of stumbling and bumbling over these things because, when you're looking for the x-intercept, you set the y variable equal to zero.
- 2.56** So it's kind of hard to keep that straight in your head. It's like, I want the x coordinate. OK, set y equal to zero. I want the y coordinate. I want the y-intercept. Set the x coordinate equal to zero. It's kind of hard to keep that in, keep that straight. So the way

I remember it is, I say, look, the x-intercept intersects the x-axis. The x-axis has the y as zero. The y-intercept intersects the y-axis, and therefore the x coordinate has to be zero. And that's how I kind of keep it straight. All right? But I agree that it's kind of, it's kind of difficult to keep that straight.

- 2.57** OK, so let's look at the slope now. What is the slope? So that's this kind of level of tilt. OK. How sloped is this line? OK, so the slope of y was equal to—what was this, $2x$ minus 4? Is the coefficient of x. Now, there's a couple things I could do here. I could just tell you, it's that way because it is. I declare that it's that way, therefore it is that way, right, which is kind of a pedantic way of saying that. Right. But at the end of the day, you should convince yourself that that's actually a reasonable thing.
- 2.58** So, let me go back to the example that we had in terms of sales, in terms of sales. Remember, we had this example where wages was equal to $100x$ plus 1,000. OK, I had \$1,000 that I made as a base salary. And every widget x that I sold, I got \$100 commission on that. So I said that that was my corresponding wage. What were the units of 100? Units of 100 said this was \$100 per widget sold. Yes? It was \$100 per widget that I sold. So I sell x widgets, I get \$100 per widget. It's a rate. This is a rate. That's the rate of my commission, \$100 per widget.
- 2.59** What is the rate in terms of a line? It is how sloped it is. How fast is it increasing or how fast is it decreasing? It is its rate of increase or its rate of decrease. OK, this line doesn't have to look like this. It could look like this. That's fine. OK, we can have, I can give you another equation that it looks down like this. OK? So that is my corresponding rate. So the rate of this line is called its slope, and that's equal to 2.
- 2.60** OK, what do I mean by its rate? Well, think about fractions for a second. You can write this as 2 divided by 1, which seems like kind of a ridiculous thing to do, but it gives you a way of visualizing how fast this line is actually going up. And what I mean by that is, the slope is given as the rise over the run. OK, the rise over the run. What do I mean by rise? I mean the change in the y coordinate versus the run, the change in the x coordinate. OK, so you think about run as being your x. You think about y as being your rise.
- 2.61** So what this is saying is the following. Look, if I start at some point on my line, I can generate another point on the line by simply going up 2 in the rise, 1, 2, and over 1 in the run. And that was exactly the point that I got here, which was 1 negative 2. If I go up again 1, 2, over 1, I get this point, up 1, 2, over 1. I'll get this one, up 1, 2, over 1. I'll get this 1. It's telling me how to generate more points on my line by simply going up my slope, up 2, over 1. OK, up 2, over 1.
- 2.62** Now, you could go up 2, over 1, or down 2, left 1, because what, 2 over 1 is the same as minus 2 over minus 1. It's the same thing. And you view the negative 2 as being go down 2, and the negative 1 as being go left 1. OK, so you could also generate points by going down 2, over 1, down 2, over 1. It's the same thing. So that's the rise over the

run. Now, what happens to get lines that don't necessarily have a slope? Like, how would I get a line that's completely vertical? Or if I had a line that was completely horizontal? Things like that, how would that work? Do they still come from things that look like this?

- 2.63** So let's analyze that a little bit and see what happens. So, to kind of recap, what did we do? We found out how to graph these things, number one. We found out how to find x-intercepts and y-intercepts associated to that. And we saw that there's a slope associated to these things. OK, all right. So what if I just give you a generic linear equation? What are my possibilities? I have $ax + cy = d$. OK, that's my generic setup. a , c , and d are allowed to be any number whatsoever.
- 2.64** So in particular, I could kind of differentiate this entire thing by saying, when are they zero and when are they not? OK, when are they zero? So I could have one case where I say $a = c = 0$. Well, then d has to equal zero, and I'm not really saying anything. If a and c are zero, the left side is zero. That means d has to be zero. I'm not really looking at anything. OK, the second one would be, a is non-zero and $c = 0$. What happens there? If c is zero, then I'm looking at $ax = d$. And if d is non-zero, or if a is non-zero, I'm allowed to divide by it. OK, you're not allowed to divide by zero. The world explodes. You're not allowed to do that.
- 2.65** OK, so x is equal to d over a . So I want to say, what does that look like geometrically? OK, that's a case that we're going to have to figure out. What about number three? What if a is zero, but c is non-zero? Well, then I get $cy = d$, and y is equal to d over c . And I want to understand, what does that look like? OK, what does that look like? And the last one, which is what we've already done, is that a and c are both non-zero. Well, then I have $ax + cy = d$. What does that look like? Those are my only possibilities here. OK?
- 2.66** a , c , and d are either zero or non-zero. And I've broken them down into the only cases that they can possibly have. OK, what do these ones look like? Let's just do concrete examples. Let's do concrete examples. To illustrate number two, let's say x is equal to 3. To illustrate number three, let's say y is equal to negative 2. And then if I have this one, let's say I have $2x - y = 4$, going back to our previous example. What do these things look like geometrically? This one is literally saying, look, the x has to remain 3 the entire time. x has to remain 3 the entire time.
- 2.67** But look, this is really like saying, $1x + 0y = 3$. Do we agree, to try and put it into this framework? OK, it's $1x + 0y = 3$. So in particular, this is saying x has to remain 3, but what can y be? It could be anything I want it to be. I could let y equal 5, y equal 10, y equals 0, y equal the square root of 2. It doesn't matter what it is. y is allowed to be anything because it's multiplied by zero, and the zero is just going to kill it. It's going to send it to zero.

- 2.68** So what does this look like geometrically? It looks like a vertical line, x is equal to 3. x is forced to be 3, but y is allowed to be anything whatsoever. If you describe points on this line, what is this point? It's 3, comma, say, 2. What's this point? It's 3, comma, say, 5, and so on. Everything is going to have x as 3, but y is allowed to be anything. OK, what about this one? y equals negative 2. Again, this is kind of like $0x$ minus $2y$. Sorry, minus— let's say this. Plus $1y$ equals minus 2. x is now allowed to be anything we want.
- 2.69** So what do I get? y is negative 2. It's going to be a horizontal line. x is allowed to be anything, but y is allowed to be negative 2. What happened with this one? With this one, when we graphed it, what did we see? We saw it did something like this. So we got a line that was askew. It had a slope associated to it. OK. It had a slope associated to it. This one doesn't have any slope. Its slope is zero. Right, it's not tilted in any way. This one kind of has like an infinite slope. It's like completely vertical. Right? But those are the two extremes. This lives somewhere in between, positive-negative slope, things like that. OK.
- 2.70** OK. So why is this relevant? Why is this relevant for us? OK. What we want to try and understand is, given an equation that looks like this, a linear equation in two variables, are there certain standard forms that I can write them in? And the answer is yes, there are. There's typically two. I'm going to talk about one of them. But there's the slope-intercept form, the slope-intercept form of a line. What does that look like? It's exactly what this one looked like when we solved for y .
- 2.71** It looked like, y is equal to mx plus b . That's what its form looks like. m is called your slope, and b is the y coordinate of the y -intercept. OK, I'm being very particular about this because I have strong feelings about this, because a lot of people will look at this and they'll say m is the slope, and b is the y -intercept. And the answer is no, that's false. The y -intercept is a point—zero, comma, something—that is the y -intercept. It is a point; it is not a number.
- 2.72** What is the y -intercept? The y -intercept is 0, comma, b . So it's understandable. Like people say, oh, b is the y -intercept. It's like, no, it's the y coordinate of the y -intercept. OK, trying to be precise with that. OK, so why is, why is all of this useful? Well, let's do this. Let's say graph, say, x minus $2y$ is equal to 3. OK. I want to graph that corresponding line. What's my strategy? Strategy is, get it into slope-intercept form first. If I can get it into slope-intercept form first, then I can just take advantage of, I know where the y -intercept is going to take place, I know what the slope is going to be, and therefore I could draw representative points on it, rather than just making a table of points.
- 2.73** So how would I do this? Again, I have the minus sign here. So I'm going to move my $2y$ over. So I'm going to add $2y$ on both sides. That's going to give me, x is equal to 3 plus $2y$. I'm going to move the 3 over. So I'm left with $2y$ is equal to x minus 3. Notice I switched the inequality or the equality. I moved the y back over to the left side. Technically speaking, I'm saying x minus 3 equals $2y$. But I want the y on this side, the x

plus minus 3 on that side. So I just reverse the order. It's an equality. So you don't have to worry about inequalities changing. And if I divide by 2, I get one half x minus 3 halves.

- 2.74** That's my slope-intercept form. OK, y equals mx plus b . And you're like, well, no, it's not, because it's minus. And I say, yes, it is, because plus and minus are literally the same thing. It's y equals m , 1 half plus b negative 1, negative 3 halves. OK? So I'm going to get m is 1 half and b is minus 3 halves. OK, now why is that useful? Because now we can look at this and say, OK, look, here's my x -axis. Here's my y -axis. I know what my y -intercept is. It's zero, comma, negative 3 halves. So I come down here and I say, OK, here's negative 1, negative 2, negative 3. Negative 3 halves is like negative 1.5. So it's like right here somewhere. That's my zero, comma, minus 3 halves.
- 2.75** OK. That's what that is. And my slope is 1, 2, one half. But that's my rise over my run. So it says, look, start at this point, go up 1 and over 2, up 1, over 2. So if I go up 1, that's here, over 2, that's this point here. I go up 1, over 2, that's this point here. And now I can say, oh, my line looks like that. OK? That's why the slope-intercept form is so useful. It allows us to immediately graph what the line looks like very quickly. OK, but again, why is this useful? Why is this useful? OK, well, let's do, let's do an example. Let's do a quick example. Comes from physics. Comes from physics.
- 2.76** Physics is incredible. I think it can kind of be a little intimidating at times, and that's completely fine. So can math. I think, so can anything. Honestly, somebody comes and talks to me about some deep literary concept, I'm going to be very intimidated about that. Right? So anything can be intimidating in the right context. All right, so let's do this for physics. OK, so in physics, here's an example. I'm going to make some kind of ridiculous assumptions, OK? So one thing that you have to do, whenever you try and model a situation, is you can't take all the variables into account. OK? You can't.
- 2.77** If I want to take this marker and throw it across the room, right, well, it's going to rotate in a particular way. It's going to be influenced by outside factors such as wind resistance. Like, minimally, right, minimally. There's some wind resistance associated to it. It's getting pulled down to the ground because of gravity and all that. But if I knew every single force that was acting on this particular marker, I could say, oh, it's going to hit in this particular location at this time in this angle. I could literally say everything about it, if I knew every single influence on the marker throughout its trajectory. Correct?
- 2.78** So what I'm going to do is, I'm gonna make the ridiculous assumption that this marker is literally just a point. It's just a point, OK? It's got no mass, it's got nothing. It is literally just a single point. And I'm going to imagine taking this marker and throwing it straight up in the air with a certain velocity associated to it. OK. I throw it up with a certain speed, and I throw it directly vertically. What's going to happen to it? Well, it's going to go up, and then it's going to come straight back down, and it's going to hit the ground. Do we agree? That's the setup. OK.

- 2.79** But I'm gonna pretend that it's just a point, like there's no mass associated to it. It's just literally a single point, in one-dimensional motion, just moving straight up and straight back down. OK? So if I do that, if I throw it up, OK, then it's going to do what? It's going to come up and then it's going to come back down. That's what's going to happen. OK, so what is the velocity of this particle? The velocity is given by the equation. And I'm going to give it to you. OK, I'm going to give it to you. The equation is the following.
- 2.80** The velocity v is equal to the acceleration of this is simply due to gravity. OK. Simply due to gravity. There's a universal constant for this. OK, well, not a universal constant, but a constant for us on the Earth. OK? It's 9.81. It's 9.81 meters per second squared. But because gravity is pointing downward, I give it a negative sign, a negative 9.8. OK, negative 9.8. And then it's going to be plus whatever my initial velocity is that I throw it with. So let's say I throw it with 5 meters per second, going straight up. OK. Let's say I throw it with 5 meters per second. That's my corresponding formula for the velocity.
- 2.81** All right, if we did a quick unit analysis of this, what is the velocity it should be in meters per second or whatever my distance is per time? So my units here would be meters per second. OK, that's fine. Acceleration is given by meters per second squared, OK. It's a meter per second per second. OK. How fast are you going per time? And I'm multiplying that by time, which will be in seconds, plus my velocity, which was meters per second. OK, so if you think about this as like meters per second second, you see, like seconds cancel and I get meters per second equals meters per second plus meters per second. OK, that seems legit. The units at least make sense.
- 2.82** I'm not saying meters per second equals meters per second per second plus meters. That wouldn't make sense. I'm saying the units make sense. If I have dollars on the left-hand side, then the right-hand side should be dollars. It shouldn't be dollars on the left-hand side and dollars per square foot on the right-hand side. That doesn't make sense. OK, the units have to make sense. So this formula is just a basic one from, from physics. The name is kinematics, but that's irrelevant. Governs the velocity of this particle.
- 2.83** OK, now how does that relate to this? Remember, variables are arbitrary. What you decide to name a variable is up to you. This is literally the same as if I had called velocity y and called time x . Those two things are indistinguishable from each other, aside from the labeling of their variables. That's it. OK, that's it. So let's try and understand what, what is the relevance of x coordinates and y coordinates in this context? Or are they relevant? And what do they mean from the physics of the whole thing? What do they mean about the physics of the entire thing?
- 2.84** OK, so let's try and find. Let's go back up to v . Let's go back up to v . Let's not think about this in terms of y and x . Let's think about it in terms of the actual variables that are given to me. I'm going to look at this as my axes. Now remember, v is playing the role of y , and t is playing the role of x . So it doesn't make sense for me to say this is my x -axis anymore. But it does make sense for me to say that this is my t -axis. And it makes

sense to say that this is my, not my y-axis anymore, but my v-axis. OK, again, v and y are the same, x and t are the same. It's just a label. OK, it's just a label.

- 2.85** All right, so how would I, how would I find my corresponding "t" intercept, x-intercept? And how would I find my "y" intercept, which is my v-intercept? OK, how do I do that? Well, remember, wherever it intersects on t, v has to be zero. V has to be zero. So wherever it intersects, let's say it intersects here. This is going to be something comma, zero, whatever that point is. So let's think about that. This is saying that v is equal to zero. Well, if I take v and substitute it in for zero, what am I actually solving for? What am I solving for?
- 2.86** I have 0 is equal to minus $9.8t$ plus 5. I'm trying to solve for t, trying to solve for t. OK, well, again, there's a lot of different ways to do this. My personal favorite, my personal, my personal favorite way of doing this is to say, look, I don't like the fact that that's a negative sign because I don't want to have to divide by a negative later. So rather than subtracting 5 on both sides, I'm going to add $9.8t$ to this side. OK?
- 2.87** So if I add 9.8, I add $9.8t$, I get $9.8t$ is equal to 5. And then I can divide by 9.8, 9.8, and get t is 5 divided by 9.8. That's what the x coordinate or the t coordinate is going to be, is 5 divided by 9.8, whatever that is, close to a half. Close to a half. If the denominator were a 10, it'd be 5 over 10. That's half. OK, so it's close to a half, somewhere in there. Now, what about the v-intercept? Remember, the v-intercept occurs at something that looks like 0, comma, something, and I need to figure out what that is. For the v-intercept, we're looking at set t equal to zero to find the v-intercept.
- 2.88** OK, so I come up to my equation and I set t equal to zero, and I get v is equal to minus 9.8 times zero plus 5. That's just 5. That's just 5. So this is zero, comma, 5. OK, now if I wanted to, I could say, well, I know this is going to be a line, so I could just connect the dots and draw this all the way like that, if I want to say, what does it look like geometrically. And that's describing the entire velocity over the entirety of the particle's motion. But you have to be a little bit careful because some of this is a little, is a little ridiculous, right?
- 2.89** But what does this actually mean? What do those two points mean, physically? Why are they relevant? OK, why are they relevant? Am I just drawing points on a board and lines on a board, or does that actually have relevance associated to it? OK, what would this point mean? This is saying when t is equal to zero, t represents time. So when time is equal to zero, the velocity is equal to 5. What does that mean? Well, presumably, time equals zero is like when I let the pen go, right? The second I let the pen go, that's t equals zero. The stopwatch starts.
- 2.90** So what does v equals 5 mean? What does the v-intercept mean? It means this is my initial velocity. It's the initial velocity at which I throw the pen up in the air. The second the pen leaves my hand at that instant in time, it is moving at 5 meters per second upward. That's relevant. That's relevant. What does this thing mean over here? This is

saying that, after a certain period of time, the velocity is equal to zero. What is that going to mean physically? Physically, it means—think about it. If I throw this up in the air, right, if I throw this up in the air, OK? Oh, pulled it off. OK, I throw it up in the air, OK? What does it mean?

2.91 Well, at some point, it has to return back down. It has to return back down. Well, think about that. The velocity goes up, positive, starts out at 5. Gravity acts upon it. It then stops, velocity zero, and then it starts coming back down. Velocity is negative because it's pointing in the negative direction. So what is this representing? This is representing its apex of motion, how high it got. So if you're interested in saying, hey, I'm going to throw a particle up in the air and I want to know, what's the maximum height it achieves? Well, you can say, I don't know. But I know it occurs at 5 divided by 9.8 seconds. I know when it happens.

2.92 And that's relevant. That's important. OK? So these particular equations, these linear equations and two variables, they show up all over the place in terms of elementary models, in terms of how we try and understand our world in very simplistic ways. I mean, this is kind of a ridiculous scenario where I say, you know, I just toss this up in the air, right? But it's a single particle. Well, that's not true. It has mass, it has a curvature. It has, like, certain things that are acting upon it. OK?

2.93 But it's an approximation to what our actual reality is. And that's ultimately what math and physics are trying to understand. How do we approximate the world in which we live? We have to make assumptions. We have to make assumptions, otherwise it's too complicated, right? If I flip a coin, technically speaking, if I know enough data, I know exactly what its outcome is going to be before it hits the ground. If I know exactly at what angle I flip it, what velocity it happens, if I know all of the forces acting on it, if I know how far it hits the ground, if I know what angle it hits the ground, and all of these different things, if I know every single ounce of data, I can flip it and say heads and then wait for it to hit the ground, and it's heads.

2.94 Or I can flip it and say tails. And before it hits the ground, I already know the outcome. It's tails. It hits the ground, it's tails. That's absurd. I don't know all of that information. So for me, flipping a coin, it's a 50-50. It's either heads or tails, because I don't have enough data. I had to make assumptions in order to try and approximate it. Similarly to these types of things, by assuming this is a single particle, I'm making somewhat ridiculous assumptions, but it's allowing me to approximate my reality. Does that make sense? And that's what we have to do when we first start modeling. But that's the introduction to linear equations of two variables.

Lecture 3: Rules of Exponents

At a glance — Rules of exponents.

- 3.1** All right, so today we're talking about the rules associated with different algebraic operations. So we've talked about expressions, we've talked about equations, single variable, two variable, graphing lines, things like that. What we want to try and introduce is, what are the rules that I'm allowed to use algebraically? Because I'm going to run into different equations that may have exponents to them. They may have certain things called roots associated to them. And I want to understand, what am I allowed to actually do with these exponents and roots and things like that?
- 3.2** Once we have that, we'll be opened up to a new world in terms of a modeling standpoint, where we can model new phenomenon throughout, in the world, using different techniques. So rather than just approximating things with lines, I can now approximate things with things that are curved and all of that. And we'll talk about some of the elementary models from them. OK, so let's get into some of the algebraic rules. And this is going to be in terms of exponents, specifically. The list of algebraic rules is a little intense. So we're going to focus on exponents.
- 3.3** So what we want to try and do is, we're going to have, x and y are just going to denote arbitrary variables. I want to try and understand, what does it mean to do things like, if I take a variable x and I raise it to the 0 power? So the thing up here is going to be called, so this thing right here is called my exponent. So x is my variable, and I'm raising it to a power.
- 3.4** So what happens if I raise it to the 0 power? Well, the answer is, this is going to be 1, provided that x is non-zero. So for example, if I said, what is 2 to the 0 power? I'd say that's 1. What is 3 to the 0 power? I'd say that's 1. What is 3 to the 0 power? I'd say that's 1. What is one-half to the 0 power? I would say that's 1. But if I said, what is 0 raised to the 0 power? The answer is, I don't know. So, but 0 raised to the 0 is equal to, I don't know.
- 3.5** As you go along further in math, you can make this rigorous, from a calculus standpoint, and 0 to the 0 is what's called an indeterminate form. It just means it's not determinable without universally. So you can give me a 0 to the 0, and I can make it equal to 0. You can give me a 0 to the 0, and I can make it 1. You can give me a 0 to the 0, and I can make it the square root of 7, whatever that is. So 0 to the 0, we don't have a definition for universally. So it's just undetermined.
- 3.6** But we have other rules as well. So for example, if I say x raised to the first power, what is that? Well, that's just x . It's just a copy of x . What is things like, what if I said x squared? How do I interpret that? Well, x squared is literally just saying x times itself. That's all it is. So this is just x multiplied by itself. And you can iterate on this. You can say, what is x to the third? This is just x times x times x and so on. What is x to the n

power, where n is some positive number? It's just n copies. There's n copies of x 's. That's all it is.

- 3.7** So for example, if I gave you 2 squared, I'd say that's 2 times 2, that's 4, 2 cubed, that's 2 times 2 times 2, that's 8, and so forth and so on. Just setting up the framework for what are exponents. Now, the next one we want to understand is, if I take x and I raise it to the m power, and I multiply that by x raised to the n power, what can I say about that? Now, I can just give you the rule, and that's fine, and there's some value to that. But again, the motto through everything is, you shouldn't believe what I say. You should convince yourself that what I'm saying is actually correct.
- 3.8** OK, so x to the m , x to the n , what is that really? Well, think about x to the m . From what we have above, x to the m is what? I'm going to have x, x, x . I'm going to have m copies of those. And then on this one, I'm going to have n copies of x . So I'm going to have x, x, x . I'm going to have n copies of those. And so how many am I going to have in total? Well, I'm going to have m of them, plus another n of them. So I'm going to m plus n copies of this. So what would that be? That would be x to the m plus n .
- 3.9** So the moral of the story is, if I have the same base, the same base, the same variable, x and x here, it's the same base, and I'm raising them to different exponents, if I multiply them, you add the exponents. That's the moral of the story. So for example, if I took 2 raised to the second power times 2 raised to the third power, what would this be? Well, I would have 2 twos here, 2 times 2. This one would be 3 twos. So how many twos do I have altogether? I have 2 to the 5, which is really 2 to the 2 plus 3. So you just add their exponents. That's one of our rules.
- 3.10** How do I handle division? Well, if I have x to the m , all over x to the n , how do I do this? Well, there's two ways to see this. OK. The first way to see this is simply just by the definition, which is x to the m minus n . So you take the exponent in the numerator expression and you subtract away the exponent in the denominator expression. OK. And there's a couple of ways that I can convince you of this. OK. One way is the following. So let's do, one of the properties up here is, how do I handle x to the minus 1 power? What does that mean?
- 3.11** The definition of this is that this is literally 1 over x . It's equal to 1 over x . It has a specific—it has a very rigorous reason as to why it is 1 over x . But x to the minus 1, what happens to this when we multiply it by x ? What happens to it? Well, we said x , from this property here, x to the first, is just x . And then the property of adding exponents means that this would be x to the 1 minus 1, which is x to the 0, which is 1. So this symbol, x to the minus first, is literally saying, it is a quantity, x to the minus first, x to the minus first, is a quantity with the following property—that when I take x to the minus first and I multiply it by x , I get 1.
- 3.12** That is the notation for it. x to the minus first is what's called the multiplicative inverse of x . What does that mean? It simply means that if I take x to the minus first times x , I

get 1. That's all it means. OK? So for example, if I said, what is 2 raised to the minus, the minus first power? I can think of that as saying, oh, there's a negative exponent. It has to be 1 over 2. That's one way to think about it, and that's completely fine.

3.13 Or I can think about it from the definition and say, what is this quantity representing? It's representing something such that when—so let's call this thing, let's call this y . If we don't know what that quantity is, we can say, what is the property that y has? It has the property that, when I multiply it by 2, I have to get 1. That's its property. It has to multiply 2, whatever its base is, it has to multiply that to 1. Well, now algebra says what? How do I solve for y ? Divide by 2. So if I divide by 2, what does that tell me? It tells me that y is equal to one-half, and that's exactly what we're trying to illustrate.

3.14 That's what we're trying to get at. OK, so then with that in mind, I can say, well, let's look at this. Let's look at this guy. x to the n is in the denominator. x to the n is in the denominator. Well, what would happen, iterating on this process up here, what would happen if I did this? What would x to the minus 2 be? Why do I leave it as just negative 1? Why don't I do x to the minus 2? Well, what is this going to be? This is x to the minus 1 times x to the minus 1 using this property. Negative 1 minus 1 is negative 2, so I can split it that way.

3.15 And this becomes what? x to the minus 1 is 1 over x , x to the minus 1 is 1 over x , so this is 1 over x squared. And you're like, OK, well, let's just keep iterating this and see what happens. So what if I did something like x to the minus n ? What would I get? x to the minus n , this would give me my 1 over x to the n . So if I look at this quantity now, if I look at this quantity, I could really write this in the following way. I could say, x to the m , over x to the n . Keep me honest here, do you agree with this, that I can write this as x to the m times 1 over x to the n ?

3.16 How do we multiply fractions? How do we multiply fractions? If I make this a fraction, how do you multiply them? You multiply the numerators, you multiply the denominator, and that's how you get your fractions, multiplied together. So if I take x to the m times 1, x to the m , 1 times x to the n , x to the n , OK, that seems plausible. But now we know that 1 over x to the n is x to the minus n . So I can replace this by writing x to the m times x to the minus n . Do we agree?

3.17 But they have the the same base. So that means I can add their exponents. So what do I get? x to the m plus minus n . I get the exact same definition that we got there. Justifying that when you're dividing out things that have the same base, it's going to be the exponent in the numerator minus the exponent in the denominator. So that's what we have there. All right. I know these are all kind of—these rules are a little sterile and esoteric and all that. So we'll get to some examples soon. I promise. All right. What else do we have?

3.18 What if I throw in a new variable? What if I say I want to take x times y , raised to the m power instead? So x and y are now my new variables. I'm going to raise their product to

m. And I want to understand, is there a formula that I can say about this? OK. The rule for this is x to the m , y to the m . OK? Provided—very rarely are you going to see this in an algebra class, OK— provided x times y is equal to y times x . OK. Otherwise, this rule is false. Otherwise, that rule is false. OK.

- 3.19** This property here says that the two variables, they commute. They commute. What do I mean by that? I mean, well, let's give concrete values. If I said x was 2 and y was 3, this is saying the left-hand side is 2 times 3, 6, fine. This is saying, is equal to 3 times 2, 6. OK, that's legit. That's fine. But is that a universal fact? Or is that just an artifact of me using integers from my arithmetic? If x and y are not just numbers, OK, 4 times 5 is 5 times 4, negative 1 times 2 is 2 times negative 1— if they're not just integers, if I'm multiplying other types of mathematical objects, am I always guaranteed that they commute— a times b is b times a ?
- 3.20** The answer is no. The answer is no. So this rule fails if this is not true. And I want to convince you of that. OK. I want to convince you of that. So what is this rule technically saying? OK, let's break it down in terms of its principles. OK, let's do an example. OK, let's do an example. What if I said, x times y cubed? x times y to the third power. Raising something to the third power, you call it xy cubed. OK, if I raised it to the second power, you'd say xy squared. And then anything basically off of those, like fourth, fifth, sixth powers and so forth, you just say xy to the fourth, xy to the fifth, xy to the sixth, and so on.
- 3.21** That's the language that we use. But what does this really mean? It means whatever this quantity is, xy , multiplied 3 copies of it. Let's write that down. That says this is going to be xy time xy times xy , yes? That's what that's going to be. Now, where the parentheses are doesn't matter. Multiplication, the parentheses can go anywhere. So in particular, I can write this as simply a string, if you will, $xyxyxy$. That's it. So why am I allowed to write this as $xxxyyy$? I don't know if I can.
- 3.22** If I know that y times x is equal to x times y , then I can switch those orders. I can move the x across the y . If I know that y times x is x times y , I can move the x and the y . I can interchange them. But that's only if this is true. If this is not true, I'm not allowed to move them. I'm not allowed to move them. So this is equal to $xxxyyy$, which is x cubed, y cubed, only if that is true. Only if that is true.
- 3.23** Now, this may seem a little weird because, for example, everything you've probably done mathematically, that's been true. And so you probably have never questioned this before. Like, oh, yeah, 2 times 3 is 3 times 2. Great. 5 times 4 is 4 times 5. Great. You probably haven't run into objects, mathematically, such that object a times object b is not equal to object b times object a . All right? And those do occur. All right?
- 3.24** An example of them is what's called a matrix, which shows up all over in mathematics and is arguably one of the most important things, and you'll see that in a linear algebra course. But matrices do not satisfy that property. You cannot take a matrix times a

matrix and then flip the order and expect it to work the same. Those are two different things, typically. So these are the, these are the, the arithmetic objects that we have and the algebraic rules. So let's look at some examples. Let's look at some examples. Oh, I have one more, actually. We did multiplication. We need division as well. So let me clear this out a little bit. We need division.

- 3.25** OK, my last property, property 6, is that what happens when I take x divided by y and I raise that to a corresponding power? What does that mean? Well, again, let's look at what this entails. This is x over y times x over y times x over y , m copies of that. And how do I multiply fractions? Well, I just multiply the numerators and then I multiply the denominators. And so what am I going to get here? I'm going to get m copies of x . So that's x to the m , divided by m copies of y . There's my last rule. There's my last rule.
- 3.26** So what I want to do now, let's just do some examples to cement this. What's my example? I want you to simplify the expression given by, let's say, 2 cubed x to the fourth, y to the fifth, all over 2 squared x to the sixth, y to the fourth. I want you to simplify that expression. So let's go through it, term by term, and see how we do this. How do we do this? Well, what we notice is, what we're going to try and do is take each of the properties that we had, this 1 through 5, 1 through 6, and try and apply them in the corresponding way.
- 3.27** So, for example, when I have x divided by y raised to the m , I have this property here, x to the m , y to the m . Remember, we had this rule that said, if I have x to the m , divided by the same base, that's x to the m minus n . That's the rule that we're going to be using a lot in this particular example. OK, so how does that work? I notice that if I look at this, I say, OK, I have this term here and this term here that have the same base. OK, so let's first split these out into their individual pieces.
- 3.28** So the first one I'm going to have is 2 cubed over 2 squared times x to the fourth all over x to the sixth times y to the fifth all over y to the fourth. So if I look at this first term, I see 2 cubed divided by 2 squared. So that fits this definition right here. Again, remember, x is an arbitrary variable. So if I specify x to be 2 , m to be 3 , and n to be 2 , I'm in this scenario. So what does it tell me? It says, look, this should be equal to 2 , raised to what power? This is my m , in this case. That's my m . This is my n , in this case. So this should be 2 to the 3 minus 2 , in that case. All right?
- 3.29** What about this one? I move to the next one. I have the same base, x to the fourth, x to the sixth. So I'm going to get an x to the 4 minus 6 . Same thing here. Same base with y . I'm going to get y to the 5 minus 4 . And then we can simplify that. 2 raised to the 3 minus 2 , 3 minus 2 is 1 , 2 to the 1 is just 2 . So I'm going to leave it alone. That's just 2 . This one becomes x to the, OK, 4 minus 6 is minus 2 , and y to the 5 minus 4 is 1 . So I can eliminate the 1 because it's just y to the 1 , it's just y . Let's leave it that one.
- 3.30** Now, the last thing I can do is I can say, look, x to the minus something, well, x to the minus 2 , remember, was equal to 1 over x squared. So what can I do? I can move this

into the denominator so I can write this as $2y$ over x squared in that case. That would be as far as I could simplify it. It's as far as I can go. It gets more complicated when, rather than everything just being multiplied together, you start adding or subtracting things downstairs as well. So you have to be very careful with the arithmetic operations that are happening here.

3.31 So let me illustrate that. If I gave you something that looks like this—let's be very careful here—so if I gave you something like $\frac{1}{2} + x$, $\frac{1}{2} + x$, and I asked you to simplify this, a very common mistake, a very common mistake, this is already simplified. This is simplified. It cannot go any further. But a common mistake, this is not equal to, by the way. OK, this is not equal to. But a very common mistake is to simply say, oh, it's just $\frac{1}{2} + \frac{1}{x}$. That's what it should be. I somehow divide the 2 into the 1, I divide the x into the 1, and all of that.

3.32 But if you think about how you add fractions, that's completely false. How do you add fractions? You need to get a common denominator between the two. If you look at this thing, how do I get a common denominator? I need this to be an x over $2x$, and I need this to be a 2 over $2x$, so they have the same denominator. And then I get $x + 2$ over $2x$. That doesn't look like that at all. So you have to be very careful in how this works. All right? You have to be very careful in how this works.

3.33 So, let's do some examples where these exponents are not just integers, 3, 4, 5, 2, 6, 4, negative 2, things like that. How do I make sense of something like 2 raised to the one-half power? How do I make sense of that? What does that even mean? Does it make, what does it mean to say 2 to the one-half power, or to the five-sixth power or to the, things like that? What does that actually mean? OK, so let's investigate that. Let's see how this works.

3.34 So this is getting into what are called radicals. So let's start fairly simple. OK, let's illustrate this with an example. So what is 4 raised to the one-half power? How would I make sense of this? Well, I could do a couple things. I could just look up the definition, and I could just have, I don't know, have something just tell me the answer. But that's not super helpful. I actually want to understand the underlying principle of the matter. So what I could do is, I could say this. I could say, well, I don't know what this is. I don't know what it is. So potentially it could be anything. Potentially, it could be anything.

3.35 In particular, that sounds like a variable. It sounds like a variable. So what I could do is, I could say the following. So here's my attempt at a solution. I could say, OK, well, let's let x equal 4 to the one-half. OK? It's unknown what x is. So let's just set x equal to 4 to the one-half. And let's see if we can solve for what x equals. Let's see if we can solve for what x equals. So in the back of my head, I'm saying, you know what? The thing that's driving me crazy that I don't understand is this exponent is a one-half. So can I somehow eliminate that power? Can I somehow eliminate that power?

- 3.36 Well, there's a rule with exponents, which says that if you take x and you raise it to the m power, and then you take that whole thing and raise it to the n power, the result is you just multiply those exponents. And so that's what we're going to exploit to figure this out, because look, what would I need to do to multiply one-half to get something that actually makes sense? So for example, what would I have to multiply one-half to be to make a 1? If I multiply one-half by 2, then I get one-half times 2 is equal to 1. That gets rid of this exponent and allows me to have some leeway in potentially solving for what x is.
- 3.37 So what I'm really saying is, I'm saying, let's take x and let's raise it to the 2 power on both sides. Let's raise it to the 2 power. And let's just see what happens. Maybe it doesn't tell me anything. Maybe it just blows up and it doesn't give me any new information, any value whatsoever. But at the very least, I mean, think about this. I get this question all the time, where people ask me, they say, "Well, I don't know what the next step is." And it's like, well, that's the story of life. You don't know what the next step is. That's the point of living life, right, is that you don't know what the next step is. You just live life and the next step will happen, right?
- 3.38 But if you don't try something, then nothing is going to change. You're not going to gain any new information. If you just sit there and do nothing, then you're literally not going to learn anything. So here's the point. Is this operation going to do anything for me? The answer is, I don't know. And that's not a bad thing. If it does do something for me, great. I actually learned something. I move forward. I solve the problem. If it doesn't do something for me, well, then I'm no different than where I was when I wasn't doing anything. So why not try something and see what happens?
- 3.39 OK, so what is x raised to the 2? So this is x squared on this side. Now, this property here, I'm using this principle here. I have a number raised to an exponent, which is then raised to an exponent. So what am I going to get? I'm going to get 4 to the one-half times 2. Well, 2 times one-half is 1. So this is just 4 to the 1, which is 4. And now I'm really happy about this because look, now I'm saying, solve this equation, which we're going to spend some time talking about how to solve equations like this.
- 3.40 But right now, I can look at this and I can say, oh, shoot, I know a number such that when I multiply it by itself, I get 4. Because what is this really saying? This is saying x times x is equal to 4. So I start scouring my mind and I say, OK, do I know a number that multiplies itself to 4? Well, 1 times 1 is 1. OK, that's not good. 2 times 2 is 4. I win. 2 times 2 is 4. So what does that tell me? I know that 2 times 2 equals 4, so x is equal to 2, x is equal to 2.
- 3.41 But what was x ? x was, by definition, the object we were trying to find, 4 to the one-half. So in particular, we just found that 4 to the one-half power is equal to 2. That's pretty miraculous. Now, there's a rebuttal to this. There's a rebuttal to this because you could have also noticed, I'm just looking for two numbers that multiply together to give

me 4. You could say, "Well, Robbie, this works. So why isn't 4 to the one-half power equal to negative 2? You told me it's equal to 2, not negative 2?"

- 3.42** And the answer is, because by doing this operation, and again, I will make this abundantly clear in another version of this, but raising something to this power lost some information for me. It lost some information for me. In particular, look, if you multiply two things together, the same two things, x and x , negative numbers will cancel. Negative 1 times negative 1 is positive 1. So by introducing a squaring operation here, I've somehow now introduced the possibility of a negative 1. It's not allowed. I'm giving too much information. I'm giving too much information, OK?
- 3.43** I did not say that I wanted x times x is equal to something like this, right? I want a single number, and that single number will be 2, all right? So this is a notion of taking a fractional exponent, and these things in particular are what are called roots, roots of these values. What is this really? Proper notation for this, fractional exponents of the form 1 over n . So these things look like, for example, if I take x to the one-half, what is that? That is the square root of x , and that's what you call it. You call it the square root of x .
- 3.44** Now, the square root is special because it's implied that, if I write the square root, then there's a 2 here. When you look up higher ones, if you take x to the one-third, to denote that it's a cube root, you take a little 3 and you just put the 3 right there. It's not like a 3 multiplying the square root or anything. It's just the location of the 3 indicating that you're taking the one-third power. Same thing, x to the fourth is the fourth root of x and so forth and so on.
- 3.45** Technically, for the square root, I should put a little 2 here. But traditionally, that 2 is not there. So you just leave it alone. You just denote this as a square root. So if we take that back to this situation right here, what I was really asking is, what is the square root of 4? And we're saying the square root of 4 is 2. That's what we were really asking. OK. So now we can ask other questions. We can say, well, for example, how would I solve the question of, what is 8 to the one-third? What is 8 to the one-third?
- 3.46** Well, think about it. What did we do over here? Over here, we said, what is 4 to the one-half? We said, well, 4 to the one-half is a solution to x squared equals 4. So then this would mean this is a solution, a solution to x cubed equals 8. That would be my way of thinking about this, without fractional exponents. If I said, what is 5 to the one-fourth? This is a solution to x to the fourth equals 5 and so on. So when I look at the n th root, if I say, what is x to the 1 over n ? That's the n th root of x , what is it? Well, it's a solution to an equation like this.
- 3.47** These equations are called polynomial equations, and we will talk about polynomials in the future. But it's just to give you a flavor for how you actually approach these types of problems. But you run into issues such as we did down here. What is the square root of 4? Square root of 4 is equal to 2. Square root of 4 is not plus or minus 2. It's not. And

I'm going to make that very precise when we talk about functions. So this is just to give you a flavor for what these fractional exponents are, what they do for us, and what we can actually do with them.

- 3.48** So do these things occur in nature? Or are these just philosophical constructs that mathematicians do to drive students crazy? I don't know. Probably a little of both. OK, probably a little of both. All right, so let's see. Let's see what we have. What do I have when I try and deal with equations involving exponents? I have equations involving exponents. So what if I gave you the following question? Find x such that 2 raised to the $3x$ plus 1 is equal to 16 to the x minus 3 .
- 3.49** A natural question you should ask is, why do I care? That's a natural question you should ask. You should ask that with anything. Why do I care? So you can imagine that 16 to the x minus 3 and 2 to the $3x$ plus 1 are modeling certain things. That's a possibility. But ultimately, what I'm interested in is, can we take the rules of exponents that we just went through, somehow apply them in this context, to resolve this equation? So I agree with you. It's fair to say, why should I care about something like this? That's not point. That's not the point.
- 3.50** The point is, can you apply basic algebraic exponential rules to come up with a value of x ? Do those rules apply in this particular context? So how could I approach this? Well, the first thing I could do is, I could say, can I somehow make this 2 any simpler than it is? The answer is like, well, not really. I can't really factor 2 down. I can't make 2 smaller. It's not like 4 , where I can take 4 is equal to 2 times 2 , or something like that, or 6 , which is 2 times 3 . I can't make 2 smaller. It's just 2 times 1 . OK. So no, I can't make that smaller.
- 3.51** What about 16 ? Can I make 16 smaller? The answer is, well, yeah, I can. 16 factors into what? I have a 4 and a 4 , right? 16 factors into 4 times 4 . But then 4 factors further. This factors into 2 times 2 , and this is 2 times 2 . So what do I notice? I notice that 16 is 2 to the fourth power, just four copies of 2 , 2 times 2 times 2 times 2 . OK. So I could simplify that a little bit. Does that help me? I don't know. Maybe. 2 to the $3x$ plus 1 is equal to 2 to the fourth now, raised to the x minus 3 .
- 3.52** You start saying, OK, this is getting out of control. 16 was fine. 16 was a legitimate number. Now you're writing it as 2 to the fourth. This is not helpful. I don't like this. And that's a fair argument. That's a fair argument. But it turns out to be the exact thing that you needed to solve the question. It turns out to be the exact thing that you needed. And how? Well, remember, we had a rule that told us, if we have some quantity raised to an exponent, we know how to handle that. You multiply the exponents. You multiply the exponents.
- 3.53** So our rule, remember, we had this rule that said, a to the m to the n is equal to a to the mn . Those two things multiply. That was our rule. So what does this become? This is 2 to the 4 times x minus 3 , which I can then use the distributive rule, the 4 goes in, to

write this as 2 to the $4x$ minus 12 . So it turns out that changing the 16 to a 2 , albeit made it look more complicated, but now what do I have? I'm looking at 2 raised to the $3x$ plus 1 is equal to 2 raised to the $4x$ minus 12 . I want to find an x for which that's true. I want to find an x for which that's true.

3.54 Think about that for a second. The base is the same here and here. Both of those are the same. So in particular, I know that this equality is going to be true if their exponents are equal to each other. If this exponent over here were an 8 and this exponent over here were an 8 , I'm simply saying 2 to the eighth power equals 2 to the eighth power. That's true. That's true. So what this allows me to do is, it allows me to say, $3x$ plus 1 is equal to $4x$ minus 12 . They have the same base, so their exponents have to be equal to each other. Their exponents have to be equal to each other.

3.55 Think about it like this. If I told you something like this, three to the y is equal to 3 to the 10 , and say, find y , what would be a solution? Well, 10 . That would work great. 3 to the 10 equals 3 to the 10 . It doesn't have much content. It's like me saying 5 equals 5 . It's like, well, facts. But it allows me to solve the corresponding equation. Set the exponents equal to each other. When the bases are equal to each other, the exponents are equal to each other. So here I have the bases, the 2 , are equal to each other. Therefore, the exponents have to be equal to each other. $3x$ plus 1 equals $4x$ minus 12 .

3.56 And now we're back into setting equations equal to each other, and we can algebraically manipulate them to solve for x . How would we do that? Ultimately, it's up to you. What I like to do is, I like to step back and say, what would be a good strategy first? Rather than just jumping in headfirst saying, OK, what's a good strategy? Well, I notice I'm going to need to gather all of my x 's together and gather all of my constants together. But if I move $3x$ to the right or I move $4x$ to the left, I'm allowed to do both. I'm allowed to do either of them.

3.57 But here's the kicker. If I move $4x$ to the left, $3x$ minus $4x$ will be a negative x . And I'm not interested in what negative x is. I'm interested in what x is. So which one should I move? I should probably move $3x$ to the right instead of moving $4x$ to the left. And that's what I'm going to do. So I'm going to subtract $3x$ on this side, subtract $3x$ on this side. This side, I get a 1 . This side, I get $4x$ minus $3x$. 4 minus 3 is 1 , so I get x minus 12 . And now I'm just going to move the 12 to the other side by adding 12 . Add 12 , add 12 . I get 13 is equal to x .

3.58 Now, one thing that I should mention is, even though this is correct, it's not proper in terms of your notation, because what were we trying to find? We were trying to find x , so I should reverse their orders. I should reverse their orders. So x is equal to 13 . That would be the proper way to write this. From a computer programming standpoint, this makes perfect sense. Because if I have a variable and I am trying to assign that variable a value, I'm trying to say x is equal to 10 or something like that, I'm assigning the variable x on the left, its assignment value, which is equal to 10 .

- 3.59** It does not make sense to say 10 equals x or 13 equals x . That's me saying I'm taking the numerical value 13, and I'm assigning it the variable value x . No, 13 is 13. It is not a variable. So even though technically this is correct, it's improper. You should write whatever you're solving for, your variable is equal to whatever numerical value it is on the right-hand side. So that's a minor thing, but just be aware of that.
- 3.60** So then I can go back and I can check. I can say, look, if I plug in 13 to both sides, which, this was the equation that we were looking at, if I plug in 13, is this true? This is the great thing about algebra, is that you can typically check your answers. So if I plug in 13 on the left-hand side, LHS, left-hand side, I'm getting 2 to the 3 times 13 plus 1. OK, what is that? 3 times that is 39 plus 1. That's 2 to the 40. It's a huge number. Versus the right-hand side, what do I get? I get 2 to the 4 times 13 minus 12. OK, this is what? 40 and 12 is 52, minus 12 is 40. So this is 2 to the 40.
- 3.61** They match. So they gave me my answer, 2 to the 40. A massive number, but it says that x is equal to 13 is the solution to this particular equation. It is the solution to this equation. OK, so you may be saying, OK, well, that's like, all right, Robbie, whatever. I don't even know what 2 to the 40 is. It's a big number. Where do these things occur in nature? Where do these things occur in my day-to-day? Why is this relevant to me? So let's think about this. Where do exponents occur?
- 3.62** Well, they occur in compound interest. They occur in compound interest. So let's see how that works with a concrete example. So let's suppose, so let's look at compound, compound interest. So what does this mean? In words, it means the following thing. So you go to a bank, and you want to give the bank a certain amount of money that they're going to hold on to, and they'll give you some incentives for it. They'll give you a checking account, credit card account, credit card with them, whatever, savings account, whatever the case may be.
- 3.63** And they effectively take out a loan from you. That's effectively what they're doing. You go to a bank and you say, "Hey, I'm going to give you \$1,000," and they're going to say, "Sweet, we're going to give you an interest rate on that for giving us your money." They're basically taking a loan from you. The only difference is that you don't get to set the terms of the interest rate. They get to set the terms of the interest rate. OK?
- 3.64** So, for example, I have a bank that I use, not for saving money side of things because the interest rate is absolutely appalling. And they will offer me a 0.1% interest rate back on my money. They will give me 0.1% on whatever dollars I give them. So for example, if I give them \$1,000 at a rate of r is equal to 0.1%. So what is this as a decimal? This is 0.001. That's their interest rate. They say, "Thank you for the money, Robbie. I will give you an interest rate of 0.001 on the money that you gave me," which means what?
- 3.65** If I give them \$1,000, they say, "Here is a dollar." It's like, "Oh, thanks. That's kind." That's a terrible return on that. And you can find other options out there. But that's how this is going to work. So let's say, hypothetically, that a bank gives you, let's say, r

% at the end of each month. At the end of each month on whatever money that you have in there. Now, typically the way these things work for banks is, they'll say, "OK, look, we will give you—", so another bank that I use gives me a 4.5% interest rate, which is massive compared to this. 4.5 compared to 0.1 is a huge difference.

- 3.66** But what they say is, we'll give you 4.5% compounded monthly, which means they're going to take the 4.5% and divide it by 12. And then at the end of each month, they're going to give me that corresponding percentage. So we're making this a little less technical. We're just saying, look, at the end of every month, your bank looks at you and says, "Look, you put in \$1,000 dollars at the beginning of the month. At the end of the month, we're going to give you $r\%$ on that \$1,000 dollars. The next month, we're going to give you $r\%$ on whatever else you have. The next month, we'll give you $r\%$ on whatever else you have," and so forth.
- 3.67** Does that make sense, how this is working? OK, so let's look at it this way. We have month 0. Month 0, I give them \$1,000. That's how much is in my account. So I have \$1,000. That's how much I have. At the end of the first month, how much do I have? Well, how much do I have? So let's do this. Let's say my amount is equal to \$1,000. At the end of the next month, or at the first month, I'm going to have a is equal to what? Well, I clearly have my \$1,000 still in there, but now the bank is going to give me a percentage of that, for letting them use my money for the month.
- 3.68** But how much are they going to give me? They're going to give me 1,000 times the interest rate that they're giving me. They're going to give me that. OK, so they gave me a dollar. They gave me a dollar. OK, well, let's factor that out. Well, what do we see? 1,000 is common to both. So if I factor out 1,000, what would need to go in there in order to make the distributive rule apply? Well, I'm clearly going to need this there, because when 1,000 comes in and multiplies that, it's got to give me this. It's got to give me that.
- 3.69** But then how do I get a 1,000? I need a 1 here. Because when I distribute, 1,000 goes into the 1. One times 1,000 is 1,000, and then 1,000 goes into the 0.001, and that gives me this. What about month 2? What about month 2? Well, now the bank is going to give me what? Think about this for a second. I have this amount in the bank. OK, I have this amount in the bank. They're also going to give me 0.001% on this. OK, so think about this. First month, I give them \$1,000. At the end of that first month, they give me a dollar. So now I have \$1,001.
- 3.70** At the end of the next month, they're going to give me \$1,001 times 0.001 plus my \$1,001. They're giving me an interest rate on that. If you work that out, what do I get? The amount is going to be the initial I gave them, times 1 plus 0.001 squared. And we keep doing that. So if I go down to month k , whatever k is, k is 12, so the end of the year, k is 24, so two years later, k is 16, right? So first year and then the next, what? I'll just pass the first quarter. What am I going to get? The amount is going to be 1,000 times 1 plus 0.001 raised to the k power. That's what that's going to be. All right?

- 3.71** So a natural question would be to say, OK, look, if I just put money in the bank and I don't touch it and they just keep compounding it over and over and over again, how long do I need to leave it in there for me to make, say, \$5,000? That would be a natural question to ask. So a natural question would be to say, how many months until we reach \$5,000? That's a natural question, right? For example, if you want to save money for whatever the case may be, whether it's, I don't know, a trip, or you have educational savings or a car or whatever the case may be.
- 3.72** You're trying to purchase something, you want to know, how long does it need to be in there until it reaches the amount that I need? In this case, it's past your lifetime. This is terrible. You're not going to get to 5,000 in a reasonable amount of time. It's going to take forever because the interest rate is so small. The interest rate is so small. But what I'm asking is, OK, well, I know that the amount that I have is given by this, and I want to know, what is x , so that that is true? How many months do I need to leave it in there in order to get 5,000?
- 3.73** And when we talk about exponential functions and logarithmic functions, later on, we'll be able to come up with a concrete way to be able to solve this. But it puts it in the framework of a real-world example for which solving with exponents is relevant. Let me give you another one. And this one has a personal story attached to it a little bit. So as I'm erasing, I'll go through that. So in 2019, I was involved in a really nasty biking accident. So I'm an avid cyclist and all of that. And a friend of mine took me mountain biking. It was great. We were on the trail for two and a half, three hours, things like that. It was fantastic.
- 3.74** We come off the trail, I go up and over my handlebars, I land on the ground and I blow my jaw out completely. So broke my jaw, fractured in multiple places, all of that. I had to be wired shut in order for the bones to heal. So they wired my jaw shut, all of that, for eight weeks. So when I crashed, I went to the emergency room and my friend was with me. And I'm sitting in the emergency room and I feel zero pain. I'm in shock. I'm not feeling anything. And the doctor comes in and they start doing their thing, suturing me up, doing all that. And the next thing I know, I feel like I'm just sinking into the bed. I feel like the bed is literally consuming me.
- 3.75** And I look at my friend and I'm like, "What is going on?" Well, my friend had spoken to the nurse and said, "Give him morphine." I did not want that. I did not want that. And so that was the feeling that I was feeling, that the bed was eating me. I look over in the nurses, plunging morphine into my IV at the time. So let's think about this. They give me an IV drip after my crash. They give me morphine to take away the pain, which I didn't want. And so now I have morphine in my system. But here we are, years later. Well, obviously, I don't have any morphine in my system anymore from that experience, right? OK.
- 3.76** How does that work? How do you know that, if you have a headache or an ache or a pain or something, you take some ibuprofen, you take 600 milligrams of ibuprofen or

whatever the case may be, how do you know that it gets out of your system at a certain point? They have all these dosages, like, you can take this amount for this period of time, but don't take this amount because it's too much. It's toxic to your liver if you take this much or something along those lines. How do you know when you can take another dose? It's based on half-life. It's based on half-life. Primitively, it's based on half-life.

3.77 I mean, it's complicated than this because your bodies are complicated. Metabolism is very complicated, things like that. But let's say, for example, you take, let's say, 600 milligrams of some substance. Substance at time 0. That substance, whatever it is, let's say it's ibuprofen. Using the example of my bike crash, it would be the morphine, whatever its half-life was, things like that, but whatever it is. Let's say you have 600 milligrams of, say, ibuprofen. And let's say that, after each hour, it is half of the amount that your initial dose. Let's make that assumption. Let's make that assumption.

3.78 So let's assume a half-life, that's what it's called, of 1 hour. So what does that mean? That means that you take 600 milligrams of whatever it is, 1 hour later, your system has 300 milligrams of it in you, 1 hour later, you have 150 milligrams in you, and so forth, and so forth, and so forth. So let's try and model that. I want to find an equation modeling the amount at time t hours. This is just to show you that exponentials are relevant and that they show up. So let's try and think about this. At t equals 0 hours, how much do you have? So this will be time, this will be the amount.

3.79 Well, at t equals 0, you have 600 milligrams in your system. One hour later, how much do we have? So here's time. So we're going to say t equals 0. At time 1 hour, I should have, well, half of it in me. At time 2 hours, I should have half of that, so I should have 150. At time 3 hours, I should have half of that, which is 75. At time 4, I should have half of that, and so forth. Do we agree? So eventually, it's going to be so small and so minuscule that it's basically not in your system at all. Do we agree? Mathematically, it would say that there's some small trace of it, but your body is efficient and it filters this stuff out. And so eventually, you have none of it in your system.

3.80 So how would you actually model that? How would you actually model this? Well, let's think about it. What am I doing to go from here to here? I'm just multiplying by one-half. Do we agree? 300 is equal to 600 times one-half. How do I go from 300 to 150? Well, again, I multiply by one-half. How do I go from 150 to 75? I multiply by one-half. So what I notice is that, well, 300 is 600 times that, 150 is 300 times a half. But I know that 300 is this guy. So this is what? 600 times one-half squared. I know that 75 is equal to 150 times a half, but 150 is this. So this is equal to 600 times one-half cubed and so forth and so on.

3.81 OK. So 75 was the amount that's in my system after 3 hours. And I saw that 75 was equal to 600, the initial amount, times one-half to the third power. So if you keep iterating this, what's going to happen? Well, the total amount at time t is equal to 600 times one-half raised to the t power. Let's check that. At t equals 0, right? When you

take whatever the substance was, 600 milligrams of it, t equals 0. OK? One-half to the 0 power, one of our exponent rules was non-zero to the 0 is 1, so this becomes 600. One hour later, when t is equal to 1, it's 600 times a half, that's 300. Two hours later, it's one-half times one-half of 600. That's 150, and so forth and so on.

3.82 OK? So this is getting into the notion of what are called functions, which are one of the most important things in all of mathematics. What is a function? So this expression over here on the right-hand side, the 600 times one-half raised to the t power, is an example of a function, which is something that we're going to explore going forward in future classes.

Lecture 4: Functions

At a glance — Functions.

- 4.1 All right. We've talked about expressions, we've talked about equations, talked about equations in two variables, linear equations, things like that. We've talked about exponentials, the algebraic rules associated to them. What we want to get into today is another concept which is vitally important with mathematics, which is the concept of a function.
- 4.2 What is a function? A function is an interesting concept, but the idea is fairly simple. The idea is the following. Is there a way to come up with a notation which takes something into it? It describes an object which takes something into it and then outputs something based on whatever that rule is that you associate to it. You can imagine something along the lines of, OK, you feed me an input, whatever that is. Maybe it's a number, maybe it's an idea, maybe it's a string of numbers or something like that. I want to do something to whatever that input is that you gave. OK.
- 4.3 So you can imagine the following situation. Let's suppose we all have these recommendation engines around us all the time, based on our user history or whatever the case may be. So you purchase something and suddenly you get these recommendations saying, "Oh, you bought this? You probably want to buy this. Oh, you'll probably want to buy this." What is going on with that? OK, what is going on? Well, the input is your purchase history or your user history or whatever the case may be. They're taking those inputs and just constantly, constantly, constantly feeding them in, over and over and over again, into something, whatever that is.
- 4.4 And then they recommend to you something on the backend of that. So let's think about that for a second. So I have an input. That's my purchase history. It goes through some black box, basically. We don't necessarily know what that is. It's called machine learning and all of that, but that's fine. But it goes through this thing, whatever it is, and it outputs a recommendation. We have this process of an input goes through something, an output. That's a very similar notion to what a function is.
- 4.5 The idea is, take a valid input and give an output. Inputs and outputs. That's really what a function is at the end of the day. There's a couple of things. Now we have this idea. We have this idea of, I want to take something in, do something to it, and give something out. All right? So as with anything, we need to put some sort of mathematical notation associated to it. OK? So let's see how we can approach that. Notation.
- 4.6 Again, I want to build this from intuition. I want you guys asking yourselves these questions so that it's not something that—I don't like the idea that mathematics is just some memorize-and-regurgitate type of idea, right? Mathematics is really creativity and creation and things along those lines. I think it's closer to an art form than it is a

science, in my personal opinion. And ultimately, you don't look at an influential painter and you tell them, "You can only paint based on these guidelines." The painter paints how they want to paint, how they feel, their emotions go into it, their life experiences go into it, and things like that.

- 4.7 They build from how they feel. They build from the questions that they ask. They build from their experiences. I want you to do something similar in terms of thinking of mathematics. If we're thinking about the concept of a function, we need to say, all right, well, if I want something that takes an input, does something to it, and then gives me an output, I need to have a notation associated to that. So let's come up with a notation associated to that. Let's say, for the notation, we're going to say, we take a valid input.
- 4.8 We don't know what that is. It's a variable. It could be anything. It could be a single number. It could be a name. It could be an address, a phone number, whatever you want it to be. Why does it have to be only one thing? It doesn't. Maybe the input is multiple things. Maybe the input is a list of GPS coordinates. That would be multiple variables and things like that. But right now, we're just trying to get the basics. Let's suppose I just take a single input, whatever that is, and I want to give an output for what that is. The valid input is this.
- 4.9 We're going to have a rule, which is called our function. Our rule, which is a function. We need to give it a name. We need to give it a name, so a notation for it. OK, well, mathematicians are not very good at naming things. We name everything the same. We call everything normal, which is a little ridiculous. It's a function, so it's like, OK, we want a notation for it, so let's call it f . Why? Because it's a function, and function starts with f . So we're going to use f as our notation. It doesn't matter what you call it. You just have to give it a notation.
- 4.10 Just like a variable, you can call the variable x , you can call it y , you can call it t , v , w , whatever you want it to be; we're going to call a function f , simply because the word function starts with f . That's our rule, f . Then we need some way of connecting the language of the function f , which is our black box, that's our rule that's doing something to whatever our valid input is. Our valid input is some x — f is going to then act on x to give me an output. What is the output going to be? The output is just going to be denoted f , so my function acting on x , so f of x . That's my output. That's what I'm going to do.
- 4.11 OK, so let's think about this. Let's just do an example. Let's just do an example. Let's suppose you want a function, which squares an input. That's what it does. It just squares an input. That's it. You feed me an input, 5, and the function should give me 5 squared, whatever that is. You feed me an input, one-half, and the function should give me one-half squared, 7 squared, 9 squared, things like that. What am I really saying? If you give me 2, the function f should give me 2 squared. That's what it should output.

- 4.12 If you give me 3, the function f should give me 3 squared. You give me any input, I'm just going to square it. That's it. How could I notate this? Well, think about it. Your input is on this side, your output is on this side. What's the common feature? Well, the common feature is just this. You're just raising things to the second power. You're squaring them. That's the common feature. So you do this long enough and you say, OK, well, if you give me an input x , then the output should just be what? It should just be x squared. And that would be the valid output, f of x is equal to x squared.
- 4.13 Now, let's check that. How would we do this? I take f and I apply it to a specific value of x , a specific value of x . So what value do you want? Let's take x is equal to, I don't know, 1. That's going to be 1 squared, which is 1. So the output is 1. What about f of 0? That's 0 squared, which is 0. What about f of one-half? That's one-half squared, which is, based on our exponent rules, this is 1 squared divided by 2 squared, which is 1 over 4, and so forth and so on. That's the notation for what functions are.
- 4.14 But I want you to think about this in terms of valid input and then output. That's really what it is. Valid input and output. And why do I keep saying valid? Why do I keep saying valid? Well, let's suppose that I give you the following. What if I say, I want a function which whatever the input is that you give me, I want to take 1 divided by that input. So you give me 2, the output will be 1 over 2, half. You give me 5, the output will be 1 over 5. Makes sense? So that's all I'm saying. Why do I want these things? Goodness knows. But let's just suppose that that's the case. Let's just suppose that's the case.
- 4.15 So why do I keep saying valid inputs? Let's think about this for a second. OK, so here's another example. Input x goes to output 1 over x . OK, so what is the function associated to that? The function associated to that is, you give me an input x , I'm going to give you an output of 1 over x . OK, that's all I want. That's the definition of the function. OK, so let's think about that. If I take x is equal to 2, its output is a half. OK, that's well defined. f of 3 is one-third. OK, that's well defined. f of 0, it's 1 over 0. I don't know what that is. I don't know what that is. We're not allowed to divide by 0.
- 4.16 OK, so what in the world is this? This is shedding light on the fact that you can't just plug whatever you want into functions without actually thinking about what your input is. That's not allowed. It makes perfect sense if you think about functions as a machine to some extent. OK? What do I mean by that? This machine, this black box, whatever it is, you feed an input into it and it does something to it and then you get an output.
- 4.17 Well, just like any machine, for example, if you have a machine that does a particular process, it takes in some particular quantity, whatever it is, it does something and gives you the output. Well, if it only takes in certain types of inputs, it wouldn't make sense to just take some completely off-the-wall thing, stick it into the machine, and expect it to do something. That doesn't make any sense. The notion of a valid input is something important. We want to try and understand, what am I allowed to do? OK, what am I allowed to do? That's one of our questions. We're going to have this running list of questions.

- 4.18 What is valid input? Is there a mathematical notion to valid input? OK, for example, in this example up here, x is equal to 0 is not a valid input. You are not allowed to plug in 0 because the output is undefined. Second question we want to ask is, what is output? What is the function actually giving me? What is the function giving me? In this case, it's just spitting out numbers. So you give me 2, it gives me a half, you give me 3, it gives me a third, and so forth. The question should be, what do I need to put into the function to get a desired outcome, a desired output?
- 4.19 If you give me this function, f of x is equal to 1 over x , and I say, I want an output which is equal to 500, well, then what would my input have to be? Does such an input even exist? Go back to the examples when we were talking about wages, and we were talking about, I want to make sure that I make \$5,000 every single month to cover my monthly expenses and the savings and all that stuff. The output that I want is 5,000. What is the input that I need in order to achieve 5,000, i.e., how many widgets do I need to sell per month in order to make \$5,000 by the end of the month?
- 4.20 So you can couch these questions that we've already gone through in terms of outputs of functions. That's the point I'm trying to make. We want to know, what is output? How do I know that? When we were talking about linear equations, we said that they have graphs associated to them. They have graphs associated to them. So a natural question would be, OK, do functions have graphs? Do they have visual representations? OK. Representations. OK. Can I graph them in a particular way? So if you give me the Cartesian coordinate system, the x and y axes, can I somehow graph them? Does that make sense? I don't know. We'll see.
- 4.21 Fourth one. This one's natural, but you have to think about it for a little bit. One of the questions is, OK, I can view a function as somehow facilitating a transaction from input to output. You give me an input, the function then does something to it, and it gives me an output. So the way I think about it in my brain is, well, OK, I start over here, I give an input, I feed it through f , some function, and it gives me an output, which is over here. The question I want to know is, can I reverse that process?
- 4.22 If I go from over here, this is my input, and it moves me to over here, which is my output, can I somehow come up with another function which then takes me from here back to where I started? Can I reverse the process? Can I reverse-engineer the function to some extent? Or reverse the process of the function? So can we "reverse", in quotes, the function? Can we do that? Those are our motivating questions for trying to understand functions.
- 4.23 In order to handle this one, we need to understand the following. We need to understand, can we do arithmetic with functions? What do I mean by arithmetic? I mean, can we add them? Can we subtract functions? Can we multiply them? Can we divide them? Can we do any other operations with them? Is that possible? Those are the questions we're going to address. Those are the questions we're going to address.

- 4.24 So let's go after the first one, valid input. What does that mean? Valid input. Well, it has a name associated to it, and that's called the domain. OK. It's called the domain of the function. Here we go. Our definition, if you will, is that the domain of a function is the entire set of valid inputs. So, i.e., the, I'm going to say, in quote, "things" you are allowed to plug into f , plug into the function. The things you're allowed to plug into the function.
- 4.25 So as an example, if I give you the function f of x is equal to x squared, the first one that we saw, you give me an input, I want to square it. That's all I want. If you start thinking about all of the numbers that you know, are there any of them that you are not allowed to square? You start running through them. You're like, OK, 1 squared is 1, 2 squared is 4, 3 squared is 9, 4 squared is 16. You're like, OK, it looks like I can square all of the integers. That seems fine. Negative 1 squared is 1, negative 2 squared is 4, negative 3 squared is 9. Those look good.
- 4.26 What if I do fractions? One-half squared is one-fourth, two-thirds squared is four-ninths, five-sixths squared is 25 over 36, and so forth and so on. You're like, OK, well, fractions seem to work. Everything that you can think of, you can plug in and you can square. The domain of f is all real numbers. That's denoted by this boldface $\mathbb{R}$. That's how that's denoted. Sometimes people will denote this by saying, well, it's anything that you can think of. So it's everything from minus infinity up to infinity thought of as an interval. It's everything in there. All right.
- 4.27 Well, then let's see the other function that we had. So we had the other function was, number 2, we had, let me denote it with another variable so I don't confuse it with f above. Again, the name of the function, it doesn't matter. The only reason why we called it f was because the word function starts with f . But it doesn't matter what you call it. So let's call it g . And we said that g of x was equal to 1 over x . And then we started going through and we said, OK, well, what are the valid things that I'm allowed to plug in? 1, sure. 2, OK. 3, good. Zero, no good. Zero is not allowed.
- 4.28 So the domain of g is all non-zero real numbers. I can plug in any x that I want, so long as x is not equal to 0, so long as x is not equal to 0. So what does that look like? What does that look like? Well, you can denote it a couple of different ways. You could say, it's minus infinity up to 0, union 0 to infinity. So this notation is a little confusing. So what is this really saying? Remember, we had the parentheses when we were talking about inequalities? Parentheses means you're not allowed to attain that value.
- 4.29 So if I go from minus infinity up to zero parentheses, I'm saying I'm not including zero. And then the $\cup$ just means that you're taking this one and this one together. It's like what you're doing is, you're saying, take the entire real number line. Here's $\mathbb{R}$. Zero is in there somewhere. Here's zero. Take out zero. Remove zero. Allow yourself to take everything from this side and everything from this side. This side is minus infinity to zero. This side is zero to infinity. And just create a symbol, which looks like a $\cup$, that says you're going to take both of those pieces. That's just the notation for it.

- 4.30 That's my domain for that particular function. Now, functions have all sorts of crazy domains, and it depends on what the function is. But the important thing is that the domain is simply the entire set of valid inputs. What you'll typically hear is that, if you ask somebody, "Hey, can you explain what is the domain of a function?" The most common answer is, "Oh, it's the x values. The domain is the x ." It's like, that doesn't make any sense. It's clear what people are trying to get at, is they're saying, "Look, a function of x is equal to an output. So the x values are the inputs and, therefore, makes sense they're going to be associated with the domain."
- 4.31 But to say that domain is x doesn't make any sense, because what if I gave you this function, h of t equals t squared? And I said, what's the domain? You're like, the x is. It's like, well, there aren't any. So what do you actually mean about the domain is the x thing? That's why I like to think about domain as, it's the valid inputs. Whatever you label your variable, it's the valid inputs that you're allowed to plug in, not just the " x " values. That's our domain.
- 4.32 What about the outputs? What can we say about that? What can we say about outputs? So let's think about this. What are the outputs? How do I make sense of that? Well, again, we're going to give a definition to it. We're going to give a definition to it, so that we have a systematic language that we can use to actually talk about these things. What is the language that we're going to give? The definition is that the set of outputs of a function f is called the range of f . It's called the range.
- 4.33 The way I like to think about this, I have a very silly brain, I think, because the way I think about things is, I need cartoonish examples as to why things are the way they are. That's what helps things stick in my brain. I like to think of f as a catapult. It's like a catapult. What is the input? It's whatever you feed into the catapult. And then what is the output? It's you launch the catapult, you launch whatever it is, and wherever it lands, that's my output. That's my output.
- 4.34 So the way I think about it is, the way I think about output and the range of a function is, I think about, OK, I'm going to feed in all of my domain values, all of my valid inputs. I feed in all of them. I just keep loading the catapult over and over and over and over again, and I just keep launching it. After I fed in every single one of the domain values, I go and I look at the target and I say, OK, what did I hit? What did the catapult hit? Everything that it hit is your range. Everything that it hit is your range. That's the idea.
- 4.35 Let's think about this. Let's think about the function which is f of x is equal to x squared. What are the outputs? What are the outputs? How do I get the range in all of this? So you give me the number 9. Can you give me an input which hits 9? Sure, 3 squared is 9. OK, so 3 squared is 9, and 3 was f of 3. So that means that 9 is a valid output. The way I think about that is, I've got 9 on some number line out here. I've got the catapult, which is x squared. I feed in 3, and it launches it over, and it hits 9.

- 4.36 Well, negative 3 also hits 9. So I've got two things that go and hit 9. That's fine. I'm just interested in what actually can get hit. Nine can get hit. OK. What about, let's say, 2? Can I hit 2? Well, can I find something that squares to give me 2? Well, from our exponent section, if I take that thing and I square it, the exponents multiply, one-half times 2 is 1, and so 2 to the one-half squared is 2. So that's legitimate.
- 4.37 OK, so what is this? This is f of 2 to the one-half. Now, remember, 2 to the one-half, we said, was a square root, so this is f of the square root of 2. That's what that's saying. So the square root of 2 squared is 2. So I can actually hit 2 as well. I can hit 2. So if you go through this, over and over and over again, you can convince yourself that, what is the range of f ? It's everything from zero up to infinity. You can hit anything from zero on to positive infinity. You can't ever hit infinity. Infinity is not a number. Infinity is a concept. Infinity is not a number. But you can hit any positive number whatsoever, including zero.
- 4.38 Zero squared is zero. That's fine. But why can't I hit a negative number? Why can't I say, for example, why can't I find negative 1 is equal to f of x ? Why can't I ever hit negative 1? When you think about that for a second, you say, OK, f of x is equal to x squared. So you're saying, does x squared ever equal negative 1? And you're like, well, I don't think so, at least not to my knowledge. All the numbers that I know don't hit negative 1. Any number that when you square it is bigger than or equal to zero—2 times 2 is 4, 5 times 5 is 25—I can't really think of a number that, when I square it, I get a negative. So I don't think that's true. I don't think that's true.
- 4.39 Now, the frustrating thing with mathematics, well, frustrating and beautiful, so it's this chimera of sorts, is that whenever you run into something where you're like, oh, that's not possible, well, then you can just come up with a theoretical way of saying, well, let's just make it possible, which is really frustrating, and it happens all the time. So there is a context in which this makes sense. This actually makes sense that there are numbers that do exist which equal negative values. But for us, we're interested in the real numbers, and real numbers do not satisfy this.
- 4.40 You cannot take a real number and square it and get something that is less than zero. That's not allowed. So that's the example of our range. Now, sometimes you'll hear people say that the range is equal to the " y " values. So when you ask somebody, "What's the domain?" They're like, "Oh, that's the x ." And you say, "What's the range?" And I go, "That's the y ." You're like, "What the heck are you talking about? What do you mean, the y ? There's no y 's running around here at all. What do you mean?"
- 4.41 Well, they mean that you can graph a function in the Cartesian plane, the xy plane, where the x axis will be your inputs, the y axis will be your outputs. So it allows us to graph these particular objects. OK, so let's see how that works. OK. So visual representation of functions. So what we're going to do is the following. We're going to say, give the output the label y . We're just going to give it the label y . And what we're going to do is the following. The graph of f is the set of points given by x , f of x in the xy

plane. OK, what the heck does that mean? Well, let's take it step by step and see how that works.

- 4.42** So let's just take a concrete example. Let's take a concrete example. So again, let's stick with the squaring function, x squared. And let's try and draw its graph. Let's try and draw its graph. OK, so step one, so draw the graph. That's what we're going to try and do. OK, so how would we attempt to do this? OK. By the way, the best way to do this is to simply just use a free graphing utility online. That's the best way to do it. OK, you just plug it in and you're done. Tells you what the graph is immediately. You don't learn a whole lot from that, aside from just plugging it in. But that's, personally, my personal choice. It's just graph something using some free utility.
- 4.43** So let's draw the graph. So the first thing I need to do is, I need to set y equal to f of x . I set y equal to f of x , because after all, everything in the Cartesian coordinate plane has an x coordinate and a y coordinate. The x coordinates are going to be my valid inputs, my valid inputs, which we already said was everything, all the real numbers. You can think of everything on this infinite line. I'm allowed to plug in. That's fine. But the outputs are going to be what? They're going to be y , which is the f of x . We're going to set y equal to f of x . That's equal to x squared.
- 4.44** Really, the only thing that we can do at this point is, we can just start plotting points and see what happens. We can just start plotting points. Just like when we were graphing linear equations in two variables, the ax plus dy equals c , or cy equals d . When we were graphing those, we just plotted points. We saw that it was a line. Same thing is going to hold here. Let's look at our table. Here's x and here's y , which is now x squared. Let's just start plugging in values.
- 4.45** If I say negative 2, negative 1, 0, 1, 2, and so on, let's see what we get. What happens when I take x is equal to negative 2 and I plug it in? Well, I'm going to get minus 2 squared. OK, well, that's negative 2 times negative 2. That's just 4. Negative 1 squared. It's negative one times negative 1. That's one. Zero squared, that's zero. One squared, that's 1. And 2 squared, that's 4, and so on. Well, there's this weird symmetry that's happening. Whenever I plug in a negative value, that's the same as its corresponding positive value. So I'm getting, like, two inputs are going to a single output. Do we agree?
- 4.46** Negative 2 and 2 go to the same place, 4. Negative 1 and 1 go to the same place, 1. Zero is the only thing that doesn't have a mirror image. Zero and minus zero are just the same number. It's just zero. But for every negative value, the corresponding positive value, they're going to the same place. OK, so let's just graph this. Let's see what happens. OK, so here's 1, 2, 1, 2, 1, 2, 3, 4, 1, 2, 3, 4. OK, zero went to zero. I got this point. One went to 1, 2 went to 4, negative 1 went to 1, negative 2 went to 4, so something like that.

- 4.47 If we were to plug in a whole bunch of points in between this, you would see that this thing looks like— it looks like that. That's what its graph is going to look like. We can associate a name to this. This thing is called a parabola. That's not very important at this point, but we can, we name everything. So we can give it a name. It's a parabola. That's great. But at the end of the day, this is giving us a technique for being able to find out, what does a function actually look like?
- 4.48 Now, what's nice is that, when you have the corresponding graph of the function, the range is staring you in the face. The range is staring you in the face. So remember, we said that the domain here, the domain of f , was equal to all real numbers, which we denoted by minus infinity to infinity. Anything on the x axis, you're allowed to plug in. And we said that the range, what was the range? We said the range of f was zero up to infinity. We said it's everything that's bigger than or equal to zero.
- 4.49 Well, you can kind of see that, because what is the range? The range is the y values that you can attain. If you give me an arbitrary value here, call it, let's say, m , whatever it is, can you give me a corresponding x value? Can you give me an x value which goes to m ? The answer is yes. The square root of m will do just fine. So we'll minus the square root of m . Those two things will map to m together.
- 4.50 But if you give me something down here, let's call this n , and I draw a horizontal line here, does that intersect the graph anywhere? The answer is, well, no. So can I ever get n ? Can I ever get negative 1, negative 2, negative 3, negative 4? The answer is no. Because if I just look at all these values down here and I draw a horizontal line, it never intersects the graph, which means they are not in the range. They're not in the range. So the range is everything from zero up to positive infinity. That's what we have.
- 4.51 In general, finding the range is complicated because you need to know something about what the shape of this graph looks like. And there's lots of different techniques for figuring this out. I think graphing it is personally my favorite technique, because I just have to punch it into a computer and it just says, "Hey, here it is." That's a nice technique. If I gave you a crazy function and just said, "Hey, what's its range?", in general, very difficult. Very difficult to find.
- 4.52 And think about that in terms of the way that we motivated functions in the beginning. In the beginning, I motivated functions by these recommendation engines. You have some sort of purchase history, whatever the case may be. That's your valid inputs. It goes through this function, whatever it is, and your output is your recommendations. Think about how difficult that would be, OK? Your recommendations are no longer like real numbers. It's not saying, "Hey, you bought this textbook. Hey, you bought these shoes. Hey, you bought this suit. We recommend that you should buy this suit or these books or something like that."
- 4.53 The output is not, "Oh, you purchased these shoes. You purchased this textbook. Your output is the number 7." It's like, that doesn't make any sense at all. Your output is a

quantity. It is a commodity. It is something that it is recommending that you buy next. So think about that. Figuring out what the range of that is is incredibly difficult. That's an incredibly difficult thing. You're trying to say, "Hey, here amongst all the books in the world, that book is your next output." That's insanely difficult, determining what your corresponding range is going to be.

4.54 In this case, we're playing with real numbers, and so it's a little bit easier to decide what that is, just based on the graph of it. But in general, this is an incredibly difficult problem. So now what do we have? We have valid inputs. That's our domain. We have the outputs. That's the range. And now we know how to graph functions simply by setting y equal to whatever the determining function is, f of x , and graphing the points x, y . So we have three of the five things that we were trying to figure out.

4.55 Let's look at the arithmetic. Let's look at the arithmetic. The arithmetic of functions. What do I mean by arithmetic? I mean plus, minus, multiplication, division, maybe something else. I don't know. Functions are weird. Numbers are nice because numbers I can do plus, minus, multiplication, division. That's fine. We're used to doing that with numbers. But is there anything else that we can do with functions? It turns out there is, which is strange. That's the hardest one, unfortunately. But let's talk about addition, subtraction, multiplication, and division. Let's talk about those things.

4.56 If I give you two functions, how do I add them? How do I add them? The notation is a little goofy, and I'll walk you through this. I'll walk you through this. But if I give you a function f and I give you a function g , whatever those two things are, and I say I want to create a new function, the new function is f plus g , that's the label I'm giving it, that's the label I'm giving it, what do I need to know? I need to know whatever that new function is, f plus g , I need to know what happens when I put an input into it? What does it output?

4.57 It's what you think it would be. I know how to get the input or the output for f when I evaluate it at x , and I know how to get the output for g when I evaluate it at x . So you just add the two together, just like normal. So as an example, let's say my running examples are f of x is going to be $2x$ plus 3 and g of x is x squared. Let's just say these are going to be my running examples. Then what would f plus g be? The running example, the running example for number one is, well, let's do for add to keep the same notation over there— f plus g of x is equal to f of x plus g of x . That's all it is.

4.58 You just take whatever f of x is, and you add whatever g of x is, combine their outputs. That is what the sum of the function is defined to be. Same thing for subtraction. If I take f minus g of x , that's just going to be f of x minus g of x . OK, so for subtraction, f minus g will be $2x$ plus 3 minus x squared. That's it. OK, that's it.

4.59 What do you think multiplication and division are going to be? OK, so if I take f times g of x , that's just going to be f of x times g of x . Multiplication is going to be f times g of x is $2x$ plus 3 times x squared. And division is no different. Division is no different. If I took

f divided by g , this is f of x divided by g of x . So division will be f divided by g is equal to $2x$ plus 3. That's f divided by x squared. That's what we get.

- 4.60 There's one more. There's one more. And this is the tricky bit. This is a new operation. We don't have this operation necessarily when it comes to arithmetic of numbers. Arithmetic of numbers, we have plus, minus, multiplication, division. Those are the four fundamental ones. You learn in grade school, I guess. I don't know, wherever you learn those things. But this new one is called composition. This one's called composition. It's by far the hardest of the operations, mainly because we're used to these operations. We're used to addition, division, multiplication, subtraction. We're used to those things. We've been doing that for a long time.
- 4.61 Composition, we aren't really introduced to it until we talk about functions. I want to motivate it in the following way. Again, you can think about functions as moving you from one place to another. You can think about them as moving you from one place to another. I start with my input that's here. I take that input, I feed it into the function f , and the function f moves me to the output, which is over here. You can think about it like that. I give you the input, which is 2. The function squares me, 2 squared is 4. It moves me over here.
- 4.62 If you think about it as a rule of motion, then it stands to reason that you can iterate this. You can iterate this, right? If I walk across the floor and I walk from here to here, well, every step that I take, I could view as being a different function. I've got one function which takes me from here to here. I've got another function which takes me from here to here. I've got another function which takes me from here to here, and so on. I can view it as just a sequence of steps which takes me from one place to another. Do we agree? OK.
- 4.63 So composition does the following thing. Its notation is a little goofy. So it's f and then a circle, g , add X . So it looks like $f \circ g$, but it's not an O , a letter O . It's just f and then symbol $\circ$, circle, g of x . Now, it looks like this. It's f and then g of x . OK, f of g of x . Now, what I want you to think about is the following. It's like these nested operations. These nested operations. The way that I think about this is, I take an input x , and this input x has to be a valid input for what function first? Where does x go into first? Does it go into f or does it go into g ? In this case, x goes into g first.
- 4.64 x needs to be a valid input in g , meaning it needs to be in the domain of g first. I take x , I plug it into g . g then does something to x , what it does. I start with x , I feed it into g , it moves it over to here. Now I'm at g of x . That's the output. Whatever that g of x is, it has to be a valid input for f , so it needs to be in the domain of f . I then take that g of x , plug it into f , and f moves it to f of g of x . That's the idea of these things.
- 4.65 What does it mean in this case for composition? What is f of g of x ? Well, let's think about this. Do I have a formula for g of x ? Sure. g of x is equal to x squared. g of x is equal to x squared. I could really replace this g of x with the formula for it. It's just x

squared. That's what it is. I now say, OK, what is this telling me to do? My input, my input is now x squared. I go and I find the formula for f , and I replace its input, wherever I see its input. I have a $2x$ plus 3. I replace that input with the corresponding input here, which is x squared.

- 4.66** What am I going to get? This tells me, take your input, double it, and then add 3. So this says, take my input, double it, and add 3. That's what my corresponding output is going to be. And that is what the composition of that function is. This is arguably one of the most important ones. It's arguably one of the most important ones because the following thing happens. So I talk about these recommendation engines. So whether your purchase history or whatever the case may be, you get a recommendation based on your user history. You get that recommendation.
- 4.67** In this case, these compositions are precisely what is happening in that process of giving you recommendations. You have an input, whatever that is, your user history, you feed it through a particular layer. That's a function. You then take that layer, you feed it into another layer. That's another function. You then take that and feed it into another layer, and you do this multiple times. Eventually, you go through enough layers, enough functions, that you get an output, whatever that output is.
- 4.68** So you have a multiple composition that you run this through, and then you finally get an output, and that output is your recommendation. This is the very, very primitive form of what's called a neural network, but it's based upon function composition. It's based on function composition, which is pretty amazing, and optimization. But it all starts here.
- 4.69** Our last one— can I undo a function? Can I reverse it? You give me an input, and it gives me a corresponding output. Is there another function which starts at your output and gives you your original input? Is there some way of inverting the process? I go from a to b . I want to know, how do I go from b back to a ? That's the idea. OK. So how to invert a function. Is it even possible to invert a function? I'm operating under the assumption that you can. Saying "how do I do something?" is implicitly assuming that you can do said thing. Maybe I can't.
- 4.70** One thing we need to understand with functions is we have to go back to the definition. So a function takes a valid input. Valid input is our domain, remember, to a unique output. Output is our range. Functions are not allowed to, say, take an input and then give you 20 things based on that input. It's not allowed to do that. You can't have a function which says, think of the squaring function. If I take in the input of 5 and I square 5, I don't get 25, 36, 79, 82, square root of 7, 0, 4, and 8. I don't get all of those things. I literally take 5, and I square it, and I get 25.
- 4.71** I get one output. That's it. I am not allowed to have multiple outputs for that. Does that make sense? I'm not allowed to have that. You might counter this. You might be like, "Well, Robbie, you just told me that recommendations, and when I watch shows and

they recommend a whole slew of shows, that's multiple outputs." I'm like, fair point, but it is still one output, I would argue. It just has multiple slots for its output. Each slot is a recommended whatever. You give me an input, your user history, and it gives me an output. The valid input is my domain, the output is my range. I want to understand, am I allowed to invert the process?

- 4.72 Well, let's go back to the function that we started with. If I have f of x , which is x squared, we saw that its graph is the parabola. It looks like this. It looks like this, roughly. I'm doing my best here. That's my parabola. Let's just start with an arbitrary value. We'll start with a value of 3. That's my input. My output is up here at this point, which is 3, comma, 3 squared. That's 9. So 9 is right here. So my function takes 3, and it sends it to 9. Well, what's the problem? It also takes negative 3, and it sends it to minus 3, comma 9. OK.
- 4.73 So think about how would you reverse this. So you have 3 goes via f to 9. You now want something which is going to take 9 and send it back to 3. Would that work? And the answer is kind of. Not at face value right now. Because think about it. Over here, I have my domain. I have the value 3. Over here, I have my range. Three gets mapped to 9. f takes 3 to 9. The problem is that it also takes minus 3 to 9. So think about it. If you want to reverse the process, you have to start from the range and go back to the domain.
- 4.74 So if I want a function which goes this way—let's call it g —if I want a function which goes that way, I need a clear, concise way of saying, where in the heck does g take 9? Somebody might argue, well, 9 should go to 3. And I would counter, well, I want it to go to minus 3. And you say, "Well, OK, well, that's fair. So what does it go to?" Well, it should go to both. Well, that doesn't make sense. That contradicts what a function does. It takes a valid input to a unique output. Unique means one output. You're not allowed to take 9 to both 3 and minus 3. Those are different values.
- 4.75 So that would not be a function anymore. It does not take an input to two different values. So then you say, OK, well, then I don't know what to do. It looks like an inverse doesn't exist. Inverting the process doesn't exist. So, g would have to take 9 to both 3 and minus 3, hence not a function. It's not a function. You cannot take a single input to two distinct outputs. That is not allowed. That violates the definition of being a function. So as stated, the inverse does not exist. So you cannot, you cannot invert f of x is equal to x squared on its domain minus infinity to infinity. And this is the key distinction.
- 4.76 You cannot invert it on the domain, the entire domain negative infinity to infinity. The reason why, geometrically speaking, is to say, if I draw a horizontal line, and that horizontal line intersects the graph of f of x equals x squared in two or more points, then I cannot invert the function because I run into this phenomenon. Where would I take 9? I would have to take it simultaneously to minus 3 and 3, and that's not allowed. So what could I do? If I wanted to invert the function, what could I do?

- 4.77 Well, I could abandon the whole concept altogether and just say, fine, we're not going to talk about inverse functions at all. We don't want to deal with them. That's not a great strategy. What would be better is to say, well, can I somehow just say, I don't want to deal with the entire domain. I don't want to deal with the entire domain. Because look, what if I just covered this side up? What if I just covered this side up? I'm not allowed to consider that side. I'm only going to consider this part of it here, from here up to there. Do I still get the entire range? Do I still get the entire range?
- 4.78 Remember, the range is, what are the y values that you can attain? And those values were from zero up to infinity. So if I only consider this branch of it, do I still hit everything on the y axis? Do I still get every y value possible? And the answer is, well, yeah. You give me, you draw a line here. Do I hit the graph? Yes. You give me this line. Did I hit the graph? Yes. This line. Did I hit the graph? Yes. But what did I gain? I no longer— none of these lines are now hitting the graph in two points. They're only hitting it one single point.
- 4.79 So if you give me the point 9, up here, I can say, where would I send it? I would have to send it to 3. So restricting the domain, so restricting the domain, restricting the domain of f of x equals x squared to the interval zero to infinity allows you to invert f . It allows you to invert the function because now, what would the graph of this look like? The graph would look like this. That's my function.
- 4.80 If you give me the value 9, and I want to say, what would g of 9 be? G of 9 should be 3. That is now the unique value that got sent to 9 under f . So f would take 3 to 9, g would take 9 to 3. I start at 3, I move to 9, I go at 9, I go back to 3. Same thing here. Four. Four would go to, would go to 2. Five, 5 would go to square root of 5. Whatever I need to get the value back. What would my function g of x be defined as? That would be my square root function. Let's check that the two get me back to the same place. Let's check that the two get me back to the same place.
- 4.81 If I take what is f of g of x ? I start with x , I plug it into g , I take the output of that, and I plug that into f . What was g of x ? That was square root of x . What was f ? Using our properties of exponents from before, that gives me my original input. I start with x , g moves x . I start with that in f , f takes that back to x . That is the notion of the inverse function. So g is the inverse function of f . And it's denoted g is equal to f upper minus 1. That's its notation. You give it that inverse notation to comply with the same concept that we had in terms of inverting exponents.
- 4.82 This does not mean 1 divided by f . It's just a notation which says this is the inverse function of f . It says that whatever f does, g undoes it. G undoes it. That's the notion of inverse functions. So there you have it. Functions are objects which take valid inputs, domains, into outputs, the range. We have arithmetic of functions. We can add, subtract, multiply, divide, and now compose. That's an additional one that we have, that we didn't have for arithmetic. And the composition allows us to say, well, what can

we actually do with functions? Can I take something to some place and then somehow bring that something back to where I started? The notion of inverse functions.

Lecture 5: Polynomials

At a glance — Polynomials.

- 5.1** All right, so today we're talking about a new class of functions which are called polynomials. So polynomials ultimately are just trying to generalize the things that we've already talked about with our linear equations and linear functions along those lines. So what's the definition that we're looking at? OK, so polynomials, it really just breaks down into two things. So poly, meaning multiple, nomials meaning names or terms or things like that. And we want to kind of start from what we already know and then just build from there.
- 5.2** So we know that we have these expressions that look like some constant times x plus another constant, where a and b are just numbers and x is our corresponding variable. We've been studying these for quite some time, and so we've called these our linear expressions, let's call them that. But what we can ask is, we can say, well, we've talked about exponents, so can I just include higher and higher and higher and higher powers of my x variable? Can I just include those?
- 5.3** And the answer is, well, sure, why not? So what if I had something that was a , I just increase the power on x by 1, and then I'm just going to tack on a single power of x and then have another constant. OK, if those are useful; if they're not, it's kind of irrelevant at this point. What we're just trying to do is say, well, can I generalize this particular expression? Can I get more, how do you say, can I get more general to some extent, right? And you can kind of see that these types of expressions, they capture this expression. How do you see that?
- 5.4** Well, what if a were 0? Well, if a were 0, then I get something that looks like constant variable plus constant, and that's exactly what I have here. So you're just trying to generalize from there. But then once you get from there, why not just kind of go crazy and keep going as high as you possibly can? OK? And that's the notion of a general polynomial. So the notation will be—I'm going to run out of letters, OK, once I get past 26. So I want to get a more general notation to be able to use this. So I'm going to start indexing things by numbers.
- 5.5** So a sub n , meaning like n could be 500 if I want. So this is just some constant, and it's indexed by n , just indicating that it's some number. But then I can just go all the way down. So $a_{n-1}x^{n-1}$, plus $a_{n-2}x^{n-2}$, and so on, all the way down to a $1x$ plus a 0. OK, so this expression right here is what we think of as a polynomial. So what are some examples of these?
- 5.6** Well, the first thing that we have to do before we start going through examples is, is there some way I can characterize these particular polynomials? Can I say that polynomials of this type have a certain property, polynomials of that type have a different property? And how can I distinguish them from each other? Well, the way you

distinguish polynomials is by their degree. OK, so the definition is going to be the degree of the polynomial here. OK? So this polynomial here, the degree of that polynomial will be the highest power of x occurring on x , which has a non-zero coefficient.

- 5.7 OK, that sounds kind of, that's a mouthful. But the degree of that polynomial is the highest power of x with non-zero coefficient. OK, so what does that really mean? Well, let's just look at examples. This is all kind of abstract. There's all these symbols on the board and things like that. Let's actually look at concrete examples and characterize things by their degree. OK, so let's look at these. All right, so we have.
- 5.8 What if I look at degree 0? Degree 0, what do these polynomials look like? Well, if I go to the definition, the definition says the degree, that's what I'm defining, is the highest power of x with a non-zero coefficient. So, if the degree is zero, and I kind of look at this expression, that means that the only power of x that I can have is zero. So I have an x to the zero term. Well, if I had an x to the zero term, what's the only thing that's going to be there? It's just going to be the constant. It's just going to be my constant. So these are constants.
- 5.9 Well, those are kind of boring. They don't really do anything. They just stay constant the entire time. Ok, what about degree 1? Degree 1, that means that the highest power of x occurring is x to the first power. So degree 1 are precisely these ones. Those are my linear expressions. OK, so those are the ones that are going to look like ax plus b .
- 5.10 Now one thing that's kind of, I don't know, I would say universal in mathematics, is when you only have a few terms, you typically label them by like an a , b , c , x , y , z , something along those lines; you actually label them with just concrete letters. But when you start getting higher than that, you're not going to label something as, like, a , b , c , d , e , f , g , whatever. You just start labeling them as, like, a_1 , a_2 , a_3 , a_4 , a_5 . It's just a notational convenience.
- 5.11 OK, so it's kind of annoying that, like the degree 0, degree 1, well, I'm giving you a definition in terms of n . So if you want, you could think of this as really using that notation as like, $a_1 x$ plus a_0 . OK? Using this notation here. All right, what about degree 2? Degree 2, these are going to look like ax^2 plus bx plus c . Or in the notation that we have down here, these are going to look like $a_2 x^2$ plus $a_1 x$ plus a_0 , and so on.
- 5.12 Now, when I'm writing these out, implicitly I'm assuming that the leading coefficient, so the coefficient in front of the highest power of x , is non-zero. I'm kind of assuming that a is non-zero, because if a were zero, what am I looking at? I'm looking at bx plus c . That's not degree 2 anymore. That's degree 1. OK? So I'm assuming that the leading coefficient is non-zero. And then let's do the last one. Degree 3. So these are things that look like $a_3 x^3$ plus $a_2 x^2$ plus $a_1 x$ plus a_0 . All right?

- 5.13** And we can keep going over and over and over. OK, so these are my polynomials. Now, some of these have names associated to them. And so we want to give those names so that we understand the language that we're talking about. So, for example, the degree 0 polynomials, I called those constants. The degree 1 polynomials, we called those the linear polynomials or the linear expressions. Degree 2 polynomials, these ones are called quadratics. We're going to spend some time with that. So those are quadratics.
- 5.14** And then the degree 3, these ones are called cubics, OK? Degree 4 would be called quartics, degree 5 would be called quintics. And then everything from degree 6 and up, like, nobody really cares. We just call them polynomials; frankly, we just call them polynomials in general, but you could put names to them if you want. You know, the sextic ones, the septic, you know, septic polynomials, the octic polynomial, whatever. We just call them polynomials once we get to a certain point, OK?
- 5.15** So now we have this new class of objects, mathematical objects. And one thing which is kind of universal in mathematics is once you have a new mathematical object, whatever that is, a linear expression, a polynomial, matrices, you know, whatever, vectors, whatever the case may be, whenever you have a new object, you want to understand what arithmetic properties those things have.
- 5.16** So, for example, when we talked about functions, functions were these objects that took a valid input and took it to an output. And we found out that we could actually do arithmetic with functions. I could take functions and add them together. I could take functions and multiply them or divide them or compose them. I can do various things with them. So now we have a new class of objects, mathematical objects, which are polynomials. And we want to ask the question, well, can I do arithmetic with them? Can I add polynomials? Can I multiply polynomials? Can I divide them? And those types of questions.
- 5.17** OK, so let's just, let's just explore that, OK? Let's write out some concrete examples and see how we would explore, like, adding them, subtracting them, multiplying, dividing, if that even makes any sense. OK, so the first two are relatively straightforward, adding and subtracting, so let's just do some examples. OK, so let, let's say, $2x$ minus 3 and $4x$ squared minus $6x$ plus 7 be polynomials. OK? So I've got two polynomials and I want to know how would we add or subtract them? OK, we want to say find, let's say, $2x$ minus 3 plus $4x$ squared minus $6x$ plus 7. We want to add the two and see what happens.
- 5.18** Well, let's think about how we could approach this. Remember when we were dealing with linear expressions like this, if I gave you like a $2x$ minus 6 plus $4x$ plus 9, and I said, how do I combine those? We just said, oh, we'll just gather the like terms. The terms that have an x , add those together or subtract those. The terms that have constants, just add those or subtract those. And so why wouldn't the same principle apply here? Right? And the answer is, it does.

- 5.19 So I look at this and I say, all right, what am I going to get? I have to step back and say, OK, what is the highest power of x occurring? OK, I stare at this long enough and I say, OK, I have an x here, a constant, an x squared, x , constant. OK. x squared, that's the highest power of x occurring. OK? Now do I have any other x squareds in my expression? The answer is no. OK, do I look at—I look at this coefficient and I say, is that a positive or is that a negative? And it's positive. So, OK, my first term is going to be $4x^2$.
- 5.20 And then I kind of cross that off to let myself know I've dealt with that term. I'm going to move on to the next degree, one less term. OK? So this is my degree 2. That's my x^2 . I now look at degree 1. So what do I have? Well, I have a $2x$. OK, I have a $2x$, and then I have a $-6x$. OK, so what is $2x$ minus $6x$? So 2 minus 6 is negative 4 . And so I'm going to get a $-4x$ here. OK? So that's gone. That's gone.
- 5.21 Now I go degree one less than that. So I have my x^2 , my x , degree 1 less than that is degree 0. Those are the constants. And I say, what do I have? I have a -3 plus positive 7 . OK, so negative 3 plus 7 is positive 4 . And I get my $+4$. And that's as far as I can go. OK, that's as far as I can go. There are no more terms. So everything's done, and cross all these off. And that's how you do it. So the moral of the story, the moral is simply just to combine like terms. OK? You combine like terms. All right, well, that's for addition. That's great.
- 5.22 Well, how would I subtract the two? OK, how would I subtract the two? So if I wanted to say $2x^2$ minus 3 minus $4x^2$ minus $6x$ plus 7 , well, the only thing that's different here is that this coefficient is negative. And if that coefficient is negative, I need to move that in using the distributive property to all three of those terms. Once I do that, because remember, addition and subtraction are basically the same operation, if I say 2 minus 4 , that's the same as 2 plus negative 4 .
- 5.23 So what I can do is, I could think of this as kind of like plus minus 1 , and then I take the negative 1 and distribute to all of those terms. And now it becomes an addition problem. And I know how to do that. OK. However you want to think about it is completely fine. I'm just going to leave the negative here. So if I have a minus, what do I have? I have a $2x^2$ minus 3 and then I have, bring the negative in. So I'm going to get a $-4x^2$. Bring the negative in there. I'm going to get a minus minus, so plus $6x$, and then minus times that, minus 7 .
- 5.24 And now we're back in the same case that we just had with the addition. OK, gather like terms. I have one of my x^2 term. So I have a $-4x^2$. That one's gone. I now have $2x$ plus 6 , so that's an $8x$. Those are gone. And negative 3 minus 7 is minus 10 . OK, so addition and subtraction, that's what we get. All right?
- 5.25 The harder of the two or the harder operations come from the multiplication and division. How does that actually work? OK, so let's start with, let's stick with this example, these two polynomials, how would I multiply them? How would I multiply

them? How does that work? And this is actually a really important technique. So multiplication of polynomials. OK? All right, so let's look at these two. How do I do that? I want to take $2x$ minus 3, and I want to multiply that by $4x$ squared minus $6x$ plus 7.

- 5.26** And I'm going to say, how do I do that? Now, classically, like, this isn't how you introduce these particular things. OK, you start from, like, smaller polynomials, and then you try and build it from there and so on. But I don't think it matters what the objects are because the underlying principle is the same across all of these. OK, you're using the distributive property. All right?
- 5.27** And so I just want to illustrate that with this one. How do you do this? The way that you do this, the method is very simple. It's kind of long and drawn out and a little, a little annoying, but the method is very simple. OK, the method is, you start with this first term. So that's my $2x$, and then you simply distribute it to every single term of the polynomial that you're multiplying it into. So you take the $2x$, you multiply it in to this term. What am I going to get?
- 5.28** The coefficients multiply. So 2 times 4 is 8. OK, great. But what is x times x squared? Well, remember, we have this rule for how to deal with these. If I take x times, right here, x times x squared, remember, our power on our exponent or our exponents rule tells us that if the bases are the same, the x 's, then the exponents just add. Right? So I'm going to get x times x squared. This is x to the first, x squared. This is x cubed. All right?
- 5.29** So if I have $1x$ and I multiply it by x squared, I get x cubed. Great. So $2x$ times $4x$, the 2 and the 4 give me an 8. The x and the x squared give me an x cubed. And now we're just going to keep doing that to all of the terms. We're going to distribute to everything. So this one came into this one. I now need to hit this one. What do I get? The coefficients 2 and negative 6 give me a minus 12. The x and the x give me an x squared. And I keep going.
- 5.30** Next one, $2x$ and 7. The 2 and the 7 give me a 14, and the x goes along for the ride. OK, you might be upset being, like, well, wait a second, you're not multiplying by an x . So how do you just get a single x back? You can kind of cheat and say like, well, look, this does have an x attached to it, just kind of in a ridiculous way. Remember, x to the zero is just 1, right? So you can think about it as, like, having an x , if you want to do that, but you don't, you don't need to. And frankly I don't think you should.
- 5.31** OK, but if that helps to try and get that principle like, oh, I'm multiplying the x 's by a constant, it's like, well, it's not going to change the x at all. The x is just going to stay the same. OK, so what have I done? I've done the $2x$ and I brought it all the way through. What do I need to do next? I look at my negative 3 and I bring that one all the way through. So negative 3 times 4 is negative 12, and the x squared goes along for the ride. I take the negative 3 multiplied by the negative 6. Negative 3 and negative 6 is 18.

- 5.32 OK, the negative signs cancel. So I get 18, and the x goes along for the ride. And then negative 3 times 7, that gives me minus 21. So that's the expansion of the multiplication. But then when I step back, I say, well, can I simplify that? Can I combine terms? And I step back and I say, OK, well, if I combine terms, remember we're using the combine like terms philosophy. I look at the x cubes and I say, are there any other x cubes that occur? And the answer is, I only see one x cubed here and everything else has lower degree.
- 5.33 OK, so I'm going to have an $8x$ cubed. I've dealt with that term. What about the squares? I see a negative 12 minus another 12, so I'm going to get a minus $24x$ squared. So that term is now gone. Do I have any degree 1 terms, the x terms? Well, I see one right here, $14x$, and I'm going to add an $18x$ to that. So that's what—20, 30, 32? $32x$, those are gone. And then the only term left is the constant, which is minus 21. OK, so minus 21.
- 5.34 I see, do I have any other terms that I can combine? The answer is no. And so this is my answer. That's my answer. OK, so multiplication is somewhat weird, but I would argue it's very natural. OK, the multiplication is, what do I do? Well, if I want to multiply these two things, I take the first term, I multiply it through everything else. I take the next term, I multiply it through everything else. That's literally just the distributive property. OK, when we're trying to, in terms of our arithmetic operations.
- 5.35 It's just kind of arduous, a little frustrating, and you can see how this can blow up and get really crazy really quickly because look at this. How many terms did this polynomial have? Well, it had two. It had the $2x$ term and the minus 3 term. How many terms did this guy have? Well, it has three, right? I've got my $4x$ squared, my minus 6, and the 7. And when I just did the distributive law, how many terms did I have? Well, when I brought the $2x$ through, I had 1, 2, 3 terms. 1, 2, 3 here, right?
- 5.36 And then when I brought the negative 3 through, I had 1, 2, 3 additional terms. I had 1, 2, 3 there. So I'm going to have 6 total terms before simplifying, right? When I simplified, I saw that I only had four terms, and that's fine. But you can see how this would be terrible if I had something like, take a polynomial here where the degree of the polynomial is, say, 9, and I have a polynomial here where, let's say the degree is equal to 10 or something along those lines. So I have these massive expressions, and I want to say, what's their product?
- 5.37 How do I multiply those two things? Well, how many am I going to have? If I have a degree 9 polynomial here, this thing potentially has 10 terms, if you think about it. Think about this one. Look, this is a degree 2 polynomial, right? The highest power of x is 2. So the degree is 2. But how many terms did I have? I had 1, 2, 3. I had 3 terms. So if I had a degree 9 polynomial, potentially I have 10 terms. Worst case, I have 10 terms, OK? So this potentially has 10 terms. This potentially has 11 terms. Without simplification, I would have 110 terms.

- 5.38** You don't do this stuff by hand, OK? Like, that's what I'm trying to illustrate. You don't do this by hand. You use some sort of computer algebra symbol to say, like, look, here's a polynomial, here's a polynomial. Tell me what their multiplication is, right? You don't do this by hand, in practice. All right? So we're illustrating this by hand because these ones are small enough that we can actually—it's tractable. We can actually do it. But in practice you don't. OK? You don't do it by hand.
- 5.39** So the question I want to say is the following. Well, look, we know how to go from this direction to this direction. You give me two polynomials. How do I multiply them? I get this. The question I want to ask is, can you reverse it? Can you reverse the process? And this is the notion of what's called factoring, which is like the bane of existence of everybody in an algebra class, right? Factoring is like the worst thing ever.
- 5.40** OK, so if I just gave you this polynomial and I said, "Factor it," what do I mean by factor? I mean, can you write this like this? And the answer in general is, that's an insanely difficult problem. But we'll handle it. OK? We'll handle it. OK, so that's our multiplication of polynomials. What about division? OK, what about division? Well, division is tough in general. But just like I said with addition and subtraction, I claim that division and multiplication are effectively the same operation.
- 5.41** OK, how do you see that? If I said I have a quantity x , and I say I want you to take x and divide it by 2, OK, fine. I just take x , divide by 2. Well, that's equivalent to saying, take x and multiply it by one-half. Those two things are identical to each other. So if I say divide something in half, it's the same as multiplying it by half. So multiplication and division are kind of like the same things like addition and subtraction were, right. If I said 2 minus 3 , that's the same thing as 2 plus minus 3 .
- 5.42** So I can think of subtraction as addition. You can also think of division as multiplication. OK? OK, so how do we handle this? So how do we handle this? Well, we need to be able to identify degrees. So here's just kind of a quick rapid-fire list of examples. If I gave you the following polynomials, 9 minus x squared plus $3x$ cubed, 2 plus x minus $7x$ to the 6 th, $4x$ squared minus $2x$ plus x to the 5 th, $7x$ squared minus $3x$ plus 1 , x squared minus x minus 1 , and so on—if I just gave you those, and I wanted you to identify the degree. OK, I want you to identify the degree.
- 5.43** The degree here is equal to what? How do I do this? Right. I look at the polynomial. And I say, OK, look at the highest power of x occurring. That's the definition of the degree. So you just scan through it, you find the highest power of x . OK, so this one has no x . This one has x squared. This one has x cubed. Three is bigger than 2 . Therefore the degree of this is equal to 3 . OK, I look at this one, this has no x . This has one x . This has 6 of them. Six is bigger than everything else. Degree is equal to 6 .
- 5.44** I look at this one, I have an x squared, an x to the first, x to the fifth. That one wins. Degree is equal to 5 . This one I see a square, a one, a constant. Two wins out. Degree is equal to 2 . Same one here, 2 , degree is equal to 2 . Those are my two quadratics. This

one is a cubic, this one is a quintic, my degree 6, and so on. The bottom one is a famous one, which has to do with the golden ratio and the Fibonacci numbers and all that kind of stuff.

- 5.45** But being able to identify the leading term, which corresponds to the term giving you the degree, is vital in order to be able to do long division. It's vital in order to be able to do long division. OK, so let's look at long division now. OK, how would I do something like this? What if I say, let's think about long division just in terms of arithmetic. OK, so let's go back to how do you think about division in the first place? How do I make sense of the quantity 17 divided by 6? How do I make sense of that?
- 5.46** Well, personally, I just plug it into a computer and it tells me a decimal and I say, "That's how I make sense of it." The computer tells me this is what it is. I don't necessarily care about like, oh, 6 goes into 17 twice with some stuff left over and then all of that. I don't necessarily care about that in my day-to-day. OK, if I'm just doing a calculation, I want to know 17 over 6, OK, the computer will tell me. It's this up to a certain number of accuracy and then I'm fine with that, more often than not.
- 5.47** How do we think about this? So we have this little way of thinking about it. I'm going to take 6 and I'm going to divide it into 17, all right? And I ask myself the following question. I say, OK, starting with 17, look at 6 and say, how many times does 6 go into 17 without going over? Without going over? So I start looking at it, and I say, OK, I've got 6. If I add 6 to itself, I have 12. If I add 6 to that, I get 18. And I can say, OK, I have to stop because 18 is bigger than 17.
- 5.48** So really, I say, OK, well, 6 goes into 17 twice. It goes in twice without going over. And then what do I do? I say, OK, let's take this. Let's multiply it by 6. That gives me 12. I'll put the 12 here, and then I'll subtract that away from 17. So I get 17 minus 12 is 5. OK, that's great. Now I say, does 6 go into 5? And the answer is, well, no, because if it did go into 5, then 2 was the wrong answer in the first place. Like, if this had been like an 8, then that means 6 goes into 8, which means that 6 could have gone into 17 one more time.
- 5.49** OK, so with ordinary arithmetic, this stops here. OK, this stops here because 6 can't go into this any further. So this thing up here is what's called your quotient. This thing is called your remainder. Remainder. And what does that mean? That means that 17 over 6 is equal to what? Your quotient, 2, plus your remainder divided by your divisor. This thing is called your divisor. That's just the language for it.
- 5.50** So if I wanted to take the number 17 divided by 6 and break it down into what is its quotient and what is its remainder, I can do that. OK, I can do that. Now, why is that useful? Maybe it's not, right? Maybe it's not. Well, it turns out it's fairly useful because what this means is, look, number one, we can check that this is actually correct simply just by adding the two fractions together. We can check.

- 5.51** OK, so just as a quick sanity check, how do you add fractions? You have 2. You think of that as 2 over 1, plus 5 over 6. How do you add fractions? Well, you need the denominators to be the same, right? This is the whole common denominator thing. You need to have a denominator that's the same. So how do I make 1 equal to 6 without changing the problem? This is like one of the most useful things in math, even though it seems like one of the most ridiculous things in math, right?
- 5.52** Is that, somehow multiplying by 1, which typically doesn't do anything, right? You multiply a quantity by 1, it doesn't change. 5 times 1 is 5, 10 times 1 is 10, 50 times 1 is 50. Multiplying by 1 doesn't do anything. But in some sense it does everything because then it would allow me to add fractions. So how do I get this to actually work? I'm going to multiply this by 1 in a very clever way, so 6 divided by 6.
- 5.53** How do the fractions combine? Well, I have to now multiply the numerators and denominators. So what does this give me? 6 times 2 is 12, that's 12 over 6 plus 5 over 6. Now the denominators are the same. So how do I add fractions? Well, the denominator stays the same and 12 plus 5 is 17. And I see, oh indeed, I got the right answer. So I know this process of long division will give me what I want.
- 5.54** All right, why is it relevant? OK, why is it relevant? Well, what happens now, if I just had multiplied this entire thing through by 6? OK, I'm trying to illustrate a principle that you typically don't run into until you get into a branch of mathematics that's called number theory. OK? Which is my, which technically is my specialty. All right? But in number theory, what we do is we kind of like bind your hands. You're not allowed to deal with the real numbers anymore. You're not allowed to have fractions, for example. You're not allowed to have square roots of numbers, for example. You're not allowed to have that anymore.
- 5.55** I'm only going to give you integers. OK? So negative 2, negative 1, 0, 1, 2, 3, the kind of the counting numbers, to some extent, with their negatives and zero. OK, I'm not going to allow you to have the number pi. I'm not going to allow you to have the number 7 over 52. I'm going to say you're only allowed integers. So, for example, in a number theoretic context, 5, 6 and 17, 6, they don't exist. They're not allowed.
- 5.56** So what I could do is, I could multiply both sides by 6 to kind of "clear" my denominators away. If I multiply this by 6, 17 over 6 times 6, the 6's cancel and I get 17, 6 times the right-hand side will be 2 times 6 plus 5. Let me put that in parentheses. Now, those are integers, and I'm happy. OK, I'm happy. All right? But it comes from long division. Now, why is this useful? This is what's called the division lemma. OK? Lemma is just a fancy word for something that's not quite considered a theorem. So it's a statement. All right?
- 5.57** But this is what's called the division lemma in number theory. And what's interesting is, Euclid did this back in his Elements when he wrote this, which I think was like 300 BC, if memory serves, somewhere in there. This led to the notion of coming up with the

greatest common divisor of two integers. So if I give you two integers, I can say, like, what are the biggest divisors that go into them, that divide into them?

- 5.58** OK, so, for example, if I give you the number, let's say 22, OK? If I give you the number 22, and I give you the number 4, and I say, what is the greatest common divisor? I could say, well, what divides them? Well, what divides 4? One divides it, 2 divides it, 4 divides it. Those are the only things that I have. And what divides 22? Well, 1 divides it, 2 divides it, 4 doesn't, 11 does, 22 does. And if I look at those sets of common divisors, which one's the largest? I would say 2. So the greatest common divisor is 2.
- 5.59** What this technique does is, it allows you to find the greatest common divisor of those two integers. And by finding the greatest common divisor, you can do a lot of different things, number theoretically. OK, so what's kind of funny about this whole thing is this, like, annoying kind of technique that you learn early on. And, like, how do you do arithmetic with numbers, which is kind of the bane of, like, any student's existence, right? Is actually something that is profound. OK? And I mean profound in a very specific way. OK?
- 5.60** So that's how you do arithmetic and long division with numbers. How do we do that with polynomials? Well, what we want to do is take the principles from arithmetic of numbers and then just abstract that to polynomials. OK, so let's see. So I'm going to keep this example over here, all right? I'm going to keep this example over here. And now we're going to do an example with polynomials. OK? All right, here's the example.
- 5.61** And this is a very important principle, just mathematically in general, is, you don't want to learn mathematics in isolation. OK? So one of my advisors, he was absolutely brilliant, and he, he taught me this principle where I would go in and I would ask him a question, and I'd say, "Look, I'm struggling to understand this particular concept." And it was just one concept that I had a difficult time with. And I'd go in and I'd say, "Can you please explain why this is the way it is?"
- 5.62** And after 45 minutes and an entire board full of just all sorts of computations and all this kind of stuff, he'd look at me and he's like, "Does it make sense?" And I look at him and I'd be like, "Well, it makes sense, but how in the world did you get there?" Like, you're connecting this to that and that to this and this to that, and like there's 20 different things on the board that's connected to different disciplines and all of this. I was like, "How in the world did you do that?"
- 5.63** And he was like, "Because you don't learn mathematics in isolation." You don't think of algebra as living in this little realm over here, and then geometry lives in this little realm over here, and then maybe pre-calculus is here, and linear algebra is here. They're all interconnected in a very beautiful way. And the more you can connect things, the easier it will stick in your mind, and you'll be able to pull from other disciplines to be able to understand new principles.

- 5.64 So that's what we're trying to get here, is I'm going to take this relatively basic principle of long division of integers, and I'm going to apply it to polynomial long division, even though the two seem like they're completely different. OK, one's red, one's blue. They look completely different, but they're actually almost identical to each other. So here's the example.
- 5.65 Let's say, find—I'm going to take $x^2 + 6x + 7$, and divide it by $x + 2$. So very similar to this question, I want to find 17 divided by 6. The objects that I'm looking at there are integers, 17 and 6. I'm then going to ask myself the same question here. The objects are now polynomials. I have a quadratic, degree 2, divided by degree 1. And I want to say, does this even make sense? Does this even make sense?
- 5.66 Well, the notation I used over here was this kind of like goofy little half—I mean, I don't even know what that symbol is called, to be brutally honest. It's just whatever this little thing is that we indicate with division. OK, so maybe I can do the same thing here. Here would be my divisor, $x + 2$. And then I'm going to draw that goofy symbol, and then I get $x^2 + 6x + 7$. OK, all right.
- 5.67 Well, OK, what did I do on this side? On this side, I said, how many times does 6 go into 17? And intuitively that kind of made sense. You're like, OK, 6, add 6 to itself, 12, add 6 to that, 18. Too big. Go back one. 12 goes in twice. So what the heck do I mean by this? Well, you have to reinterpret by what you mean, 6 goes into 17. What you're really asking yourself is, can you find a number that multiplies 6 into something that doesn't go above 17? That's really what you're asking, right? Can you find a number such that when you multiply by 6, you don't go above 17?
- 5.68 And the answer is, well, 1 will do it, but so will 2, and 2 is the maximum that'll do it. So I'm going to take 2. So you're going to ask yourself the same question here. But what you do is you identify the leading terms. You identify the leading terms and you ask yourself the questions. Number one, what is the leading term of the quadratic, $x^2 + 6x + 7$? It's x^2 . What is the leading term $x + 2$? It's x . You ask yourself the question, can you multiply x by something, a polynomial, to get x^2 ?
- 5.69 What would I need to multiply x by to get x^2 ? Well, in order to get x^2 from x , I need to multiply by x . That's your analog of 2, in the division case; that's your analog of 2. OK, now what do I do with that quantity? I take this x . Then what did I do with this 2? I took the 2, multiplied by 6, that gave me 12, I put it here and then I subtracted. So what am I going to do here? I'm going to take this x , multiply it by $x + 2$, put it here and subtract.
- 5.70 OK, so if I take x , multiply it by this, what do I get? I get an x^2 . OK, so what are we doing? x times $x + 2$, that's polynomial multiplication. What does this give me? $x^2 + 2x$. Great. So I get $x^2 + 2x$ and I subtract. How do I subtract polynomials? Well, I combine like terms. So I'm going to take x^2 minus x^2 . What happens? Those disappear. They have to disappear, because if they

don't, then the thing that you found up here wasn't good enough. That's the whole point of choosing this particular object.

- 5.71** This object was chosen so that when I multiply it, I actually wipe out that x squared so I can decrease the degree. You're going to go term by term and eliminate the x squareds, eliminate the x 's, eliminate the constants if you can. That's what you're going to do. What do I get here? $6x$ minus $2x$, that's $4x$. And then 7 minus, this is kind of like a zero if you want, just a placeholder. 7 minus zero is just 7 . So you bring the 7 down, OK, you bring the 7 down.
- 5.72** And now you repeat the same process, but you repeat the process with your divisor, x plus 2 , and the polynomial, $4x$ plus 7 . So now you say, OK, what is the leading term of $4x$ plus 7 ? That's my $4x$. What is the leading term of your divisor? Well, that hasn't changed. It's still x . And you ask yourself the following question. What do I need to multiply x by to get $4x$? Well, 4 . OK, so that's my next thing. Plus four. That's what I put up there.
- 5.73** I then rerun the same thing. Take this 4 , multiply it through. If I multiply it through, what do I get? 4 times x plus 2 . Distributive property tells me that's $4x$ plus 8 . Remember, the 4 has to come into both. I then write the $4x$ plus 8 here and I subtract. You're just repeating the same process that you did in the first step. And you go until you can't go anymore. So $4x$ minus $4x$ is 0 , 7 minus 8 is negative 1 , and now we're done. And now we're done.
- 5.74** And you may be like, well, OK, wait a second. What do you mean, we're done? Right. Well, when I got here, why was I done with 5 ? Why was I done with 5 ? Well, if I come back to this one and I ask myself the question, can you find a number that multiplies 6 into something that's not quite bigger than 17 , you could say, can you find a number that multiplies 6 into 5 ? And you sit there and you're like, well, I can, but it has to be a fraction, right? Can you find a number that multiplies 6 into 5 ? Sure. Six times 5 over 6 .
- 5.75** The 6 's cancel, and I get 5 . Great. But is 5 over 6 an integer? No, it's a fraction. So am I allowed to say, like, there's a 5 over 6 hanging out over here? And the answer is no, you're not. Similarly, over here, you're asking the question now, can you find something that multiplies x into negative 1 ? And the answer is sure. Just take x times minus 1 over x . The x 's would cancel, and I would get negative 1 . Great.
- 5.76** But the question is, is negative 1 over x , is it a polynomial? And the answer is no. Polynomials only have powers of x 's. X squared, x to the first, x to the tenth. It doesn't have powers, x to the negative 1 , x to the negative 2 , 1 over x 's, things like that. It doesn't have that. So when the degree of this thing here, it's a constant, so its degree is 0 — when that degree becomes less than this degree, you're done. You're done. You cannot do long division any further.
- 5.77** So this thing then becomes your remainder. That's your remainder. This is your quotient, and this is your divisor, exactly like we had over here. OK, so what does that

mean at the end of the day? Well, what did it mean over here? It meant that 17 divided by 6 was equal to my quotient, plus my remainder, divided by my divisor. That's what it means. So what would that mean over here? That means that the thing that I was interested in, $x^2 + 6x + 7$ divided by $x + 2$, should be equal to—this should be equal to my quotient, $x + 4$.

- 5.78** I'm going to have to erase this. The remainder is negative 1, and the divisor was $x + 2$. Plus my remainder over my divisor, just like I had with the arithmetic of the integers. OK, now, because this is a negative here, what I'm really going to do is, I'm going to write this as $x + 4 - 1$ over $x + 2$. That's what I get. OK, but how do I know if that's right? What did we do over here? We said, well, if I have this expression, then let's just combine the fractions and check to see if I actually got the right answer. OK, well, let's do the same thing over here.
- 5.79** How would you combine these as fractions? Well, think of the $x + 4$ as like $x + 4$ divided by 1. In order to get a common denominator, I need these two guys to be the same. So, well, 1 and $x + 2$, how do I make them the same? I need $x + 2$ in the denominator. So you multiply by 1 in a clever way, just like I did over here. I multiplied by 1 in a clever way, 6 divided by 6 . Exact same principles apply. I now have the same denominator.
- 5.80** So what do I get? I get $x + 4$ times $x + 2$ minus 1 in the numerator, divided by $x + 2$. And now I do the distributive law for multiplication of polynomials, OK? So what you're kind of learning with all of this is that all of these arithmetic principles that we have, addition, subtraction, multiplication and division, they're all intricately interwoven when doing all of these different things. In order to do division, I needed to understand degree. In order to do degrees, I needed to understand multiplication of polynomials. In order to understand that, I need to be able to do— how do I do the distributive law here?
- 5.81** How do I combine like terms? How do I do all of that? Everything is intricately connected. So what do I get? How do I do this again? Remember, x times x , that's my x^2 . x times positive 2 is a plus $2x$, 4 times x plus $4x$, 4 times 2, that's an 8, minus 1, because that's the backend term. All over $x + 2$. Combine like terms. What do we get? We're going to have x^2 . That's this term. I see a $2x$ plus a $4x$. So I get a plus $6x$. That's this term and this term. I see an 8 minus a 1, that's a plus 7 divided by $x + 2$. And that's exactly what we had up here. OK, that's exactly what we had up here. All right. OK.
- 5.82** The arithmetic of polynomials. Now, why in the world is this useful? Why is this useful? Remember, I gave you this example on this side of the board, where I gave you a degree 1 polynomial. I think it was like $2x + 3$. And I said, multiply that by a degree 2 polynomial, a quadratic. It was like $4x^2$ plus something. I forget exactly what it was. And we factored it out and we saw we got this term, this thing with four terms, that was a cubic, degree 3.

- 5.83** And I said, well, that's great. We went from this side, as it was already factored into its individual components. We expanded it to get its expanded form. And I said, what if I wanted to ask the reverse question? Here's the expanded form. I want to know, how did it factor? Where did it come from? What were the multiplication factors of it? OK. It's the same exact question arithmetically, if I ask you the following question, which is a very deep question and a very hard question, even though it seems very simple to articulate.
- 5.84** It turns out that this is one of the most important problems in the world today, surprisingly, is how do I factor things? How do I factor things? OK, if I gave you the number 100. OK. OK, great. The number 100 has meaning in a lot of different contexts. OK. It's a number, great. It's positive, great. If it's money amount, great, all the better. It has a lot of different meanings. But I say, OK, what is it made up of? What is it made up of? Well, I could say, OK, well, 100 is a 50. Well, let's do. This is a 2 and a 50, right? And then 50 is what, a 2 and a 25, and 25 is a 5 and a 5.
- 5.85** And I say, can I go any further? Can I factor it any further? And the answer is, well, in the integers, no, I can't. So what is this? This is 2 squared times 5 squared. OK, great. And that's all I can kind of do. All right? If you want to go even further, you can say, this is 2 times 2 times 5 times 5. And you can say, well, you've kind of broken up 100 into its individual constituent parts. Two 2's and two 5's, all multiplied together. OK, well, that seems simple enough. That seems simple enough.
- 5.86** OK, I give you a number factored into pieces. It's like, all right, it turns out that that's insanely hard. And that's kind of the premise of what's called RSA in terms of RSA encryption and how different things are encrypted. Using RSA security is basically asking, how do I take large numbers—and by large, I mean insanely large. I mean, you're talking trillions of digits— OK, how could I take a number like that and break it up like this? Is that even possible? And the answer is, well, not necessarily.
- 5.87** If you have an infinite amount of compute power, if you have an infinite amount of time, then it's like, sure, I can figure it out. But if the answer is, I just need to make sure that it's hard enough so that I can use that corresponding encryption technique to encrypt something so I can actually do some sort of transaction, where if somebody's trying to break that encryption, I can do the transaction before they can break the, before they can break it. That way, the transaction is secure. That's kind of the premise behind RSA security, OK? RSA encryption.
- 5.88** It's kind of the, this is, this is like, really bleak, OK? This is really bleak, but it's kind of like you're hiking, OK? You're hiking. The concept of security, right, at the end of the day, the concept of security is like, you can't guarantee that something is 100% secure. You can't, right? Whatever your password is, whatever your encryption is, whatever the case may be, you can't say, oh, my password is 125 characters long, therefore, I am guaranteed that I'm secure. It's like, no, you're not.

- 5.89** Somebody could literally randomly guess those 125 characters correctly and happen to break your password, right? You cannot say with absolute certainty that somebody cannot just pull that out of thin air and do it, right? That's not the case. OK? So the concept of security is kind of like, you're as secure as you can delay the ability of somebody who's trying to break that security, so that you can do the transaction before they break the security. Does that make sense? Right?
- 5.90** It's kind of the idea of, and this is, like, kind of dark, but it's kind of the idea that you are, you're walking with your friends and you're hiking with your friends up in the mountains, right? And you roll up on a grizzly bear, right? It's like, you're safe from the grizzly bear so long as you are the fat—you are not the slowest runner, right, to some extent. That's really grim, OK? That's really grim. But to some extent, like, you are safe from the bear as long as you're not the slowest runner, right?
- 5.91** OK. That's the same idea with security, OK? You're safe from doing a transaction so long as whatever encryption you're using takes long enough for an attacker to actually break that code, OK? And it comes from this. It comes from taking numbers and breaking them down into what's called their prime factorization. So for polynomials, we want to ask the same type of question. Can I factor polynomials? OK. Can polynomials factor? OK.
- 5.92** All I mean by that is, can I take a polynomial and break it up, just like I take a number and I break it up? Well, how do I break up numbers? I divide. How did I say that 10 was 2 and 50? I said, well, 2 divides 100. How did I say that 50 was 2 and 25? I said, well, 2 divides 50. How did I say 25? Well, 5 divides 25. And then eventually I get to a point where I say, well, nothing else divides 5, aside from 1 and 5, because 5 is what's called a prime number. And so then I'm done.
- 5.93** So I can ask the same questions with polynomials. So let's say, for example, how would I take the polynomial $2x$ plus 6? Can I break it down any further? Can I break it down any further? Can I say something that's somehow common to everything and break that down? Well, what I notice is that I have a 2, and I have a 6. And I can say, do 2 and 6 share any common things? I can say, well, they're both divisible by 2. They're both divisible by 2.
- 5.94** So what I could do is, I could take a 2 out and say, well, what's going to be left on the inside? In order to get a $2x$, I need to have an x here from the distributive property, right? And then in order to get a 6, what do I need? I need a 3. OK, can I do something else? Like, what if I had something like, so here's example 1. Example 2, what if I had something that was like $3x$ squared plus $6x$? I can say, well, can I take out terms? Can I take out terms?
- 5.95** And I can look at this. And I can say, well, what do they have in common? What do they have in common? I can say, well, this one has a 3 and an x squared. This one has a 6 and an x . So they both have an x . So I keep that in mind. And then both of the

coefficients, 3 and 6, they have a 3 in common. So what can I take out? I can take out a 3 and an x . And what's left on the inside? Well, in order to get a $3x$ squared, this has to be an x by the distributive property.

- 5.96** And in order to get a $6x$, this has to be a 2 again by the distributive property. And then can I factor anything else out? OK, so another question we want to ask is, well, that's all fine and good, but those ones are relatively straightforward in the sense that I can just look at this term, look at this term, and say, what do they have in common, if anything? Maybe they don't have anything in common, you know. What if I gave you something that was like $2x$ plus 5 and I said, "OK, factor it."
- 5.97** You'd be like, "Well, I can't, because 2 and 5 don't share anything other than 1. And the $2x$ has an x and the 5 doesn't have an x . So there's really nothing in common between them." OK, now I can counter this by saying factor out something. And this is a technique that I wish was emphasized more because it's incredibly useful in mathematics. But on the surface of it, if you asked somebody, right, if you just ask somebody randomly on the street, you never do this, right? Like, you never get actual questions on the street about mathematics. Like, no one ever does this.
- 5.98** But hypothetically, let's suppose you go up to somebody on the street and you say, "Hey, I want you to factor out 7 from this number." And they look at it and they're like, "You're out of your mind." Like, number one, you're asking me a question on the street about math, which is insane. But number 2, $2x$ plus 5, neither one of those have a 7. I claim that you can still do it. How would you do it? OK, factor out a 7. What do I need on the inside?
- 5.99** Well, in order to get a $2x$, when I multiply by 7, I just give myself a $2x$, but when I multiply by 7, I have to kill the 7. So just divide by 7. If I have a 5, how do I get the 5? Well, I want 5, but I have to multiply by 7 by the distributive law. So just divide by 7. You can factor anything out of anything. It just has to be in the right context. OK, it doesn't make any sense to say, factor 7 out of this, in terms of, find out what they have in common.
- 5.100** But from a simple, basic principles, mathematically, you can say factor out the 7 because I can simply just divide by 7. OK, so that's kind of superficial. It's useful when you take things like calculus and optimization and things like that. But let's suppose I give you this following, the following question. What if I give you something that it seems, like, none of the terms seem to have anything to do with each other. I can't factor anything out, but it actually does factor.
- 5.101** OK, what if I gave you something like this? x^2 plus $6x$ plus 8. OK, I can stare at that for a while and I can say, OK, wait a second. So, x^2 only has two x 's and a 1 for its coefficient. OK, $6x$ has a 6 and an x . OK, so those two things together share an x . But then the 8 doesn't have an x . So it's not like I can factor an x out because they have some sort of commonality there. So, OK, it doesn't seem like I can factor those out. But

then I look at the coefficients and I say the coefficient is 1 here. The coefficient is 6, this one is 8.

- 5.102** Well, those don't share any commonalities either. So I would kind of step back and be like, "I can't factor it," when in fact you can. The way that this factors is x plus 2 times x plus 4. And we can check that, just using the distributive property. Distributive property says what? Take this x , multiply it into x . I get an x squared. Take this x , multiply it into 4, I get $4x$. Take this 2, multiply by x , I get $2x$. Take this 2, multiply by that, I get an 8, and I get x squared plus $6x$ plus 8, which is exactly what I wanted.
- 5.103** The issue, though, is that that's not obvious. It's not staring you in the face kind of like this one was, where it's like, look, they both have a 3 in common. They both have an x . Factor out $3x$. This is saying, these don't look like they have anything in common. How in the world would I ever come up with those? And that's something that we're going to explore when we talk about the quadratic formula and completing the square.
- 5.104** Is there any way to prove that x to the power of 0 is 1?
- 5.105** Is there any way to prove that x to the power of 0 is equal to 1? It's typically taken as a definition that when you raise something to the 0 power, you get 1. You can, you can accept it as a definition. That's completely fine. But ultimately you're trying to say, look, how do we make sense of exponents? Well, what am I allowed to raise things to? Well, a natural question would be, OK, if I take x and I raise it to a number, right, if I take x and I raise it to something, the question is, well, what is m allowed to be?
- 5.106** Is m allowed to be integers like negative 1, negative 2, 0, 1, 2, 3, so forth? Is it allowed to be fractions, things like that? Is it allowed to be, like, what is it actually allowed to be? And it turns out, you can take this to be anything, provided you know what x is. OK? So we saw, for example, that, like, 0 to the 0 is indeterminate. It doesn't make any sense, right? So definitionally, we typically just take x to the 0 to equal 1, unless x is equal to 0, in which case it doesn't make any sense. OK?
- 5.107** It's really just a convention that we make for taking an element, raising it to the zero, equaling 1 in that case. But we have to be, we constantly have to be cautious as to, like, what is x ? So, for example, when we talked about, like, x to the one-half, we said, that's the square root of x , right? That's the square root of x . Now, what makes sense to be in the square root of x ? You're allowed to plug in square root of 0, that's 0. Square root of 1 is 1. Square root of 4 is 2.
- 5.108** Square root of 2 is like 1.4 and change, right? What's the square root of negative 3? It's like, well, does that even make sense? What's the square root of anything along those lines? So the way that you make sense of this object is you say, what is—so, for example, what is the square root of 5? What is the square root of 5? Well, I could go to a computer and I could say, what's the square root of 5? Well, it's in between 2 and 3, because 2 squared is 4, 3 squared is 9.

- 5.109** Five lives in between 4 and 9. So it's somewhere in between 2 and 3. How do you really approach this? You say, OK, let's set that equal to x and then let's say, square both sides. If I square both sides, I get x squared is equal to 5, and I can say, OK, the square root of 5 is a solution to this equation. It's a solution to this equation. And then I can go down and say, how do I find a solution to this equation? I could graph it, I could quadratic formula, I could do all sorts of different things associated to it. So you can ask the same principle in general.
- 5.110** If I say, x to the 0, how do I decide what is x to the 0? Well, I could say, well, there's a corresponding equation which says that x to the 0 is equal to some quantity, that satisfies a certain equation, and 1 actually does it. One is the number that satisfies that. So that's what you define x to the 0 to be. So does that answer that question? Cool.

Lecture 6: Quadratics

At a glance — Quadratics.

- 6.1** Now that we have a new class of mathematical objects to consider, the polynomials, we want to try and mimic what we did when we first started talking about linear expressions. We started with an expression of the form, say, ax plus b , and we asked ourselves the questions, do I have arithmetic operations associated to those? Can I add them? How would I combine like terms? Things like that.
- 6.2** And then we started asking ourselves, well, what if I turn this into an equation? What if I said, OK, this is equal to zero; can I somehow "solve" for x ? Is there an assignment of the variable x , which actually makes this equality true? And we spent some time trying to figure that out, right? So for example, we have $2x$ plus 3. We said, how would I determine what the solution is? And we said, all right, well, if I subtract 3 on both sides, I end up getting $2x$ is equal to minus 3. And then I said, OK, I can divide by 2, divide by 2, 2 is canceled, and I get x is equal to minus three-halves.
- 6.3** And then just as a sanity check, you should always go back and take this and plug it in and see if you actually get the correct answer, and you do. OK, great. So in some way, we've entirely classified solutions to this type of equation. They come in two flavors. a and b are just arbitrary numbers, and so, at the end of the day, well, a is either zero or it's not. If a is equal to zero, then this term is not here, which means b has to be zero. If b is non-zero, there's no solution to this whatsoever.
- 6.4** There's no solution to $0x$ plus 2 equals 0. That's never true. There's no assignment of x that I can think of which will actually make that true. So really, we have, well, for this type of equation, we have the solution type categorized by the following thing. If a is zero, a equals zero, well, then b equals zero. That's the only thing that can happen. But I'm not really saying a solution in that case, because a solution means a variable assignment for x . So I'm going to get rid of this case when a can be zero, because that's not saying anything about when x is equal to something, I get a solution to this. That's not changing the variable to actually get an answer.
- 6.5** But when a is non-zero, if a is non-zero, then what could I do? Well, I get ax is equal to minus b , and so x is minus b over a . That would be my corresponding solution type. And looking at that with this example, that's exactly what I got. In this case, my a is 2 and b is 3. And I saw that my solution was negative b divided by a . That's exactly what we got. So that's all fine and good. But now let's put it in the context of polynomials. Let's put it in the context of polynomials.
- 6.6** Polynomials, this has degree 1. So degree equal to 1, polynomials, look like ax plus b , where a is non-zero. OK, again, a is non-zero. If a were zero, there's no power of x . That wouldn't be a degree 1 thing, that would be a degree 0 thing. And we're interested in classifying solutions to polynomials. So we're going to assume that a is non-zero. And

we just saw that the solutions to $ax + b = 0$ are of the form $x = -b/a$. So we've classified them perfectly.

- 6.7** We said in any degree 1 polynomial whatsoever, when I set that degree 1 polynomial equal to zero, I can guarantee you that's what its solution is going to be. OK, great. Let's move on to the next degree. We've solved degree 1. Let's look at degree 2. Degree 3. What do degree 2 polynomials look like? Well, these are the category of quadratics, quadratic polynomials. So quadratic polynomials look like $ax^2 + bx + c$, where, again, a is non-zero because if a were zero, then there's no x^2 , and it's not degree 2 anymore. It's degree 1 or less, depending on what b is. OK.
- 6.8** So then we say, all right, well, how in the world would I solve solutions to $ax^2 + bx + c = 0$? How would I approach that? So we're trying to follow the same principles that we did for degree 1 polynomials. We're just going to look at it now as one degree higher. And maybe I can solve it. Maybe I can classify them. Maybe I can actually say, oh, x equals something concrete. And I can say this is what all of the solutions are going to look like. Or maybe I can't. Maybe I can't do anything.
- 6.9** And this is really what mathematics is about. Oftentimes I get asked the question, how do mathematicians come up with the next math? How do they come up with the next theory and all of that? And the answer is, well, they experiment. And that's not a really good answer, but honestly, it's the best answer that I have for that question. There's this misconception that mathematicians just sit in their office with a pad and paper, and they're just sitting there thinking, and they're like, "Oh, and here's this beautiful new theory." And it's like, that's not realistic.
- 6.10** Maybe some branches of mathematics, there's aspects of it which are that way. But ultimately, mathematics is done much like science is done. Science is done by what? This is what we know. This is what has been experimentally verified. I have a question, and I want to see if I can somehow expand our knowledge that we currently know by answering that question. How am I going to do that? Well, I'm going to look at simple examples. I'm going to look at, are other people working on this? Has there been any other progress made on this? Things those lines.
- 6.11** And then I'm going to see if I can somehow expand our current knowledge by even the slightest amount. It's very rare that you're going to actually sit down and be like, oh, here's our current knowledge. I have a question which is going to create this entirely new discipline which nobody has ever thought about ever, in the history of humankind. And it's going to revolutionize everything. It's like, that's fantasy land. Mostly, science is incremental increases in knowledge until you can put the entire thing in a large whole. And that's similar to how mathematics is done.
- 6.12** So how can I approach this question? Well, if I go along those lines and have that mantra, I can say, well, OK, let's look at some small test case examples. Let's look at some small test case examples. So I have three parameters here. I have a , b , and c . I

know a is non-zero, but b and c can be whatever the heck I want them to be. They can be anything, any number whatsoever. So why don't I make my life somewhat easier and just say, well, let's just pretend b is zero for a minute. Let's pretend b is zero. Can I say something?

- 6.13** OK, so let's start off with test examples. And this is really how I wish, what I want to convey to you in terms of how to approach mathematics, in terms of how to approach mathematics. Don't be afraid to experiment with small test examples. You don't need to know the finish line from the start line when you start a problem. You're running a marathon, you don't need to know what mile 26.2 looks like. What you need to know is, what does mile 1 look like? What does mile 2 look like? What does mile 7 look like? Because those are your incremental steps to get to your ultimate goal of what does 26.2 look like?
- 6.14** You don't necessarily need to know the full classification of what the solutions are before you start. You can start with little test examples and gain some knowledge along the way. OK, so let's start with some test examples. I'm going to make the assumption, we're going to assume that a is 1 and b is zero. a is 1 and b is zero. OK, so if that's the case, what do my quadratics look like? If I go in and I replace the a is 1, b is zero, I am now looking at quadratic equations that look like that.
- 6.15** Much simpler than just arbitrary quadratics, where this could be like $120x^2$ plus $742x$ minus 3,822. This is much simpler than that. And maybe I can glean something from this, or maybe I can find a pattern that says something to the effect of like, "Oh, Robbie, if you do this, that's a generic principle that you can then abstract to the general case." And then I have a shot of being able to understand it from a more general framework. Does that make sense? OK.
- 6.16** So let's start playing around with values of c . OK, well, let's say c is equal to just say negative 1, for example. If c is equal to negative 1, I'm looking at x^2 minus 1 is equal to zero. Now, it's very tempting to do the following, but I would caution you against doing this. I would caution you against doing this because you could fall into a very poorly understood and poorly explained oversight regarding mathematical principles.
- 6.17** It's very tempting to say, OK, just like we did up here, how did I solve for x ? I tried to isolate x , get x on one side of the equation and get everything else on the other side. And whatever's on the other side, that's my solution. x is equal to that. OK, that's tempting. So let's try that. If I just add 1 to both sides, add 1, I just get x^2 is equal to 1. OK. It's very natural to say, well, can I somehow get rid of the root here or this exponent here, this 2.
- 6.18** And when we talked about exponents, we said we had a rule that did this, right? When I have x to the m power and I raise it to the n power, that's equal to x to the m times n . OK? So I can ask myself the question, can I somehow raise this to some power,

whatever that is, and then do the same thing on this side? Maybe that'll work. OK? Is there something that I can replace this question mark with, such that using this rule, when I multiply it in by 2, the 2 goes to a 1? The answer is, well, divide by 2, to some extent, right? Two divided by 2 is 1.

- 6.19** But remember, multiplication and division are basically one and the same. And so rather than divide by 2, I'm going to multiply by one-half. OK, so I'm going to replace this with one-half. What am I effectively doing? I'm saying, OK, take the square root of x squared, and that's the square root of 1. It's effectively what I'm doing. The problem with this is that the number one mistake, the number one mistake that is made, is the following. People look at this. So this is wrong, by the way. But in order to show you what's right, I have to show you what's wrong. There has to be a juxtaposition of right and wrong.
- 6.20** If you don't know what's wrong, then it's hard to know what's right. Does that make sense? That's just a general principle in life. OK. So what is wrong here? People will typically do the following. Oh, the square root of x squared, those things cancel; therefore, I just get an x . According to the rule, that's what we would have. x to the m to the n is just x to the mn , and so I get rid of that. And then people say, well, the square root of 1, because the elements that satisfy something squared is equal to 1, there are two of them.
- 6.21** And what are they? Well, 1 times 1 is 1. That's true. But also, negative 1 times negative 1 is 1. So here's the mistake that people make. People typically go, oh, the square root of 1, that's just plus or minus 1. And that's false. That's completely false. And I can prove to you that that's false. I can prove to you that that's false. Why is that false? Well, this thing here is really, think about it as a function. Remember, when we talked about functions, we had f of x is the square root of x .
- 6.22** So really, what I'm saying is that the square root of 1 is the value of this function at 1. If I plug in 1, that's what I get. Do we agree? Square root of 1 is the value of that function. But what did we say the property of functions had to be? If you give me a valid input, I am only allowed to give you one output. OK, so the question is, well, which output is it? If you are claiming that the square root of 1 is plus or minus 1, then you are claiming that the function, the square root of x , gives me two outputs. It gives me a 1 and a minus 1, which contradicts the fact that it is a function.
- 6.23** So which is it? The answer is that the square root of 1 is equal to 1. The square root of 1 is equal to 1. So then you step back and you say, OK, well, then, Robbie, then what is this actually saying here? Because if you're telling me that x is equal to just positive 1, then where in the heck is the other 1 coming from? Because I know that negative 1 times negative 1 is also equal to 1. So I know that negative 1 is a solution to this equation. And what I am saying is that there is a big difference between saying, what are solutions to an equation versus what is the value of a function?

- 6.24 Those two things are vastly different from each other. They're interconnected, but they are vastly different from each other. So what I propose is that we think about this differently. We think about this differently. Rather than the same mentality that we had over here, where x was equal through some quantity—we get x on one side, we get everything else on the other side—rather than thinking about it that way, I want to think about it in the following way.
- 6.25 Rather than moving the 1 to that side, can I somehow factor this polynomial? Can I factor it? And why do I care about that? Why do I care about that? OK, here's why. Here's why. If I look at the polynomial x squared minus 1, and I'm trying to factor this thing on the left-hand side, I can look at it and I can say, OK, do they have any common terms? Well, x squared and negative 1, they don't have any common terms. There's no common x 's I can pull out. There's no common numbers that I can pull out, anything like that.
- 6.26 But that doesn't mean it doesn't factor. How can I factor this? Maybe it factors something like this. I've got some polynomial times some polynomial equals zero. Maybe that's the case. And so I try and reverse-engineer it. I say, OK, well, how would I get an x squared? If I had an x here and an x here, when I multiply the two together, I'd get an x squared. OK, so that's a good starting point. But then the question is, what goes here and what goes here? What goes there and there?
- 6.27 And the way that you can think about this question is the following. As long as this coefficient of x squared is a 1, whatever goes in this slot and whatever goes in that slot, they have to have the following properties. When you multiply this thing times this thing, what should it be? It should be that number. Do we agree? When you multiply this box and this box, it should give me that number. That should be my constant because everything else is going to have what?
- 6.28 x times box, that'll have an x . Box times x , that'll have an x . The only thing that's going to give me number times number, which will give me the negative 1, is those two products. I have to look at negative 1 and say, OK, negative 1, how do I break it down? Well, negative 1 looks like what? 1 times minus 1. Those give me candidates for what could go in those particular spots. And let's see, let's just check. OK, so if I did an x plus 1, that's corresponding to this one, times an x minus 1, that's corresponding to this one—does that actually work?
- 6.29 Well, the other property you have to keep in mind is, remember, general quadratics with coefficient 1 look something like, let's use d here. They look something like this. Well, if this thing factors as x plus or minus something times x plus or minus something, then what do I need to know? Those two things multiplied together have to give me that number. And when I add those two numbers together, that has to give me that number. So if I look at this, I say, look, this polynomial, x squared minus 1, really looks like x squared plus $0x$ minus 1.

- 6.30 And I notice that 1 and minus 1, they multiply to give me negative 1, and they add to give me zero. So I know that this is going to work. And if we just do a quick sanity check, x times x is x squared, x times minus x is minus x , 1 times x is plus x , 1 times minus 1 is minus 1, those cancel, and I get what I wanted, x squared minus 1.
- 6.31 Now, why is this a better way to think about this particular problem? Because number one, exponents aren't playing a role at all. I'm not taking a square root on both sides, and so that's causing some issues. All I'm simply saying is now, that is equal to zero. And why is that a beautiful thing? Well, let's think about it. x is a number. Whatever x is, I'm trying to find out what is x , but at the end of the day, x is a number. So if I take the number, say, 5. Just choose a random number, say 5, and I plug it in, I'm going to get 6 here and 4 here. Clearly, that doesn't work because 6 times 4 is equal to 24, not zero.
- 6.32 But the principle I'm trying to get through is, I have a number times another number is equal to zero. When is that possible? When is that possible? If I have the product of two things, two numbers, real numbers, is equal to zero, when can that be true? When can that be true? And the answer is, well, one of them has to be zero. One of them has to be zero, at least in the real numbers that's the case. In general, that's not true. That's not true. There are situations in the world, the real world, not just fantasy math land, but in the real world, where the product of two things equals zero, but neither one of them is zero.
- 6.33 And I'll give you an example of this. However, in the real numbers, what we've been talking about this entire time, if you have the product of two things and that's equal to zero, one of them has to be zero. So we can then say, so either x plus 1 equals 0, or x minus 1 is equal to 0. And now we're back in the degree 1 situation where we know how to classify those perfectly, and we can say so either x equals minus 1 or x is equal to 1. And we get both of our solutions, rather than doing sloppy, lazy mathematics and saying that the square root of a number is plus or minus the number, which is false.
- 6.34 This is the importance of factoring and why factoring matters, is it allows us to solve equations algebraically without having to resort to making false assumptions that square roots somehow give us plus or minus two answers. The square root is always a positive number. The square root of 9 is 3. The square root of 16 is 4. The square root of 25 is 5. Not plus or minus 3, not plus or minus 4, not plus or minus 5. The square root is a positive number. It's what it is.
- 6.35 But coming back into quadratic equations, because it's quadratic and I'm squaring it, I know the negatives would cancel. So I also want to incorporate the negative values. But I can't say that the square root is plus or minus something. That's not true, mathematically. So we could generalize this. C was equal to negative 1, but any negative value of c , we now know what the solutions are going to be. The solutions are going to be a minus the square root of c and a plus the square root of c , will be my corresponding answers. OK?

- 6.36** So let's talk about two techniques for coming up with solutions to these things. OK? Two techniques. Let's go back to the general quadratic formula and come up with a way of doing this. Our general quadratic equation. So we have what's called the quadratic formula and the technique to prove the quadratic formula, which is complete the square. The quadratic formula tells you what the solutions to your general quadratic equation is going to be. So our general quadratic equation looks like $ax^2 + bx + c = 0$, and it tells us precisely exactly what the solutions are going to be.
- 6.37** Now, how do we come up with this? There's this notion of complete the square. I'm going to give this to you in a way which, typically complete the square is presented to you for solving quadratic equations. But I would claim that that's missing the mark, because you don't need an equation to be able to factor something. I don't need an equation to be able to say that $x^2 - 1$ is equal to $(x - 1)(x + 1)$. There's no equation saying $x^2 - 1 = 0$. I'm not trying to say that it equals zero. I'm simply saying I can break this down and factor it into terms.
- 6.38** Now, if I'm interested in when is it equal to zero, then this is relevant. This is super helpful. But if you just give me $x^2 + 7x + 10$, and I don't say that I want to know when is that equal to zero, I just want to know, can you factor it down? Just like the number 100. Can you take the number 100 and break it down into its constituent parts? Well, there's no equation here, but I can still write this as $(x + 5)(x + 2)$, right? Two numbers that multiply together to give me 10, 5 and 2, and they add to give me 7— 5 plus 2 is 7.
- 6.39** So I want to teach you to complete the square without thinking about, set the equation equal to zero. Set the equation equal to zero. Try and solve the equation. OK. So complete the square in general. We have two cases. Case number one, a is equal to 1. This is the "easier" case. Case number two will be when a is not equal to 1. But let's go after case number one. So my quadratic looks like $x^2 + bx + c$. That's what it looks like. All right. How do you complete the square? Here's the method. Here's the method.
- 6.40** Step one. Step number one. Make sure that this coefficient is equal to 1. That's like step zero. Make sure that that coefficient is equal to 1. If this coefficient not equal to 1, make it 1. How do you do that? You factor it out. You factor it out. So once you've done that, step one is to identify this term. Identify that term. So in this case, it's just b , plus b . So step one, you compute b divided by 2. Whatever b is, you just divide it by 2. That's it. That's step one.
- 6.41** Step two, you square step one. Step two, you compute $(b/2)^2$. So that's what? $b^2/4$. b over 2 times b over 2 is $b^2/4$, divided by 2 times 2, that's $b^2/4$. Step three, you do one of the most important tricks in all of mathematics, which seems completely ridiculous, and is one of those things that you would never see. It's like those, when your friend sends you a picture, they send a

picture and they're just like, "Oh, do you see it?" And you look at the picture and you're like, "It's just like a birthday party or something like that."

6.42 And then you look in the background and there's somebody making some crazy face or something like that. And you're like, "Oh, that's really funny." You don't see it until it's pointed out to you. It's one of those things. It's like, oh, you don't see it until you see it. Step three is this tried-and-true technique of mathematics, which you wouldn't think to do until someone tells you to do it. And what you do is, you add zero in a clever way. Add zero. And you're like, what the heck are you talking about? Like, add zero? OK, great.

6.43 Two plus 0, it's still 2. Big deal. But it turns out to be exactly what you need it to be. Now, what is the zero you're going to add? In step two, you computed b^2 over 4. The zero that you add is b^2 over 4 minus b^2 over 4. Clearly zero. Whatever b^2 over 4 is, let's say it's the number 5, 5 minus 5 is 0. That's fine. Why in the heck do you do this? Because it's precisely what you need in order to turn this into a square. It's exactly what you need in order to turn this into a square. OK, so let's look at it. So let's follow the process.

6.44 OK, so step one, I compute b over 2. Check. Step two, I compute b over 2 squared, which is b^2 over 4. OK, check. Step three. OK, so we add zero. OK, now I can actually grab the quadratic that I had and add zero to it. And here's how I'm going to do it. You add zero in the following way, x^2 plus bx plus c . What I'm going to do, c , I'm just going to push to the far right-hand side of this equation. I'm just going to push it to the right. I'm going to add zero right here, b^2 over 4 minus b^2 over 4 plus c . Here's zero. There's zero.

6.45 Why do I do that? Put your parentheses right here. That's a perfect square. What is it? This is now x plus what you got in step one, b over 2. The whole thing squared, and then what's left is out here. Minus b^2 over 4 plus c . Why is that the case? Well, work it out. Work it out and see what happens. OK, what is this term right there? I have x plus b over 2 times x plus b over 2. That's what something squared is. It's the quantity times itself, x plus b over 2 times x plus b over 2.

6.46 If I take x times x , I get x^2 plus x times that, I get a b over $2x$, plus b over $2x$, again, plus b^2 over 4. But what is half of b plus half of b ? That's all of b . So these two things together give me x^2 plus bx plus b^2 over 4. And that's exactly what this quantity is. So now it's a perfect square. And this thing is called the complete the square form. So this is the completed square of the quadratic. That's what it is.

6.47 Now, that's the case when a is equal to 1. That's the case when a is equal to 1. If a is not equal to 1, so let's say there's a 4 here, what would I do? Well, I factor out the 4 first. Now the coefficient is a 1, and I'm back in case one. So how would I do that? How would I do the general case? Here we go. General case. So case two, a is not equal to 1. So I have ax^2 plus bx plus c . The first thing I do is, I just factor out a , just like I

gave you earlier, how can I factor out things? You're like, "Well, Robbie, you can't pull an a out unless b and c have the same factor."

- 6.48** It's like, false. Yes, I can. As long as I don't break math, I can do whatever I want. If you give me something that's, for example, $2x^2 + 3x + 6$, 2, 3, and 6 don't share any common factors other than 1, but I can still factor 2 out. $2x^2 + 3x + 6$ becomes $2(x^2 + \frac{3}{2}x + 3)$. I guess they'll factor it out. That's fine. OK, so factor out a. What am I left with in here? In order to get ax^2 , that has to be x^2 . In order to get bx , well, it has to be a b. But when a comes in to multiply, I've got to kill a. So I've got to divide out by a, so that the a's cancel.
- 6.49** And then in order to get c, when a comes in, well, I have to eliminate a as well. So I just divide by a. But then here's the point. This is case one. That coefficient is now 1, and you can run the exact same procedure. OK, so that's the abstract nonsense. Let's actually get brass tacks. Let's actually do this in practice now. Let's do some examples. It's one thing to—so I'm an avid cyclist. OK, I love cycling. I think it's awesome. Personally, I think it's one of the hardest sports on the planet.
- 6.50** It's one thing to watch the Tour de France and watch all 21 grueling stages of this event. It's another to get on a bike and ride 100 miles yourself. OK, so it's one thing to have a theoretical understanding of how complete the square works. It's another entirely to actually get on the bike and ride. Does that make sense? So let's actually figure this out. OK. So I want you to complete the square with the following. And notice that none of these are going to be involved with solve this equation. Set this quadratic equal to zero and solve it. It's literally just saying, here's a quadratic, factor it. That's it. OK?
- 6.51** OK, number one, $x^2 + 4x + 3$. $x^2 + 4x + 3$. What's my procedure? The procedure is, look at, identify the first coefficient is a 1. OK, good. It's a 1. I don't need to factor anything out first. Step two, identify this coefficient. It's a 4. OK, so I compute 4 divided by 2. That's the b over 2. That's a 2. Great. That's step one. OK, so step one. Step two, square that. That's 4. Step three, add zero in a clever way by adding 4 and subtracting 4. So this becomes $x^2 + 4x + 4 - 4 + 3$. Your parentheses go here. That'll be a perfect square.
- 6.52** What is it going to be? It's going to be $x + \text{whatever you found in step one}$, quantity squared. So this becomes $(x + 2)^2 - 4 + 3$. Be very careful with this, by the way. When you're doing arithmetic like this, it's very tempting to be like, oh, it's 4 plus 3, that's 7. And then the negative comes along. It's like, no, that's not what you're doing. You're doing minus 4 plus 3. The minus is attached to the 4. It's not simply, oh, minus 4 plus 3. That would be this, which it's not. The order of operations matters, and the objects that you're doing this with matters. So be careful with that.
- 6.53** So this is minus 4 plus 3. Minus 4 plus 3 is minus 1. And there he is. There's your completed square. Done. Number two, let's do one that doesn't have a coefficient of 1.

So let's say $4x^2 + 8x + 5$. I don't know, 5. I don't care what the numbers are. The theory holds. The theory holds in general. So it doesn't matter what the numbers are. As long as I follow the procedure, I'll get to where I want to go.

- 6.54** Step zero. That needs to be a 1, to factor it out. Take out the 4. What do I have? I have an x^2 . I need an x^2 here. What do I need in order to, when I distribute the 4 in, what do I need in order to make that an $8x$? Well, I need a $2x$. Plus $2x$. When I take the 4 in to get a 5, well, in order to get a 5, I need a five-fourths. You need a 5 divided by 4. Four doesn't go into 5 evenly, so I just give myself the fraction, 5 divided by 4, 4 multiplied by that gives me 5. I'm happy.
- 6.55** And now I'm back in case one because the coefficient is a 1. OK, so what do I do? I identify this term. It's a plus 2. So I come over here. OK, step one, I compute b over 2. That's a 1. Step two, I square that. So I take 1 squared, which is 1, it doesn't do anything. So then I add zero in a clever way. But how do I add zero? I'm not going to add zero out here because out here, 4 isn't being touched. You just pretend that this 4 doesn't exist. This is what you're completing the square on. So you add 0 in here.
- 6.56** So what do I get? 4 times x^2 plus $2x$ plus one minus 1 plus five-fourths. Does that make sense? You add zero in that way. My parentheses now go there. That's my perfect square. That should be x plus my b over 2. My b over 2 that I got in step one was 1, so this thing is going to look like 4 times x plus 1 squared. OK, negative one plus five-fourths, what is that? Negative 1 plus five-fourths is really negative 4 over 4 plus five-fourths. That's really just a quarter. That's just a quarter, so that's fine. OK. So this gives me plus one-quarter.
- 6.57** Now we have these nested parentheses. How does that work? How does that work? Well, we have distribution. This comes into this term, this comes into that term. What do you end up getting? This is 4 times x plus 1 squared plus, what's 4 times a quarter? Well, 4 quarters is 1. And there's your completed square. It's done. Now, why is this relevant? You're like, "OK, great, Robbie. You're literally just saying the same thing to some extent. You're saying, here's a polynomial. Here's the same polynomial just in a different font."
- 6.58** I'm giving you this polynomial, and I'm saying I can write the same polynomial, it just looks different. It just looks different. It's like I'm writing the same essay in one font, and then I copy and paste the essay and put it in a different font and say, "Look, they're different." It's like, yes and no. The font looks different. Their structure looks different. Are they the same thing? Yeah, of course they are. So then the question that's begging to be asked is, why in the heck are you doing this?
- 6.59** And the answer is, when I finally decide I want to know, where do quadratics equal zero? When I want to finally answer that question, complete the square is how you prove the quadratic formula. The quadratic formula is this all-purpose tool that solves every possible quadratic equation on planet Earth. But you need to complete the

square to be able to get there. All right? So let's look at this. I'm going to give you the quadratic formula. I leave it to you to work out the details by completing the square in general, which you have all the tools to be able to do now. OK?

- 6.60** But I'm going to give you the quadratic formula. We'll be able to classify all solutions to quadratics, which is pretty cool. It's also kind of sad because, well, and I'll mention that in a minute, I'll mention why it's sad. Remind me that I need to refer to why I'm sad about this. The quadratic formula says the following thing. To solve $ax^2 + bx + c = 0$, where again, a is non-zero, a is non-zero, because we're thinking of actual quadratics. If a is zero, it's not a quadratic. It's not degree 2.
- 6.61** The quadratic formula says your solutions x will be the following things. And there's all these ridiculous songs that people make to this to actually remember it, and that's completely fine. I think "pop goes the weasel" is a really good one that people use. It's hilarious to watch all these memes of people doing this. I'm not going to do that. All right. So x is equal to, when it looks like this form, you write this as minus b . So you negate this b , whatever it is. If b is 5, this is negative 5. If b is negative 5, this is positive 5. You just negate it.
- 6.62** You then get plus and minus. So you're writing both solutions to this, both solutions. The plus or minus is incorporating both solutions. Plus or minus the square root of $b^2 - 4ac$ all over $2a$. That's what it is. That's what the quadratic formula says. Let's use it. Let's use it. Remember, we had one that said $x^2 - 1 = 0$, and we found that the solution was plus or minus 1. And then I went on this rant about how the square root of 1 is only 1, not plus or minus 1, which is a fact.
- 6.63** But now that I have this tool, I can simply just plug it in. But I have to first identify what is a , what is b , what is c , for this particular thing. So in this case, what is a ? That's the coefficient of x^2 , so a is equal to 1. What is b ? Well, there's no term for x , so b is equal to zero. And what is c ? c is equal to, remember, this is plus c . Since this is a minus, c has to be negative 1. That's what a , b , and c are in this example. So it tells me that my solution should be, x is equal to—OK, let's just follow it through.
- 6.64** Negative b , so negative zero, plus or minus the square root of $b^2 - 4ac$, zero squared, minus 4 times a times c all over 2 times a . That's what it says. OK, let's go through and simplify. So negative zero is just zero, so I'm not going to worry about that. This is plus or minus the square root of zero squared is zero. That's fine. What is negative 4 times 1? That's negative 4. And negative 4 times negative 1 is positive 4, all over 2 times 1 is 2. And so I get plus or minus. Remember, the square root of 4 is 2. So I get plus or minus 2 divided by 2.
- 6.65** So this is 2 over 2, or minus 2 over 2, i.e., 1 or minus 1. Those are my solutions. And that's exactly what we said they were when we factored it. Quadratic formula got them for me. That's fantastic. OK, well, let's do it for one that—the famous one that I mentioned, the Fibonacci one, the golden ratio and all of this. OK, let's look at that.

Let's solve this one. So I just erased the formula, so now you got to go by memory. OK, I'll help you.

- 6.66** $x^2 - x - 1 = 0$. So again, I need to identify a, that's 1. B, that's minus 1, and C is minus 1 as well. Just the coefficients of everything. So a is 1, b is negative 1, c is negative 1. The solutions via the quadratic formula, this gives x is equal to $\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. What is negative b? So this is minus minus 1. Plus or minus the square root of minus 1 squared, that's b squared, minus 4 times a times c. All over 2 times a. Do we agree? Just plugging it in directly, using the formula.
- 6.67** Negative negative 1 is positive 1, plus or minus the square root. OK, this is a 1. Negative 4 times 1 times negative 1, that's the same one we had over there. So that's plus 4 all over 2. So this is 1 plus or minus the square root of 5 over 2. So my solutions are 1 plus root 5 over 2, or 1 minus root 5 over 2. Those are the two solutions that satisfy this guy, whatever those are. This one is the special one, and I'll give it a name. Greek letter phi, P-H-I. This is called the golden ratio. Golden ratio.
- 6.68** Why is it called the golden ratio? Because it's special. It's very special. And there's a lot of weird things about the golden ratio that show up. So for example, if you look at a sunflower, a fully grown sunflower, and you look at how the sunflower seeds arrange themselves, if you look at it, it's kind of trippy, because you stare at it and you have some of them that are going counterclockwise and some of them that are going clockwise and things like that. If you look at the ratio of those spirals, the golden ratio is sitting in that.
- 6.69** If you take an apple and you cut it and you look at the arrangement of the seeds inside of the apple, the golden ratio is there. If you look at Greek architecture and you look at the Parthenon, for example, and you look at these giant columns and you look at the ratios to the distances versus how tall they are, magically, that's there. Why? Who knows? Maybe there's something associated to aesthetics associated to it.
- 6.70** If you look at, what is it? Leonardo da Vinci's, what is it, the Vitruvian Man, I think? Something along those lines. If you look at that, there is the golden spiral sitting inside of that as well. Some say that the golden ratio has something to do with beauty in terms of the aesthetics of things. And that's why our eyes naturally gravitate towards creating things that have this in its corresponding ratio. Ultimately, who knows?
- 6.71** You can even drive yourself crazy and go down to the beach and throw a Frisbee. And depending on how the Frisbee hits wet sand, when it actually hits the ground and carves out a spiral, the spiral that it carves out has the golden ratio sitting inside of it. So I have no idea why this is true. I have no clue why this is true. But at the end of the day, we now have a formula for solving quadratic equations. We now know how to classify solutions to quadratic equations.
- 6.72** Now, one thing we have to be very careful about is, look, we're taking a square root here. So if this square root has a negative number underneath it, then the solutions to

the quadratic are not real numbers. They're what are called complex numbers. Sometimes people call them imaginary numbers. I think that's really unfortunate. But the story behind imaginary numbers is the following.

- 6.73** Back in the day, mathematicians would give each other problem sets, and they would challenge another mathematician to say, "Hey, here's a bunch of problems. Can you solve this? Here's a bunch of problems. Can you solve this?" And there was some "ranking" amongst mathematicians. It's like, "Oh, I solved six out of seven of your problems. You only solved two out of seven of my problems. So somehow I'm better than you as a mathematician."
- 6.74** So in the 16th century, imaginary numbers came to be, and they came from this quadratic formula side of things, where you're looking at taking the square roots of negative numbers, and what does that actually mean? And people were like, well, back then, numbers had to be something that—they existed. They were real. You could draw on the ground. You could get your compass and your straightedge, and you say, look, this is the number, the square root of 2. It's this particular length on a triangle that I can actually draw in the sand. The square root of 2 represents that.
- 6.75** Or the square root of 5 is represented by this particular length. So when somebody came up and said, "Oh, the solution to this quadratic is this number, and it's got this square root of a negative number," they're like, "That's imaginary. You can't do that. That's ridiculous." Hence why people call them imaginary numbers. But they're not imaginary numbers. The way you get an airplane off of the ground is using imaginary numbers, and airplanes are very much real. So you have to be careful with that language.
- 6.76** So they're technically called complex numbers, but people still call them imaginary numbers because you can't draw in the sand what the square root of negative 1 looks like, necessarily. All right. So again, why is this relevant for us? Well, let's do an example from physics. Let's do an example from physics. Let's suppose I want to launch a firework into the air. OK. And my job is quality control to some extent. I want to make sure that, when I shoot a firework up in the air, that it's going to go high enough to where, when it explodes into all of its colors, that it's safely away from any observers on the ground. That seems like a reasonable thing to do.
- 6.77** So I'm going to model this again in a very simplistic way. I'm going to assume that the firework is a point particle, whatever it is. That's not true. When you actually think of fireworks, you actually launch a firework, it has mass. It is an object. But I'm going to assume that it's a point particle. So it's a point particle. I'm going to launch it from the ground. I'm going to launch it from the ground. And I'm going to launch this up at, let's say, I don't know, let's say, 130 feet per second. That's going to be its initial velocity. It's going to shoot straight up off the ground at 130 feet per second.

- 6.78 And then it's going to go up to a certain point, and then it's going to burst at that particular point, and I'm just going to assume that that's double whatever my initial velocity was. So I launch it at 130 feet per second, I'm just going to assume that this is 260 feet. Because for whatever reason, I've deemed that once it's at 260 feet, if it explodes, it's fine. It's safe. The question I want to ask is, when I model this particular behavior, number one, what is the model?
- 6.79 So this is coming from physics in a section called kinematics, which assumes constant acceleration, which is not super realistic, because acceleration, if I launch it with a velocity of 130 feet per second, I have gravity acting on it, which is going to give me some sort of negative force pulling me down. But I'm also going to have air resistance as well. So it's not really a great example for this. But what is the corresponding model that I have for this? The model is the following.
- 6.80 So whatever the height is, whatever the height is, that's given by $16t^2$, sorry, minus $16t^2$, plus my initial velocity. So let me call that $v_{\text{sub zero}} \times t$. This is just my initial velocity. That's what that is. That's what I ended up getting. The acceleration due to gravity in feet per second squared is negative 32. So this coefficient being one-half, the kinematic equation that I'm using is one-half acceleration, t^2 plus initial velocity plus initial height. That's technically what it is. That's neither here nor there. We're just going to say that this is my corresponding equation.
- 6.81 Now, what am I interested in? I'm interested in the firework going off at 260 feet. So that's h , that's the height. So I know that h is equal to 260 feet. That's going to go there. So I look at 260 is equal to minus $16t^2$. My initial velocity is 130, so that's plus $130t$. And what is the question I'm asking? I want to ask the question, at what time is this equation true? The question is, find t such that that equation is true.
- 6.82 Why would that be relevant? Well, think about fireworks displays. The timing at which they go off is kind of important. You don't want to have a situation where you're launching fireworks in a particular way, and as they go off, you're trying to spell out a name in the sky or you're trying to do something along those lines, if that's not timed very well. Timing is key. So you need to know that, OK, based on whatever the fireworks composition is, if it gets to 260 feet, it's going to explode at that point. OK, so how long does it take to go from the ground up to 260 feet, so it can explode, so that I can then time the fireworks display? That's relevant.
- 6.83 But I claim that, well, this is just a quadratic equation. How do I see that? Let's move these guys to the side. Again, I'm biased because I have a negative coefficient here. I don't like negative coefficients, and so I'm going to add the $16t^2$ squared over, instead of just doing the one operation of moving the negative 260 over to the left. So if I move both of those over, what do I get? $16t^2$ squared minus $130t$ plus 260 is equal to zero. OK, that's what I have.

- 6.84** Why is that a quadratic equation? Well, remember, variables are arbitrary. Whether this is a t or an x , it doesn't matter. This looks like $ax^2 + bx + c = 0$. This is a quadratic. So what is a here? A is 16. What is b ? It's minus 130. What is c ? That's 260. So what does a quadratic formula tell me? Quadratic formula tells me what? It says that t , that's my variable. If this were x , I'd say x here, but my variable is t . T is equal to what? Negative b . So I get positive 130, plus or minus the square root of negative b squared, sorry, b squared, minus 4 times a . A is 16, times c , which is 260, all over 2 times a .
- 6.85** Goodness knows what that number is. OK? Go find your favorite free online calculator, whatever it is, plug that in, and you're going to get the values of t for that. OK? And that will give you the time, the times that you need to be considering, in order to say, when does that firework actually explode? Again, this is a very simplistic model with all of this. I'm assuming that it doesn't have any mass. That's an unrealistic thing. If you have a projectile that you're launching into the air, it's very different for me to take this marker and throw it up in the air than it is to grab this podium and launch it up in the air.
- 6.86** There's two very different problems associated to that. So mass matters, but you have to start somewhere when modeling the real world. And a good place to start is to just say, well, suppose it doesn't have mass. Suppose it's just a single particle and you want to launch it up with a certain velocity and a certain height. What can you say? Then once you know that, you can then increase the complexity of the problem by increasing, like, talking about the mass of the object, talking about wind resistance associated to it. Maybe it's a windy day when you're doing your fireworks display. All of these things matter.
- 6.87** Now, let me just briefly go back. What have we been able to accomplish? We've been able to say, when I have degree equal to 1, things that look like $ax + b$, I can completely talk about what their solutions are. We've characterized those. When I have degree equal to 2, quadratics, we now have the quadratic formula which completely classifies those solutions. So a natural question would be, let's just keep going. What if I look at degree 3, so I look at cubics, or a degree 4, quartics, or degree 5, quintics, things like that. Are there "quadratic" formulas?
- 6.88** So a cubic formula, a quartic formula, a quintic formula similar to the quadratic formula? And the answer is yes for cubics and quartics, so degree 3 and degree 4, and it's provably true that there isn't for degree 5 or above, in terms of elementary algebraic operations, square roots and things like that. It's provably true. That's a huge result. Galois was the one who actually proved that result. There's a deep branch of mathematics called Galois theory, which is used to actually prove that particular result.
- 6.89** So why I said that this makes me sad is, it's embarrassing to look at this and say, look, that's degree 1. Its graph is a line. Quadratic, it's a parabola. It looks like this or like this upside down, one of the two. Cubics have a certain shape, quartics have a certain

shape, and things like that. But the second we get to degree five, suddenly everything's out. That's embarrassing. Degree 5 is incredibly small, yet mathematics can't even handle that. There's no formula for it.

6.90 It's, to me, a testament to how difficult and how intricately detailed mathematics is, is that even degree 5 or above, we don't have a shot of being able to do that. But at the very least, we have complete the square and the quadratic formula, which allows us to classify solutions to quadratic equations.

Lecture 7: Exponents & Logarithms

At a glance — Exponents and logarithms.

- 7.1** All right, so we've talked about solving for the solutions to quadratic equations, and we saw that completing the square and the quadratic formula gave us the techniques essentially to solve that. OK, we're going to pivot a little bit, and we're going to bring in a new class of functions. We've kind of mentioned some of this a little bit when we talked about properties of exponents. But we're going to bring in exponential functions, all right? Exponential functions and their corresponding analogs, all right, which are called logarithms, kind of the bane of everybody's existence.
- 7.2** Nobody really likes logarithms. They're just this messy, nasty thing that kind of has this really negative sense to it. And I kind of want to show you that they occur very naturally, OK? And the way I want to show you that they occur naturally is, remember when we were talking about functions, we said a function takes in a valid input, so something from the domain, it inputs it into the function f , whatever it is, and then that function f outputs f of x , whatever the output was. OK?
- 7.3** So we had examples of squaring things. So you give me a 2, I square it, I get a 4. You give me a 5, I square it, I get a 25, things like that. And then we have this question of, like, well, OK, if you give me the number 2 and I square it and I get 4, can you come up with a function that then takes 4 back to 2? You give me the number 3, I square it, I get 9, and I want a function which takes 9 back to 3. That's the notion of the inverse of the function. Can I somehow reverse the corresponding function you gave me?
- 7.4** OK, and we saw that, well, that's tricky and depends on the domain. OK? For the squaring function, the squaring function, f of x is x squared, has the domain everything. You can plug in anything you want, from negative infinity to infinity. But we saw that, well, 2 and negative 2 both went to 4. So if I'm standing at 4 saying, well, where should 4 go? It's like, well, does it go to 2? Does it go to negative 2? The answer is, well, I don't know. So I have to restrict my domain in order to allow for the inverse function to exist. OK?
- 7.5** So the exponential functions and logarithms will be inverses to each other in a very precise way. But how do they actually show up? OK, so there's this very kind of like, I don't know, it's an ancient problem to some extent, which is given in the following way. OK, so here's kind of the motivation for exponential functions. So exponential and logarithmic functions is the following. So let's suppose you have a chessboard or a checkerboard or whatever. OK, you have a chessboard. And so how many squares do you have on a chessboard?
- 7.6** OK, so you have this little square, all right? And correct me if I'm wrong, but I believe there are 8, right? So I have 1, 2, 3, 4. So I need to subdivide each of these into 2, OK? So now I have 8 this way. So 1, 2, 3, 4, 5, 6, 7, 8. Good. And then I have to do the same

the other way. It's an 8-by-8 grid, right? So this is like this. And then I subdivide like this. OK, so let's suppose I have my chessboard and I'm going to—the way the problem is posed is the following, is that I am going to place a grain of rice in the upper left corner of the chessboard and then I'm going to move left to right, OK?

7.7 So I'm going to go here, here, here, here, here, here, and then back down, here, here, here, here, here, until I work my way all the way down to the bottom, OK? With the following property, every time I move to another square, I put on that square double the amount of rice grains that I had in the previous square. OK? That's the game. So it's like, I'm going to have one here, one grain of rice. I put one grain of rice in there. I then double it, I move to the next one. So that's two. I then double that. So I'm going to have four and then eight and then 16 and then 32 and then 64 and then 128 by the time I get to the last row.

7.8 And then I jump down to the next row and I'm going to have 256 and 512 and 1,024 and 2,048 or whatever it is. And that's about as far as my brain goes, OK? And I just keep doubling over and over and over again. And then the question is, how many grains of rice are in the last square? So I go left to right all the way down the whole way until I get to the—I guess you could call this like the 64th square. OK? I get to the 64th square and I want to know, how many grains of rice are on there? OK?

7.9 Well, you can kind of imagine, like, this is going to be a massive amount of rice, right? Like, I already have 128 just going across the first row. Then I'm going to have 256, 512, 1,024, 2,048, you know, 4,096. Like, all of that, it's just going to be, like, astronomically huge. I'm doubling it every single time. So what is this process of doubling every single time? Well, it's a notion of an exponential function. OK, so let's think about this. OK, I have square one, I've got square two, square three, square four, and so on.

7.10 On square one, I have one grain of rice. On square two, I have 2 times 1 grain of rice. On square 3, I have 2 squared times 1 grain of rice, 2 cubed times 1 grain of rice, and so forth and so on, right? 1, 2, 4, 8, 16, and so on. I just keep doubling. So let's think about this. This is really like 2 to the 0. Do we agree? Two to the 0 is 1. One times 1 is 1. So that's what I have there. So what's the pattern? This is one of the best things that you can develop to understand mathematics, is understanding patterns so that you can then take that pattern, abstract it out to a generalized idea, and then prove something about that generalized idea, whether it's correct or incorrect.

7.11 So in square one, I have 2 to the zero one. Square two, I have 2 to the 1 times 1. Square three, I have 2 to the 2 times 1. In square four, I have 2 to the 3 times 1. So it looks like whatever the number is in the square, I'm getting one less power of 2 times 1. Would we agree? If we went one more, OK, what would square five be? Square five, well, based on the pattern that we just saw is whatever the number is of the square, it's like the exponent is one less than that. So that's 2 to the 4 minus 1. So I would say this is probably 2 to the 5 minus 1. That's 2 to the fourth times 1, and go from there.

- 7.12** And let's check that against our intuition. Well, we said it was, OK, square one was 1. OK, that's 1. Square two was 2. OK, that's 2. Square three was 4. That's 4. Square four was 8. OK, 2 times 2 times 2 is 8. Square five is 16. Two times 2 times 2 times 2 is 16. OK, it looks like it's working out. So we keep doing this over and over and over until I get all the way down here to square 64, the very last one. OK, I've got 8, 16, 24, 32, 40, 48, 56, 60, 64. I get to the very last square, 64. What should I have?
- 7.13** Well, based on this pattern, it should be one less than that. So this should be 2 to the 63 times 1, which is an astronomically large number. OK, 2 to the 63rd power. All right. Just absolutely massive. OK. You can't fit that on there, obviously. Right, OK, so what is this really getting at? This constant doubling is getting at the notion of what's called an exponential function. OK, so an exponential function.
- 7.14** We've kind of already seen this when we did the interest rate example, where we said, if I give you an interest rate of like 0.1% and I give you that interest rate at the end of every single month, OK, we had a function which was exponential. It just didn't grow like this, unfortunately, because the interest rate was so small, but it was still an exponential function. So there's kind of this misunderstanding, to some extent, saying that, oh, exponential functions, they grow without bound. And it's like, in the right context, that's true.
- 7.15** But the rate at which they grow is very different and depends on a lot of factors. OK, so an exponential function looks like this. It's a function f of x . So we can give it a name, f of x . That's f as the function. You're taking your x , which is an input, plugging into f , and it gives you something that looks like b to the x . OK. Where b is a number, and I'm going to give you a restriction on what that number can be. OK.
- 7.16** Where 0 is less than b is less than 1 . OK, so we're using our inequalities again. What does this mean? It just means that b is a number that's in between 0 and 1 . And the fact that I don't have inequalities like less than or equal to, or less than or equal to, means you're not allowed to be zero and you're not allowed to be 1 . OK, so I could be one-half, for example. That's fine. That lives in between zero and 1 . I could be one-eighth. That's fine. Or I could be my interest rate, 0.1% , which is 0.001 , which is 1 over $1,000$. That's in here. OK, that's totally fine.
- 7.17** Where b is in between this or b is greater than 1 . For example, in the doubling example with the grains of rice, b is equal to 2 . I'm just doubling every single time. OK, I'm just doubling every single time. So if I wanted to come up with a formula that looks like this, I wanted to come up with a function which models the grain of rice example, OK, so the rice example has function given by f of x is equal to 2 to the x minus 1 . That's what it is. Where x is going to represent the number of the square on which you are on.
- 7.18** So if x is 1 , square one, I'm getting 2 to the 1 minus 1 , that's 1 , 2 to the 0 . If x is 2 , that's square two. I'm saying 2 to the 2 minus 1 , that's 2 . If x is 3 , 2 to the 3 minus 2 , that's 2 squared, that's 4 . And that's exactly capturing the grains of rice that we have in each, in

each one. OK? And we quickly see, if we were to graph this, OK, this one does have extreme growth. This one does have extreme growth. But if we were to graph this, what we notice is the following. Look, when x is equal to 1, I get f of 1 is equal to 2 to the 0, which is 1. OK?

- 7.19** So let's try and graph that. If x is equal to 1, then my output is 1. So I know I go through that point. I know I go through that point. If x is 2, what do I get? OK, if x is 2, so let's just make a quick little table. So x and then f of x , which is 2 to the x minus 1, if x is 1, I get 1. If x is 2, I get 2. If x is 3, I get 4, and so on. We agree, OK? So at 2, I get 2. OK, I get this. At 3, I get 4. OK? So up here at 4. So here, I'm up here. If I graph this, it's going to look something like that. Again, choose your favorite online graphing utility, whatever the case is, go in and plug this in, right? Save your money and plug this into this little free calculator and you get this awesome little graph.
- 7.20** And it tells you, look, here's what it looks like, OK? The value of doing that is that now I could say, well, I don't care if I'm on just a 60, or an 8-by-8 grid; the chess example is now out the window. Now I can just say, oh, this is a function in and of itself. OK? You can, you can argue with me and you can say, look, this doesn't look like an exponential function. You told me exponential functions look like a base raised to the x power. And I'm like, OK, that's fair. It's a fair argument, because now I'm giving you something that says 2 raised to the x minus 1. OK?
- 7.21** So let me prove to you that these two things are equivalent. OK, how would I do this? Remember our properties of exponents, we had this. If I have x to the m plus n , this is x to the m , x to the n —we had that rule for exponents, right? We just add the exponents. We can break that apart as a product. OK, so really, this looks like what? 2 to the x times 2 to the negative 1. Like that. And what is 2 to the negative 1? That's just one-half. OK, so this is really just one-half times 2 to the x .
- 7.22** OK, so there's my "exponential" function that looks like this. I'm just scaling it by multiplying it by a half. OK? Big deal. If I have the function x squared, it's not the same function as $5x$ squared, but it's the same kind of family of functions, right? It still has the x squared kind of flavor to it. So even though this is slightly different, it's a little bit of a shift of that, it's still an exponential function. But now I can just study this as a function. I don't need to worry about that there's some underlying problem with this grid.
- 7.23** I could say, shoot, what happens on the 4,532nd square? How many grains of rice are on there? Evaluate the function at that point. Right. Gives me more versatility. So the beautiful thing, though, is that the graph of an exponential function in general, OK, in general, so if I look at b to the x , where b is either positive or in between 0 and 1, I want to understand what their graphs look like and what are some of the salient features to those graphs that I can then kind of exploit to see if inverses exist or they don't? OK, so let's look at that. Let's see what we have.

- 7.24 So that's kind of the motivating example that exponential functions exist. OK. But people need to be very careful when they talk about exponential functions. They see an example like this and they say, oh, that, that, like, blows up to infinity very quickly. That gets massive really fast. Right, and that's kind of the narrative that people have with exponential growth. They'll say, like, oh, this thing is growing exponentially, or this thing is growing, or it's decaying exponentially or something. And people are like, oh, it's either going off to infinity really fast or it's going to zero really quickly.
- 7.25 And it's like, well, compound interest is an exponential function, technically. And like, I don't know about you, but, like, my bank account doesn't look like that, where I simply go to the next square and suddenly I have 8 figures that just got deposited in my account, in terms of interest. That would be nice, but that's not what's happening. But it's still an exponential function. It's just the rate of growth is dependent upon that interest rate. So you have to be a little bit careful about this.
- 7.26 All right, so what do the graphs of exponential functions look like? OK, so they're graphs. Well, they depend. So if b is greater than 1, the graphs will look like this. So this is the graph of y equals b to the x . OK, remember how we graph functions? Like, if this is my function f of x equals b to the x , we replace f with y and we think of the output as being the y coordinate when we graph. OK, so I have y is equal to b to the x . If b is positive, what do I notice? I always notice that when I plug in x is 0, I get 1. So in particular, it's always going to go through this point, 0, comma, 1.
- 7.27 That was the, the question that was asked, right? Like, when I raise something to the zero power, OK, y is equal to 1. OK. So I always know it's going to be that case. But when b is bigger than 1, this thing blows up. Just like the rice example, when b was 2. OK, so this thing is going to look like this. Now, the way I'm drawing is it makes it seem, like, really close. It suddenly blows up to infinity. But this thing could be very low for a little while, and then at some point, then it takes off really quickly. OK? So you have to be a little bit careful.
- 7.28 Versus, remember, the other case that we had, was when b was in between 0 and 1. Again, I get 0, 1. That's always going to be there. But all it does is it takes this and it just mirrors it across the y , the y axis. That's all it does. OK, so this thing looks like, it looks like this. That's what this graph looks like. Now, I can kind of convince you that that's the case. OK, so for example, if like, b is equal to 2 here, versus b is equal to one-half, let's try and look at those two things independently of each other. OK?
- 7.29 If I look at the thing that looks like y equals 2 to the x , then I notice that, as x gets large to the right-hand side, this exponent is getting huge, and this thing is going to blow up to infinity. It's going to go off as large as we can think of. OK, but notice that over here, if I look at y is equal to one-half to the x , well, what's that really doing? Properties of exponents. Remember, if I have a over b and I raise it to the power, that's a to this power divided by b to that power. So what is this? This is 1 over 2 to the x . That's what that is.

- 7.30** Because what is—OK, so I'll do one step. 1 to the x over 2 to the x , that's 1 over 2 to the x . OK, 1 times 1 times 1 times 1 x times is just 1. That's fine. But look, if you're telling me that as x goes large this way, 2 to the x goes arbitrarily huge. OK, so let's just think of a number. OK, so let's suppose 2 to the x is like 1 billion. All right, well, then 2 to the x is 1 billion. And this is 1 divided by 1 billion, which is really small. And so in particular, the graph, it makes sense that the graph should be getting close to the x axis, should be getting smaller.
- 7.31** OK, one-half, one-fourth, one-eighth, one-sixteenth, 1 over 32, 1 over 64, 1 over 128 and so forth. It's getting smaller and smaller and smaller and smaller. Still positive, bigger than zero, but it's getting small. OK, so what can I say about these graphs? Well, let's think about the inverse function. The inverse is where you have a valid input, you take it into the function, it gives you a valid output or it gives you an output. OK, can I take that output and then give myself back, to get myself back to the input? And the answer is yes.
- 7.32** And the reason why is, if I draw any horizontal line whatsoever, how many times does that horizontal line intersect the graph? I could even draw them down here, by the way. These ones are irrelevant because they don't intersect the graph. But how many times does one of these red lines intersect the graph? And the answer is, at most once. At most once. So think about that. What does that mean? This one intersects the graph in this point. So let's think about it, function theoretically.
- 7.33** That means that this was the domain value, the x value, that the function b to the x sent to this value here. So what would my inverse function do? My inverse function would say, take this value back to this value here. And there's one and only one way to do that because of these horizontal lines only intersecting in one point, OK? So what that means is that this is called the horizontal line test. So exponential functions are invertible. OK? They're invertible. They satisfy this horizontal line test.
- 7.34** The horizontal line test simply says, if you draw a whole family of horizontal lines, do any of the horizontal lines intersect the graph in two or more points? And if the answer is no, they only intersect the graph in one or less points, then your function is invertible. There is an inverse. There is something that will take you from x to f of x and from f of x back to x . There is something I can define to do that.
- 7.35** Now, what is that thing that does the inverse? OK, we need a notation to be able to talk about these things. So this is a notation for inverse functions, OK? This is just abstract nonsense, OK? We need a notation to be able to—I mean, it's like I said at the very beginning of this whole thing. Mathematics is a language, OK? I think mathematics is, personally, I believe that it is part of the sciences, even though people would argue that it's not, because it's not necessarily like an experimental kind of thing. Right? You're not in a lab doing an experiment to prove a particular thing.

- 7.36 I would argue against that. I think mathematics is a science. But ultimately, I think mathematics is more akin to art and beauty than anything else. Because the creativity necessary for mathematics is very akin to the creativity in somebody who's sculpting or somebody who is painting or somebody that is that. You dare to ask the question, can I create something like this? And then you pursue that with abandon to see if you can actually create said thing. Right? I think it's more akin to art than it is to science, personally. I know people would disagree with me on that, and that's fine.
- 7.37 But all the same, we need a notation to explain, what do I mean by an inverse function? What do I mean by something that reverses what f does? It does the opposite. What do I mean by that? I need a notation to actually designate, this notation means, this is what an inverse function is, OK? And so what is the notation? Notation is the following. So if f is invertible, invertible, OK, that doesn't look right. OK, that's fine. If f is invertible, that's fine, if f is invertible, then its inverse is f upper minus 1. OK?
- 7.38 And you might look at that and be like, OK, that's confusing because that's an exponent, right? And it's like, no, it's not an exponent. I am literally just giving it a name. That is it. The reason why there is a minus sign, minus 1 up there, is to indicate things like this. Look, if I gave you the number 3 and I said, "What is its inverse?" What I'm asking is, what is the number that multiplies 3 to 1? That's what I'm asking. And so if I think about this as an equation, I have 3 times y equals 1, solve for y , I'd say y is equal to one-third. OK?
- 7.39 So we conflate this notion of having a minus 1 in the exponent by saying that, oh, this means 1 over f , and it's like, no, no, no, no, no. That's not what I'm saying. I'm simply just giving this a designation, saying that, what is the inverse of 3, multiplicatively? It's 3 inverse. What is it? Well, that just happens to be 1 over 3. That's fine. But that's what's happening over the real numbers. These are functions. We're not talking about real numbers, we're talking about functions. So what am I going to say?
- 7.40 I'm going to use the same notation to say it's an inverse. But this does not mean 1 over f . It does not mean that. It's just a label associated to it. OK, now what is the property that inverse functions have? The property is the following, and this is the key. OK? So if y is equal to f of x , then f inverse of y is equal to x . That's the property that the inverse function has. OK? So think about this for a second. Suppose I have my domain, OK, my domain of f on this side. And I have my range of f here. Let's suppose I have that, OK?
- 7.41 And so what does f do? f takes a point over here, x , and it sends it to a point here, y equals f of x . That's what f does. It takes your valid input of x and it gives me an output of y . OK, what does the inverse function do? Well, it should take whatever that valid, whatever that output was of y , and it should give me back x . So what should it do? If I take y and I do this via f inverse, what should x equal? x should be, what is f inverse of y ? That's what it should do. OK?

- 7.42 So that's my defining characteristic. OK, so now I'm going to take the two things and put them together. I'm going to say, on the one hand, I know that the exponential functions are invertible because they satisfy the horizontal line test. One input, one output. OK, fine. What is its inverse? I'm going to give them a name, and their name is a logarithm. That's what they are. OK? OK, so definition. The inverse to f of x is equal to b to the x . OK. Is a logarithm base b . OK, logarithm base b , denoted f inverse of whatever you want to use. You want to use y , you want to use x , I don't care what variable you want to use. It doesn't matter.
- 7.43 I'm not going to use x so that it doesn't confuse it with the original x . OK, so I'll use a different variable. But f inverse of y , what's the notation? It's $\log$, sub b of y . Whatever the input was for f , that's what it is. And the way you read that is, it's $\log$ base b of y . That's how you read it. $\log$ base b of y . OK, so just as a silly example, for the function f of x equals 2 to the x — this is our doubling function—the inverse would be f inverse of y is equal to $\log$ base 2 of y . That's all it is.
- 7.44 OK, now what's the property that it has to satisfy? Well, if I do f inverse composed with f —OK, so remember, this is function composition, f inverse of f of x —if I do the function composition, what should I do? What I'm really saying is, if I start with x , I apply f to it. It gives me 2 to the x , and then I apply f inverse to it, I should be back at x . OK, so what should this equal? That should be x on the nose, but in particular, we know formulas for these particular objects. So what is this saying? This thing is saying, what is f inverse of, what is f of x ? This is 2 to the x .
- 7.45 And what is f inverse of its input? That is $\log$ base 2 of whatever its input is. That's y , 2 to the x . And I'm saying that's equal to x . $\log$ base 2 of 2 to the x is equal to x . So that's a property that we're going to hold on to. OK, so we're going to list the properties of logarithms up here. Going to list the properties of logarithms. Properties of $\log$. OK, number one, if I were to generalize this to f of x is equal to b to the x , what would the property be? Well, this property says, $\log$ base 2 of 2 to the x —as long as those two things are the same, all I do is I get the exponent.
- 7.46 So I'm going to have $\log$ base b of b to the x is x . So I'm going to get $\log$ base b of b to the x is equal to x . OK, that's one of the properties that we have. Now, what I want you to think about is kind of the following. You're like, OK, well, that's great. You're pushing symbols on the board. Fantastic. Like, why do I care? Why do I care? Well, let me explain why you should care. OK, we did examples in the exponential section where we had things like this. We said, I want you to solve for x in something that looks like, something that looks like that. Do we agree?
- 7.47 We had things that looked like that. And we said, solve for x . And we said, OK, how do I do this? I'm going to try and write 16 as a power of 2 . So I can write this as 2 to the 4 th. That's raised to the x plus 2 . This is now 2 to the x minus 1 . And I'm going to write this as 2 to the x minus 1 equals 2 to the 4 times x is $4x$, 4 times 2 is 8 . I'm going to look at that. And then we made this amazing kind of leap, and we said, look, if these two bases

are the same, 2 and 2, well, then I can just set the exponents equal to each other and I can say $x - 1 = 4x + 8$.

7.48 And that tells me what? If I move the x over, move the 8 over, I get $3x$ is equal to minus 9 and x is equal to minus 3. We say that's what the solution would be. OK. It's kind of a miracle because I just came up with that off the top of my head, and the fact that the number worked out to be a beautiful negative 3, I'm really happy with. OK, so, but that was a fluke. That was a complete fluke. Why? Because the 2's happen to be the same. The 2's happen to be the same. I can break this incredibly easily using the following thing. OK, I can break this in the following way.

7.49 And I'll show you how, like, fragile this example is. OK, this example is a fluke. Why? Change the 16 to a 3. It's the smallest change I'm going to make. I'm now looking at 2 to the $x - 1$ is equal to 3 to the $x + 2$. What was our technique with the 16? Oh, write 16 as a power of 2. OK. Write 3 as a power of 2. Good luck with that. So where do I go? What do I do at this point? It's like, "Ah, I can't solve it. OK, fine." No, that's not an answer to anything, right? Just throw your hands up, OK, I can't do it. At the very least, if you can't do something, you should be able to explain why you can't do it. You should be able to justify why that's not possible. Otherwise it's plausible that it's possible.

7.50 OK, so this property right here using logarithms is giving you an amazing tool to say, look, what, ultimately, what am I trying to do? I'm trying to get the exponent out, down, so that I can actually use principles of algebra to solve for my variables. I can't solve for the variables until they're down, like this, right? I can't do that. So what is the logarithm saying? It's saying, look, as long as you take the same base of the logarithm, this guy can come out. It comes out front.

7.51 OK, so what does that mean? That means if I take log base 2 of 2 to the $x - 1$, well, whatever I do to one side, I have to do to the other, because I have to balance the equation, right? So I can't say, like, take log base 2 on the left and log base 3 on the right. That's not the same operation. But I can take log base 2 of 3 to the $x + 2$. And what is this property telling me right here? OK, when you take logarithms, it doesn't matter. So in order for this equality to be true, that this to actually equal x , the bases have to be the same.

7.52 But if I took something that was log base b of c to the x , what happens? The exponent still comes out as an x . But now I have log base b of c , and this is just a number. That's all it is. I still spring the exponent out of the exponent. I can still bring it down. I just can't say that this is 1. I can't say that this is 1 anymore. So what do I get? This thing here gives me an $x - 1$ times log base 2 of 2. That quantity is equal to 1. That's what this principle is telling me here. If x is 1, log base b of b is 1. But then this quantity is what? This is $x + 2$ times log base 2 of 3. Now this is not equal to 1, but it is a number. It is a number.

- 7.53 I can go into my favorite online calculator, click it in. It'll give me some ridiculous decimal that probably has meaning someplace, but at the end of the day, it is a number. It is a number. And so what can I say now? I'm looking at x minus 1 is equal to x plus 2 times a number. Yes, the number looks scary. It's got letters and numbers and subscripts and parentheses and all of that. But at the end of the day, it's a number. It's a number. So if I gave you an equation like this and I said, "Solve for it," you already know how to do it. You already know how to do it.
- 7.54 If this number were, say, 6, for example, you know how to solve it. Distribute the 6 in, gather the x 's on one side, gather the constants on the other, and solve for x . What does it matter if this looks like log base 2 of 3? It's still a number. You can still solve it. And so now something that originally looked completely impossible because the base for 3, 3 to the x plus 2, doesn't look like a power of 2, is suddenly possible. So logarithms have taken this, this circle of problems that we could solve, things that look like 2 to the x minus 1, 16 to the x plus 2, and it's enlarged the circle, saying, here's a larger class of functions that we can satisfy that we can solve. OK.
- 7.55 All right. So that's property one. What's property two? Well, why do I do f inverse after I do f ? Why don't I switch their roles and do f inverse first and then do f ? So why don't I take b raised to the log base b of x ? What do I get? I'm still going to get x . Think about it in terms of the picture. I start with x , I apply f , I get f of x , I apply f inverse, I go back to x . Well, why don't I start over here and first apply f inverse and then apply f ? What happens? Well, I'm still in the same place. So if I exponentiate a logarithm or I logarithm an exponential, I didn't do anything, as long as their bases are the same.
- 7.56 OK, now we want to say, OK, you spent an entire time talking about properties of exponents, properties of exponents. You had things like x to the m times x to the n is equal to x to the m plus n . So there was some sort of relationship between multiplication and addition. When I multiply x to the m and x to the n , I got x to the m plus n . Some relationship between multiplication and addition in the exponential-type world. Does the same thing hold in the logarithm world? The answer is yes.
- 7.57 If I take the logarithm base b of x times y , a multiplication, this becomes the logarithm base b of x plus—let me erase this— plus the logarithm base b of y . The logarithm applied to a product is the sum of the logarithms. That's how I remember these things. I have a very simplistic way of remembering things. I know that sounds a little ridiculous, but ultimately I need something in the simplest form possible. I think it was Einstein who said, "A theory should be as simple as possible and no simpler." Something to that effect, which I think is a beautiful statement, right?
- 7.58 So the way I kind of remember these are kind of like the, the JFK quote, right? Like, "Ask not what your country can do for you, but what you can do for your country" kind of thing. So that's how, that's literally how I remember these things. I feel like I've got JFK in the back of my head when I'm thinking about these properties. I'm saying, what? The logarithm of a product is the sum of the logarithms. The logarithm of a quotient is

the difference of the logarithms. Things like that. I don't know why that sticks in my head. It doesn't make a lot of sense when you actually verbalize it. But for whatever reason, it sticks for me. If it sticks for you, that's great.

7.59 OK, how do I take the logarithm of x divided by y ? Well, again, we have this other product, product that said x to the m over x to the n was equal to x to the m minus n . So there's a relationship between division and subtraction. So what's the analog in the logarithmic world? The analog is division turns into a difference. The difference. There's one more property, and it is, what happens when I take x and I raise it to a power? What happens when I take x and I raise it to a power? Again, look, this is x . This is not necessarily b . If x were b , I know what the answer is. The answer is just r . It's just the exponent because that's property one. Log base b of b to the r is r .

7.60 I know that, but what if x is not equal to b ? Just like I did over here, I took log base 2 of not- b to the x plus 2. What happens? This r comes out front. This is r log base b of x . OK, that's what we have. That's great. Well, that's all fine and great. So again, who cares? Well, I gave you an example right here that says why we care. We can now solve new types of exponential problems. That's a good thing. That's a good thing. Going back to the grain of rice example, going back to the grain of rice example. Well, actually, you know what? Let's do. OK, maybe I should stick to the grain of rice example.

7.61 If I stick to the grain of rice example, I could ask the following question. The grain of rice question was f of x is equal to 2 to the x minus 1 . That was how many grains of rice was on square x . Right? That was the formula. What if I said—you've got an infinite grid, infinitely many squares—at what point, at what square does the number of grains of rice exceed, say, 10 to the 50 th power? OK, so here's the question. On which square does the number of grains of rice exceed? OK, we'll do, what is it, the number of particles in the universe or something like that? It's like 10 to the 88 th, something like that, right?

7.62 Beginning of the Big Bang, I think that's the number. I forget exactly what the number is. But we'll just say for argument that it's 10 to the 88 th. On which square does the number of grains of rice exceed 10 to the 88 th? OK. So what am I really asking? I'm saying, OK, what is x such that this, which models the number of grains of rice, is bigger than or equal to 10 to the 88 th? That's what I'm really asking. OK, so what is x such that 2 to the x minus 1 is greater than 10 to the 88 ? That's what we're trying to figure out. So we're bringing everything together now.

7.63 Now, we have inequalities, we have equations, we have functions, we have logarithms, we have all that kind of stuff, and we want to try and solve these corresponding things. We're bringing it all together. Well, how would I do that? Well, I need to bring this down. I need to bring that down. How do I do that? Logarithms do that. OK, so I'm going to take log base 2 on the left side, and I'm going to take log base 2 on the right side. But now the question is, well, what happens to the inequality? It's not an equality.

So maybe the inequality, like, we had examples with inequalities that, when I divided by a negative, it somehow flipped the inequality.

- 7.64** So it's like, maybe applying logarithms does something goofy as well. OK, well, let's think about that. What do logarithms look like? One property that inverse functions satisfy is the following. If this is the original function, b to the x , OK, so implicitly, I'm assuming b is bigger than 1 here because I'm drawing it this way. What you imagine is, on this line, which is y equals x , the graph of the logarithm function is just a mirror image down here. You imagine, like, this line y equals x is a mirror. And the corresponding thing on this side is what that's going to be.
- 7.65** So what is it going to look like? It's going to look like something like this. This is $\log$ base b of x . That's what its graph looks like. OK, so let's think about that for a second. What does that mean? I'm taking two quantities. This quantity is less than this quantity. OK, so let's say this quantity over here is my 10 to the 88 th, and that is less than this quantity, which is my 2 to the x minus 1 . OK, then what about the logarithms? The logarithm is this point. The other logarithm is this point. What do I notice? The $\log$ of that is less than the $\log$ of this. This one is higher.
- 7.66** So in particular logarithms, $\log$ base 2 preserves the inequality. If a is less than b , then $\log a$ is less than $\log b$. That's true. It preserves the inequality. But now what happens? By property, let's see, property one, $\log$ base 2 of 2 to the anything is just that anything back. OK, and that's greater than, I mean, goodness knows what this is. The best thing I can do is take the 88 out front, by the property here, $\log$ base b of something to the r , the r can get pulled out front.
- 7.67** So all I have to do is add 1 to that side, and it says that as long as the squares number is greater than 88 , $\log$ base 2 of 10 plus 1 , you will exceed—the number of grains of rice will exceed 10 to the 80 th. That's great. Why is that useful? Well, there's a lot of models out there which will model certain behaviors, such as radioactive decay. You have a certain radioactive isotope, it decays predictably. It has a half life. Just like the example that we gave with the half life of taking, like, Tylenol, a certain drug, right? It decays at a certain rate at a certain time. So it cuts in half at a certain time.
- 7.68** You want to understand that if you have a particular object that is irradiated and you know how much there is, and you know how fast it decays in half life, which is an exponential, you want to understand, when does the radiation amount get to a safe threshold? It's exactly what this is getting at. It's exactly what this is getting at. OK? So let's do some, let's do some problems on this, OK? One thing that we have to be careful about is kind of abstract nonsense, is that, if I look at the domain of f , that is equal to the range of f inverse, and vice versa. If I look at the range of f , this is equal to the domain of f inverse. OK?
- 7.69** The argument is kind of the following. Let's say this group is the domain and this group is the range. So a function would go from the domain to the range. But what does the

inverse function do? The inverse function goes from the range to the domain. And so in particular, your domain is the range of the inverse, and your range is the domain of the original function. OK, does that make sense? Like, they, they just play off of each other. The domain of the original function is the range of the inverse, and the range of the function is the domain of the inverse. OK?

7.70 So in particular, the domain of the log function, log base b , is equal to what? Well, that has to be from zero to infinity. Why is it zero to infinity? Well, look at the exponential function. It never touches zero, but it gets everything above zero. That's its range. It goes from zero to infinity. So if the range of the original function is zero to infinity, then that means that needs to be the domain of the inverse function. The inverse function is the log base b . So logarithm has domain zero to infinity.

7.71 Now, what was the domain of the exponential function? It was everything. You could plug in anything to an exponential function. So in particular, the range of log base b is equal to everything, everything from minus infinity to infinity. And that pans out with its graph, right? Its graph of the logarithm looks like, looks like this. What am I allowed to plug in? This is y equals log base b of x . I'm allowed to plug in anything from zero on. That's my domain. And what are the y values that I get? I get everything from minus infinity all the way up to positive infinity. OK?

7.72 So with that in mind, let's try and solve some equations. Keeping with the theme that we have is that when we have a new branch of functions, we want to solve equations associated to that new branch of functions. So sticking with that theme, let's do the following. I want you to write as a single logarithm. OK? The beautiful thing about logarithms and the properties that they have is that you can use those properties to combine logarithms in a very precise way. OK, so let's suppose I give you the following equation.

7.73 I'm going to give you log base 2 of x plus log base 2 of—let's see, what do I want? I want x minus 2 is equal to 3. OK, let's suppose I give you that. I'm going to ask you two questions. Question number one is, I want you to write the left-hand side as a single logarithm. OK? So that's number one. Number two is that I want you to solve the equation. I want you to give me an x value that solves this equation. OK, number two, solve the equation if possible. Maybe it's not solvable. OK, so how do we do this?

7.74 Well, we use the properties of logarithms. And so I have to step back and say, OK, number one, do I remember what the properties are? Number two, can I apply them correctly? Well, let's see what we have. First thing I notice is, I have log base 2, log base 2. That's observation number one. They're the same base logarithm. OK, I then think back and I'm like, wait a second, I remember something that said log base b of something plus log base b of something—there was something that related plus and multiplication.

- 7.75** There's one of the properties that we have that said that plus and multiplication were related in a certain way. And I go back and I review that. That says that log base 2 of x plus log base 2 of x minus 2 is equal to log base 2 of what? The product of whatever's inside the parentheses. So this is x times x minus 2. So I can write it as a single logarithm. That's the solution to one. This logarithm, log 2 log base 2 of x plus log base 2 of x minus 2, is log base 2 of x times x minus 2. OK, so that's fine.
- 7.76** So I can rewrite this now as saying, OK, this is log base 2 of x times x minus 2 is equal to 3. OK, so how do I do that? Well, logarithms and exponentials are inverses to one another. So if I have an equation that has a logarithm, then there's a corresponding equation which has an exponential, and how does that work? It works in the following way. If log base b of x is equal to y , then— here's the trick— b raised to the y equals x . b raised to the y is equal to x .
- 7.77** OK, why is that true? OK, again, you should never just accept anything that I say as being true. You should convince yourself that it's actually true. OK, why is that true? OK, well, let's start, let's pretend that we don't have the left-hand side. Let's pretend we have b to the y equals x . Let's take the logarithm base b on both sides. What happens? If I take log base b of b to the y is equal to log base b of x , what do I get? Well, there's a property of logarithms that says if I take log base b of b to the y , that's just equal to y . And what's the right-hand side? That's log base b of x . That's literally what I have here.
- 7.78** That's how you undo them. That's how they're related to one another. OK, so if I apply that and I say, OK, look, I have a log base b of blank equals blank, how do I turn that into an exponential? The property is what? It's the same as 2 raised to the 3 equals that. OK. So that equation right here, star, let me make an asterisk. OK, so the asterisk is really equivalent to saying 2 to the third is equal to x times x minus 2. Using that property. Well, now I should be really happy. Well, I mean, it's math, so you know, like, within reason.
- 7.79** OK, so now I should be happy, though, because look, 2 to the third is 8. If I distribute the x , I get x times x is x squared, x times negative 2 is minus $2x$. If I move the 8 to the other side, I have x squared minus $2x$ minus 8 is equal to 0. That's a quadratic. And we have a quadratic formula which will give me the solution to that immediately. Fantastic. Or I could ask myself, does it factor? I could say, does it factor? Now remember, factoring is tricky because if it factors like this, I would need an x here and an x here for sure. Because x times x needs to give me the x squared term there. That definitely has to be true.
- 7.80** But then, remember, whatever goes on the next side, those two numbers have to multiply together to give me negative 8, and they have to add together to give me negative 2. So you're sitting there and you're like, OK, I have negative 8. How do I factor that? That's a 4 and a minus 2. It's also a negative 4 times a 2. It's also 8 times minus 1. It's also minus 8 times 1. Those are kind of like my first candidates that I'm going to try

in terms of factorization. All of those work in terms of the multiplication, that this times this is negative 8.

- 7.81** But which of them satisfy that, when you add them, you get negative 2? Well, negative 8 plus 1 is negative 7, so that's out. Eight plus negative 1 is positive 7, so that's out. Four plus negative 2 is 2, not negative 2, so that's out. But negative 4 plus 2 is negative 2. Those are my factors. So this becomes x minus 4, x plus 2. OK.
- 7.82** But then we step back and we say, that's fantastic. Because I now have a number, x minus 4 times a number x plus 2 equals zero. Because we're in the real numbers, that means that one of them is zero. So that means that either x minus 4 is zero, or x plus 2 is zero. So either x minus 4 equals zero, or x plus 2 is equal to zero. So x equals 4 or x equals minus 2. Now here is the important point that you have to be aware of, OK? It's very tempting to get lost in the weeds and just say like, oh, OK, x equals 4 and x equals negative 2.
- 7.83** Those are my solutions. Those are my solutions to my original problem, which was trying to solve, what? It was trying to solve log base 2 of x plus log base 2 of x minus 2 is equal to 3. Yes, that's what I was trying to solve. I did a lot of algebraic operations and function theoretic operations to get this down to this quadratic formula, to get down to the quadratic. Now, the quadratic, just in isolation, has two solutions, 4 and negative 2. But are those solutions to this? That's the kicker.
- 7.84** Because what is the domain of the corresponding function? The domain of the corresponding function for log base 2 is everything that's bigger than 0. Remember the domain considerations. The domain of the log base 2 function is equal to zero to infinity, which means that the only valid inputs that I'm allowed to plug into either this or this, whatever the result is, has to be bigger than zero. So in particular, check your answers. If x is equal to 4 and I plug that in, is that a valid input? Sure, 4 is bigger than zero. If I plug in 4 there, is it valid? Sure, 4 minus 2 is 2, 2 is bigger than zero. That's legit. So I can definitely say that one is a solution.
- 7.85** What about x is equal to negative 2? I take x is equal to negative 2 and I plug it in here. That's an invalid input because my domain does not include negative numbers. So immediately I can say, no, that is not a solution. That is the most overlooked thing when solving equations with logarithms, is going back to the beginning and saying, do the answers that I got in isolation solving a quadratic equation, which seemingly has nothing to do with the logarithm solving this equation, does it actually recapture the solution that I wanted?
- 7.86** So the moral of the story is, when you're doing these types of equations, be careful. Apply the principles correctly, but always go back and make sure that your solutions are correct in the context from which they came.

Lecture 8: Linear Equation Systems

At a glance — Systems of linear equations.

- 8.1** All right, so today what we're going to talk about is, we're going back to our two-variable linear equations. So those are the things that looked like, for example, $2x$ minus $3y$ is equal to 5 . And when we started with these, we said, well, is there a way to geometrically understand them? What do they look like? And we saw that we could write it in slope intercept form, y equals mx plus b . We said the slope and the y coordinate of the y intercept are given by all of these different things. And we said that they look like lines.
- 8.2** OK, so what we want to try and understand now is, well, what if instead I change my perspective a little bit? Rather than saying I have one equation with the two unknowns, x and y , I want to say, what if I give myself another equation, let's say, x plus y equals 3 ? And I want to consider both of those simultaneously, rather than individually. And I want to ask myself the following question. Does there exist an x and y value which simultaneously satisfy both equations? Again, maybe this is relevant, maybe it's not.
- 8.3** Turns out to be incredibly relevant, especially in a lot of kind of modern technology in some way, in a very precise way, actually. So what I'm thinking of now is, I'm thinking of these as two equations that are somehow tethered together, where rather than simply saying, what are the solutions of $2x$ minus $3y$ equals 5 , now I'm saying, what are the solutions of $2x$ minus $3y$ equals 5 and x plus y is equal to 3 , taken together? OK, taken together, if that even makes sense.
- 8.4** So, for example, I could stare at this equation all day long and just cover this one up, and I can say, oh, if x is 3 and y is 0 , that satisfies this. 3 plus 0 is 3 . Great. If x is 2 and y is 1 , that's also a solution. If x is 1 and y is 2 , if x is 0 and y is 3 , if x is 5 and y is negative 2 , I could sit here all day long and just list out a bunch of those. OK?
- 8.5** But if I take a sample point here—let's take $3, 0$ —that's a sample solution to this. x is 3 , y is 0 . It works. Does it work for this one? Well, 2 times 3 is 6 minus 3 times 0 , 6 minus 0 is not 5 . So then I'm like, OK, well, that's not a solution. But I look at this one and I say, well, $2, 1$ was a solution. If I plug that in, what do I get? I get 2 times 2 is 4 , minus 3 is 1 , not five. OK, that's not a solution. I want to ask myself the question, do they actually have a solution or not? Maybe they do, maybe they don't. OK.
- 8.6** Well, how could I approach a question like this? How could I attack the solution? Well, we spent a lot of time trying to understand these geometrically. We said that they're lines. OK? So maybe there's something to be said about the geometry that will shed light on a possible solution or not. So let's see what we can do. OK, let's attempt to graph the equations. Let's attempt to graph them. OK, now remember, how did we do that? We said, solve for y in terms of x to get our slope intercept form, y equals mx plus B . So we're going to do that for both of these.

- 8.7** OK, so what am I going to do? Again, the negative coefficient of the y . So the trick is not to move the $2x$ over, it's to move this over and move this over. That's going to be the trick. So I get $2x$ minus $3y$ is equal to 5 . I'm going to add $3y$. Add $3y$. I get $2x$ is equal to 5 plus $3y$. Subtract 5 , subtract 5 , I get $2x$ minus 5 is equal to $3y$. Divide by 3 , divide by 3 , y is equal to $\frac{2}{3}x$ minus $\frac{5}{3}$. OK, so that's equation one. Let's give it a name. That's equation one.
- 8.8** Let's do the same thing for equation two. It's a little simpler. All I have to do is move x over. So for equation two, I get x plus y equals 3 . Subtract x , subtract x , I get y is equal to $-x$ plus 3 . That's equation two. OK, well, I can graph them now. Let's see what that looks like. OK. I have an x . I have a y . Let's start with this one. OK, so my slope is equal to $\frac{2}{3}$, and my y intercept—remember, the y intercept is where you intersect the y axis, which means x is zero. OK, so my y intercept is zero, comma, whatever b is. In this case, it's $-\frac{5}{3}$. So that's just shy of negative 2 . Right? It's negative 1.67 , blah, blah, or 667 , whatever.
- 8.9** OK, so I get negative 1 , negative 2 , negative 3 , 4 , 5 . One, 2 , 3 , 4 . One, 2 , 3 , 4 , 5 . One, 2 , 3 , 4 , 5 . OK, well, hopefully this will help. Now, let's see. I have negative 5 over 3 , so that's negative 1 and change. So negative 1 and change. That's like right here, maybe. And my slope says go up 2 and to the right 3 . So I'm going to go up 1 , 2 , over 1 , 2 , 3 . So a point somewhere right there. And then I'm going to connect the dots with the line, roughly. OK. Again, this is just an approximation. Maybe it's going to shed light on something. Maybe it's not. I don't know.
- 8.10** OK, this one, what is my slope? Slope is equal to -1 . That's this coefficient of my x . And the y intercept is equal to 0 , 3 . So this one says go up 1 , 2 , 3 here. And the slope says, go down 1 , over 1 , down 1 , over 1 , down 1 , over 1 , something like this. And connect those two dots. OK, so this one is the graph of 2 . This one is the graph of 1 . All right, so what does that mean? This graph, the points on this line for 1 are coordinates x , y that satisfy this equation, i.e., they satisfy this. Yes, they satisfy equation one.
- 8.11** The points on this line are coordinates xy , which satisfy this equation. Are there any points that satisfy both? And the answer is yes, that one satisfies both one and two, satisfies both. Algebraically, what does that mean? It means that this point, whatever its coordinates are, x , y , they satisfy both this equation and this equation. Therefore, it is a solution to both. OK, so a solution to both simultaneously exists. Well, that's great. I mean, it's one thing to say that something exists. It's another thing entirely to say what it is, right?
- 8.12** It's one thing to say the Loch Ness monster exists. It's another to pull the Loch Ness monster out and be like, look, here it is. Right? Those are two very different things. Asserting that something exists versus actually finding it are two very different things. OK, so it satisfies both one and two. How would I figure that out? How would I figure that out? Well, here's the trick. What is a defining characteristic of this point? These are

where the two lines intersect, which means that their x and y coordinates are equal to each other at that point.

- 8.13** Well, what is the y coordinate to this guy at that point? The y coordinate there is this, y is equal to negative x plus 3. What is the y coordinate at this point for this line? It's this. Those two y coordinates have to equal each other. They have to equal each other. So what does that mean? I can set this y equal to this y , and what does that give me? That gives me $-x + 3 = \frac{2}{3}x - \frac{5}{3}$. Now, if you sit with that for a second, you're like, oh, that's like the first thing we did. I take a linear expression, I set it equal to a linear expression, and I solve for x .
- 8.14** OK, now here's a trick. I think this is an often overlooked trick. I don't really like fractions all that much. They kind of freak me out. OK? So what I'm going to do is, I'm going to say, can I get rid of these fractions somehow? Well, I know if I were to multiply this by 3, I could cancel the 3's in the denominator. But whatever I do on one side, I have to do to the other side. So what I'm going to do is, I'm going to say multiply both sides by 3. I haven't changed the problem.
- 8.15** I'm allowed to do one thing to one side and one thing to the other. That's again the balance technique that we have, right? If I add 5 to one side, I add 5 to the other side. I haven't changed the balance of anything. If I multiply one side by 3, I multiply the other side by 3. I haven't changed the balance. The equation is still the same. I change what it looks like, but I haven't changed its solution set, and I haven't changed what—I haven't changed its solution set. And I haven't changed the original kind of equation. It just looks different.
- 8.16** OK, so if I take that 3 into both, what do I get? $-3x + 9 = \frac{2}{3}x - \frac{5}{3}$, bring the 3 in, I get $2x$. Bring the 3 in, I get -5 . That I'm more comfortable with because the fractions are gone now. And so now I try and solve for x . Well, what would my technique be? There's a ton of things you can do here. Personally, I would say, look, if I subtract the $2x$, I'm going to get a negative. I don't like negatives. So I'm going to add $3x$ on both sides. What do I get here? I get a 9, I get a $5x - 5$. I add the 5 over. What do I get? I get a 14 is equal to $5x$. Divide by the 5.
- 8.17** But remember, you don't define your variable on the right-hand side; you define the variable on the left. So if I divide by 5, what am I really getting? I'm getting x is equal to 14 divided by 5. What does that mean? Well, that means that x is equal to 14 divided by 5. This point has x coordinate 14 divided by 5. That's what it means. So then the question is, well, what is the y coordinate? And you're like, "OK, well, Robbie, what y do I plug it into? Do I plug it into this one or do I plug it into this one?" And the answer is yes, it doesn't matter which one you plug it into, because look, how did we set these two things equal to? We said they're equal to each other.
- 8.18** Which means that when I take x is equal to 14 over 5 and I plug it in here, which is the one we're going to plug it into, because it's a little simpler—I'm not going to have to

deal with this fraction of two-thirds and all that—we set this equal to this. So it doesn't matter which x I plug or which y I plug it into. They're going to give me the same value. OK, so let's see what happens. Let's clear this off a little bit. That's my algebra.

- 8.19** So x is equal to $14/5$. OK, let's plug it into equation two. So I'm going to get y is equal to $-14/5 + 3$. OK, how do I add fractions? This is like $3/1$. So I need a common denominator, which is a 5. So I'm going to multiply by $5/5$. That's my trick for getting a common denominator. I get $-14/5 + 15/5$, negative 14 plus 15. Again, remember, this is not saying $-14 + 15$. It's not minus 29. It's negative 14 as a quantity, plus 15. So that's going to give me a $1/5$.
- 8.20** Now, intuitively, that kind of makes sense because look, what is the y coordinate? It's really close to zero. It's positive, but really close to zero. And what is one-fifth? That's what, like 0.2? Something like that? Right. It's pretty close to zero, but positive. But it's one-fifth. That right there is the unique solution to both of these equations. And again, as a sanity check, you should plug it in and see what happens.
- 8.21** OK, if I take x , that's $14/5 + y$. y is one-fifth. What is that? That's $15/5$, which is, oh, 3. Sweet. Satisfies that one. What if I plug it into this one? $2x$. So $2x$ is, what, $28/5$? I have a double 14. So $28/5$, minus what? Three times y , so that's, what, $3/5$? OK, $28/5 - 3/5$ is $25/5$. Five goes into 25 five times, and bingo, it satisfies both. OK? So maybe there's something here. Maybe there's something we can say concretely. All right? We want to try and understand how this works, OK?
- 8.22** So let's talk about language. Let's talk about language first. It doesn't look like it right now, and you have no reason to believe me, that what I'm saying is actually true. But the GPS you use on your phones uses this. The GPS you use on your phones uses this. All right, so maybe it's got some relevance. We use those occasionally. So let's see. The first language that we need is that this thing over here is what is called a system of linear equations, OK? A system of linear equations. So a system of linear equations is a collection of linear equations. OK?
- 8.23** So a system of linear equations. It's just a collection of linear equations in some number of variables. OK? In some number of variables. OK, I'll just abbreviate that. System of equations. A solution to a system of equations is an assignment of the variables which satisfies every equation in the system simultaneously. That's what a solution is. OK? So we want to try and understand, how do you classify solution types? How do you classify solution types?
- 8.24** OK, if I give you a system of linear equations, in this case, I have two equations in two variables. That's kind of like the ideal situation, when you have the same number of equations as you have variables. OK, but I don't have to give you the same number of equations as you have variables, right? I can give you four or five equations in two variables, right? I could just write something like $x - y = 6$, $3x + 2y = 7$,

12y minus 7y equals 8. And now I have five equations and two variables. OK? That can happen.

- 8.25** So I want to classify the solution types. Well, how did we find the solution to these in the first place? Well, we said, oh, they're lines. And so we graphed the lines, and we said, if the lines intersect, we said that that intersection point was my solution. OK? So if I only have two linear equations, think about that geometrically. What are the possibilities that you can have? If you only have two, classify solution types for two linear equations in two variables. What can I have? Well, what can lines look like?
- 8.26** I can have the case that I had where I have a line that looks like this and a line that looks like this. Here's my y axis. Here's my x axis. Here's line one and line two, right? I could have something that looks like that. They intersect, right? They intersect. What else could happen? They intersect in one point. What else could happen? Well, I have an x, I have a y. I could have a line where line one is equal to line two. They're literally the same line, right? How can you see that?
- 8.27** This one could be like, the first line is $x + y = 6$. And this one is literally the same equation, just multiplied by 5, $5x + 5y = 30$. Those are literally the same thing. They look different. So I can present them to you and say, "Hey, look, here's one equation, here's a second equation," and you're like, "That's the same thing." What are you talking about? You gave me one equation. They look different, but they're the same. It's the same line. So they intersect everywhere, right? They're literally the same line.
- 8.28** Or what else can happen? Line one, line two. I have parallel. I just have two lines that are parallel to each other. They never intersect each other. They never intersect each other. And if you sit there for a little while, you can kind of convince yourself that this is all you can have, because each equation gives you a line. And so what are the only possibilities? Either the lines intersect in one point and only one point. Either they are parallel or either they are the same line. Those are the only things that you can get.
- 8.29** So now think back to the geometry. If I have one point, this is a unique solution, unique meaning one and only one solution. OK, one and only one solution. This one, they intersect everywhere. So how many points are on this line? Well, I have infinitely many, infinitely many solutions. Every point on the line is a solution. Right? Think about that. In this example, where these two are literally the same line, what is a solution to this? 6, 0 is a solution. But if I plug in 6, 0, 5 times 6 is 30 plus 0 is 30. Oh, cool. So 6,0 satisfies this. 6,0 satisfies this. What about 1, 5? OK, 5 plus 25 is 30. It satisfies it.
- 8.30** OK, so I've given you two solutions, but every point on the line is a solution. There's infinitely many. Whereas something like this would be like this. It'd be like $x + y = 5$, and then $x + y = 10$. Those two lines are parallel to each other. They have the same slope, which is negative 1, but their y intercepts are different. One is 5, one is, one is 0, 5, and the other 0, 10. Right? And so, if I find a solution to this 5, 0,

well, 5, 0, no, doesn't work. 4 plus 1? 4 plus 1, nope, doesn't work. If I'm declaring that the symbol x plus y is 5, I can't declare that the same symbol is also simultaneously equal to 10. I can't claim x plus y is 5 and x plus y is 10. That's never true.

- 8.31** Right? So this is never going to have a solution. They are never going to intersect. Now, in general, for linear systems of however many variables you have, these are the solution types that you can have, and the only solution types that you can have—a unique one, none, or infinitely many. It's very different from, like, the quadratics that we saw. The quadratics that we saw, we said, what? Quadratics can have two solutions, right? We can see that quadratics only have one solution. That's possible. Or no solutions. That's also possible.
- 8.32** For linear systems, that's impossible. You either have a unique solution, none, or infinitely many. OK, so let's talk about how do you solve linear systems? How do you solve linear systems? Well, using this as an example, using this as an example, how did we solve it? One method, so solution methods, OK, the method we used was substitution. Do we agree? What did we do? We looked at it, we said, OK, let's solve one of the variables. Let's solve one of the equations for y and let's take that y and let's plug it into the other equation so that that eliminates that variable y , and now we can solve for it in terms of x .
- 8.33** That's literally what we did. OK, so for example, if I say this one, y is equal to negative x plus 3, then I can take that y , I can substitute it in there and get $2x$ minus 3 times y . What is y ? Minus x plus 3, that's equal to 5. OK, be careful with the negative. Remember, we talked about this in terms of distribution. When I distribute in, I distribute the negative 3. So this is a $2x$, negative 3 times negative x is plus $3x$. Negative 3 times 3 is minus 9, is equal to 5. What do I get? $5x$ is equal to, move that over, 14. x is equal to 14 over 5. That's exactly what we got.
- 8.34** I can then take that x , substitute it back in there for x rather, to figure out what y is. And y would be equal to one-fifth. That's the method of substitution. It's kind of clunky if I'm being honest, because you can imagine—take this whole entire thing. I give you 10 equations and two unknowns, and I'm going to say, hey, solve for y and plug it into the other nine equations and hope that all of the x 's are going to give you the same value. It's like, that's fantasy land. That is not going to happen on accident. OK, so substitution is a little clunky.
- 8.35** There's a better method for approaching this, and it's called elimination. So what you want to think about is, you want to eliminate one of your variables, OK? And that introduces the concept of a matrix, which is one of the most important concepts in all of mathematics. OK? So I'm giving you a primer into a branch of mathematics called linear algebra, which is arguably the most important branch of mathematics out there, in my opinion. OK? So let's talk about the method of elimination. Elimination.

- 8.36** The idea is simple enough. I have two equations and two unknowns. I want to eliminate one of the variables. It doesn't matter which one you choose to eliminate. So you write the equations with their variables lined up. You write them with them lined up, and you ask yourself the following question. Is there a variable which is easier to eliminate than others? Is there a variable that's easier to eliminate than others? That's a subjective question. I could ask you that question and you're like, "Oh, yeah, x is the one to eliminate." Great.
- 8.37** And I ask somebody else and they're like, "No, I like y . I want to eliminate y ." There's no right or wrong answer here. OK, you're choosing a variable to eliminate. I'm going to choose to eliminate y . Let me explain why. OK, pardon the pun. Let me explain why. I see stacked above this y a negative 3, and this is a positive y . So if I multiply this equation by positive 3 on the left and on the right, this is going to give me a $3y$ here. And if I simply just add the equations, negative $3y$ and plus $3y$ will cancel.
- 8.38** But there's nothing wrong with saying, "Look, Robbie, the same argument can be made. You have a $2x$ here. You have an x here. Why not just multiply this by negative 2 on both sides? And then the negative $2x$ and the $2x$ can cancel. And then you're done, in the same way." And it's like, fair point. You can do that, too. All right, so what I'm going to do is, I'm going to multiply this equation by 3 on both sides. I'm going to multiply this by 3, multiply this side by 3.
- 8.39** Have I changed the equation? Well, no, because whatever I do to one side, I do to the other side and then that's totally fine. OK, still balanced. I haven't changed anything. So I get $2x$ minus $3y$ is equal to 5, and $3x$ plus $3y$ is equal to 9. That's my new set of equations. Well, "new." They look new, but they haven't changed their geometry. They're still the same two lines. Still the same two lines. But now if I just add them, if I add them, what do I get? $2x$ plus $3x$ is $5x$, negative $3y$ plus $3y$ is $0y$, and 5 plus 9 is 14 . So I get $5x$ is equal to 14 , x is equal to 14 , divided by 5 .
- 8.40** Surprise, surprise. It's the same answer that we got when we did substitution. If it didn't give me the same answer as substitution, then this would probably be a bogus method, right? We want methods that are coherent across the solution set. We don't want one method that gives me 7 over 5 for x and another method that gives me 14 over 5 for x . It's like, those two are incongruent to each other, right? That's not working.
- 8.41** Now, once I have x is equal to 14 over 5 , I can go back up, choose whichever one I want, plug it into that one, plug it into that one, and solve for y . That'll give me what I'm interested in. OK? OK, so why, like, why do I care about this? Why should I care about something like this? Here's one reason. OK, here's one reason. What if I ask you the following question? What if I ask you the following question? I want to know, can I, if I give you three points, three points, can I fit a parabola through those three points? A quadratic. If I give you three points, can I put a quadratic through those three points? OK, let's see.

- 8.42** So if I give you the following points, let me say here's x , here's y . We're going to start off with a relatively easy example, but this can be abstracted to anything. OK, but you got to start off with something relatively simple. I'm going to give you three points. I'm going to give you this point, the origin, $0, 0$. I'm going to give you this point, which is 1 , oops, I'm going to give you this point, which is right here, 1 , minus 1 . And I'm going to give you this point here, which is $2, 0$. OK, those are 3 points.
- 8.43** And I want to come up with a quadratic which specifically goes through those three points. OK, so I want a quadratic. So it's going to be a parabola. It's not bad. OK, I want a quadratic that goes through those three points. How would I approach this question? Well, the first thing I'm going to do is, I'm going to say, well, do I know what a general quadratic looks like? Do I know what a general quadratic looks like? OK, well, we've talked about quadratics. We've said that they look like $ax^2 + bx + c$. That's what they look like.
- 8.44** So if I say, OK, well, x is my input, y is my output, then all of these points on here need to satisfy that equation. Do we agree? Any point that I choose on here needs to satisfy this corresponding equation. In particular, the three points that I know, $0, 0$, 1 , negative 1 , and $2, 0$, have to satisfy this. That has to be satisfied. OK, so what happens when I plug in the point $0, 0$? What do I get? On the left-hand side, I get y is zero. On the right-hand side, I get a times zero plus b times zero plus c . That tells me c is equal to zero. OK, so c is equal to zero. All right?
- 8.45** If I plug in the point 1 , minus 1 , what do I get? That tells me that minus 1 , that's y , is equal to a times 1 , so that's a . b times 1 , that's b . And then c just goes along for the ride. OK, and if I take in the point, what was the other one, $2, 0$? It says that y is zero. OK, x is 2 . So 2 squared is 4 , so this is $4a$. x is 2 , so that's $2b$ plus c . OK, how in the world is this connected to anything? Well, here's three equations and three unknowns, a , b , and c . This is a linear system of equations.
- 8.46** What are my equations? I have $4a + 2b + c$ is zero, that's this one. I have $a + b + c$ is minus 1 . That's this one. And I have c is equal to zero. That's like the best one, because it tells me immediately what c has to be. c has to be zero. OK, so this, on the surface of it, is a system of one, two, three equations, linear equations, in one, two, three unknowns.
- 8.47** Now, just by happenstance, because I went through the origin, c is zero. So what can I really do? I could say, oh, sweet, I already know the value of c . So why don't I substitute that in there and substitute that in there? Then I can take my system of three equations and turn it into a system of two equations by eliminating one of the variables like we just did. Eliminating c , I get $4a + 2b$ is equal to zero, and $a + b$ is equal to minus 1 . Now I'm back in the situation that I had initially. Two equations and two unknowns. My unknowns are a and b . They're a and b .

- 8.48** So what do you want to do? Which one do you want to eliminate? Do you want to eliminate a , or do you want to eliminate b ? I personally don't care. It doesn't matter which one you approach. So let's just choose one. Let's eliminate b . OK, how would I eliminate b here? I'm going to multiply both sides of the equation by negative 2 so that I can cancel it with the plus 2 above it. OK, so what do I get here?
- 8.49** This little arrow is just an implication symbol. So I'm saying $4a + 2b = 0$. I'm just leaving that one alone. $4a + 2b$ is equal to zero. OK, I'm going to multiply both sides by negative 2. So $-2(4a + 2b) = -2(0)$ is equal to $-8a - 4b = 0$. OK, I'm multiplying both sides of the second equation by negative 2. That's all I'm doing. OK, well, now let's add everything together. Let's add them up. $4a + (-8a) + 2b + (-4b) = 0 + 0$. So $-4a - 2b = 0$. So I'm not going to write it. And $-4a + 4a + 2b - 2b = 0 + 0$. So $0 = 0$. That's a miracle. a is a very nice number. Divide by 2, we get a is equal to 1.
- 8.50** If a is equal to 1, plug it into either one of the equations. The number one question that you get is, well, which equation should you plug it into? It's like, does it matter? No, because think about it geometrically. You're looking at where they intersect. Their coordinates are literally the same. So it doesn't matter which equation you plug it into. They have to be the same. That's the whole point. So if I take $a = 1$ and I plug it into this one—this is kind of the easier of the two to plug it in—I get 1, so I'm going to get $1 + b = -1$. Subtract the 1 over, I get $b = -2$. I get B is equal to negative 2.
- 8.51** And we already said c was 0. That was given to me. So what's the equation of that quadratic? This quadratic has the form y is equal to $1x^2 - 2x + 0$, otherwise known as $x^2 - 2x$. That is what the equation is for that guy. OK, so you're still like, OK, so this is still all just fake, right? Like, this isn't worth anything. Like, what's the point? Well, think about it. What's the point? Parabolas are curved. Think of curved surfaces that you're interested in. Contact lenses, for example. You want flexible contact lenses. That has a curved shape to it. Maybe you have three points that you're trying to fit a particular shape of contact lens to.
- 8.52** OK. Satellite dishes are great, similar to a contact lens. Same kind of idea. You have test points. You're trying to fit a particular curve through that test point. OK, what about automobiles, right? What about cars? You have the hood of a car, has a certain curvature associated to it. You want to be able to manufacture or machine a particular curvature for that particular hood. Same basic principle. OK, this is incredibly relevant, is what I'm trying to say. Right now, it looks like we're just pushing symbols around on a board and all of this. You do this in a computer algebra system. OK? You don't do this by hand.
- 8.53** But if you're trying to machine a particular hood, you have to train—on a car—you're trying to take a machine and train it on how to machine out whatever the curvature is on that hood. What do you do? You take a bunch of test points and you say, "I want the curve to look like this." So you try and fit a curve that way. You can then train your

machine to actually carve out the metal the way it's supposed to go, bend it a particular way. OK, that's why this is useful. All right? It all starts with linear algebra. Linear algebra is amazing. And that's the next step from our basics and fundamentals of algebra.

8.54 OK, so let's think about another example. Let's think about another example. I want to, I want to flip the script a little bit. OK, so we've been asking the question, if I give you an equation of a line, you can generate data, which is points along that line, right? If I give you the line, let's say $2x$ plus y is equal to 1, if I give you the line, you can generate points that live on that line. You can graph the line. You can say, oh, the points look like this. That's fine.

8.55 What if I flip it on you and I say, OK, I want you—I'm going to give you data. I'm going to give you a whole bunch of dots in the xy plane. And I'm going to ask you to give me a line which gets as close to those dots as possible. That's the problem. OK, so here's the setup. I'm going to give you a whole bunch of dots, OK? I'm giving you this one, this one, this one, this one, this one, this one, this one, maybe this one here, maybe there's one down here, things like that. I'm going to give you a whole bunch of dots.

8.56 Now, the way that I've drawn this, these dots don't live on one single line. That's pretty clear, OK? If they lived on one line, that's a pretty goofy looking line, OK? So I can't say, OK, find a line which goes through every single red dot. That's not solvable. But what I can do is, I can say, OK, can you find a line that does something, something like that, such that this line is "closest" to the red dot for every red dot, meaning that if I were to look at another line, say one that looks like this, OK, comes from top left to bottom right, if I look at something like that, that somehow the error between the black line and the red dot is larger than the errors that I get for these guys, OK?

8.57 Things along those lines. How would you approach a question like this? I know that what this line looks like, this line looks like y is equal to mx plus b . So what I'm trying to understand is, I'm trying to "learn" the parameters m and b . Yes, I'm trying to learn, what does m have to be, what is the slope of the line, and what is the y intercept coordinate supposed to be? B . That's what I'm trying to understand. I'm trying to learn those parameters and I'm using learn in a very precise way here. Can I do that? And the answer is, yes, you can. Yes, you can.

8.58 All right, so how do you even approach a question like this? You can say, well, if I have these points, whatever they are, I can say they have an x and a y coordinate, yes? So I can say, well, can I plug them in and the equality is going to hold? The answer is, well, no, of course not, because they don't all live on the line, for example. So what I can do is, I can say, OK, for each of these points, let's call this one, let's say for the sake of argument, this is 3 comma, 3 comma, 2, maybe something like that. This point up here is maybe 1, comma, 4 and so on. I'm just trying to illustrate the principles underlying all of this.

- 8.59** OK, it doesn't make sense to say, oh, I can plug this in and I can get 2 is equal to $3m$ plus b . That's not going to be true because this point does not live on the line. So that's not true. But what can I do? Well, I can say, let's modify this a little bit to look at y plus some error is equal to mx plus b . Well, that is going to be the error term. That's going to be an error term. OK, what do I mean by that? Look, this point $3, 2$ does not live on the line. So there's going to be some distance e between those two. That's the error.
- 8.60** OK, so what I'm really saying is, I can say, oh, well, the error is equal to what? It's mx plus b minus y . That's the difference that these points have in their y directions. So I'm going to get a whole bunch of things like this. I'm going to get a whole bunch of things that look like the error 1 , error 2 , error 3 , all the way down, where I'm going to get an error for every single point that I have. For example, this point here, OK, this will be my error, say, 1 . This point here, when I plug it in, what am I going to get? I'm going to get that my error looks like x is 3 .
- 8.61** So I'm going to get a $3m$ plus b minus y , which is 2 . That's the error for that particular point. Let's say this one here is error 2 , error 2 . So error 2 is telling me that x is now 1 . So I'm going to get m plus b minus y minus 4 and so on. And I continue in this way. I continue in this way. So now I have a list of numbers that involve the parameters I'm trying to learn, m and b . And what is the question that's being asked of me? I'm asking to find the equation of this line which is closest to every single one of those points. So in particular, I'm saying, this list of errors I want to be as small as possible. That's what I'm asking.
- 8.62** So what I want to do is, I want to minimize the list of errors. That's the idea. That's the idea. What do I mean by minimize? I want to make it as small as possible. What do I mean by small as possible? Well, I want all of these numbers, when I combine them in a particular way, to be minuscule, tiny, as small as humanly possible. How do I do that? You need optimization. You need to understand rates of change. You need to understand things along those lines. You need to understand calculus on some level, OK, in order to optimize this.
- 8.63** So we don't have that. So what can I say? Well, maybe I can describe the objects at play here. Maybe I can describe the objects. These objects, these little vertical lists of entries, are what are called vectors, OK? These list of numbers are called vectors. So this is a vector. OK? All right, so let's try and take this to an example that we can actually get our hands on. It's not as abstract. It's just, you give me, say, three points. They don't live on a line. And you want that, you want to find the equation of best fit, OK? That's what we're trying to get at.
- 8.64** This line that we're looking for, this y equals mx plus b , is called the line of best fit, OK? This is called linear regression, is what the name of all this is. But we're trying to just, we're trying to just discover this organically with natural questions that we can ask and then discover the mathematics along the way. That's the idea. OK, so let's talk about matrices and vectors. Matrices and vectors. I'm going to go back to the system of

equations that we had where I had two equations and two unknowns, OK? Not the same one. I'm going to give you a different one.

- 8.65** But let's suppose I gave you a system that looks something like x minus y is equal to 3 and then x plus y is equal to, let's say, 5. I don't care. Let's suppose I gave you that. When asking you to solve the system, OK, remember, when we plug in for all of these things, I get a whole bunch, I get a massive system of equations. I get a bunch of equations in the variables m and b that I'm trying to learn, OK? So I need a mathematical framework in which to handle systems of equations. And that's where matrices and vectors come into play. OK?
- 8.66** So let's illustrate it with this guy. OK, so what I'm going to do is, I'm going to create a vector. I'm going to create two vectors. The vector is going to look like the first one. I'm just going to put the variables of interest in them. So I'm going to put little brackets around them, and I'm just going to list them as x , y . The other one that I'm going to look at is, I'm going to look at this side here. I have these numbers here. So I have 3, 5. OK? That's what I have.
- 8.67** Those are vectors. Vectors are literally just an array of something, numbers, whatever type you're interested in. Strings, data type, things like that. It's just a list, an array of something. In this case, they're just numbers. The matrix, the matrix is given by looking at the coefficients of your variables. OK? So you consider the coefficients of your variables. That's what you do. So you have to kind of order your variables accordingly. All right? So you kind of stack all of your x 's, you stack all of your y 's, you make sure they're all lined up nicely.
- 8.68** And the matrix that you get is this little block where, how many rows am I going to have? The number of rows is equal to the number of equations you have. So I'm going to have two equations, so two rows. And how many columns am I going to have? Well, the columns are going to be corresponding to the number of variables that you have. OK, and so the way I like to think about it is that my first column is given by the coefficients of my first variable, which is x . So what are my coefficients? I have a 1 and a 1. So I'm going to have a 1 and a 1. Those are the entries that go in there, and then move over to my next variables.
- 8.69** So my next column. And the coefficients are negative 1 and positive 1. So I get a minus 1 and a positive 1. That's what I get. OK, so now we have the mathematical framework of matrices and vectors, and we can write this system in terms of matrices and vectors. How do we do that? OK, so this is matrix-vector form. OK?
- 8.70** The way you do this is, you take your matrix, 1 minus 1, 1, 1. You then put the vector xy next to it and that will be equal to the vector on the right-hand side. 3, 5. 3, 5. OK? And this is very, like, this is very precise, OK? Like, what this actually, what this actually means. And we're building to this error situation that we had earlier, OK? Because we need the notation to be able to evaluate that. OK, so how do I actually think about this?

How do I take a matrix? Let's call this matrix capital A, OK? Times this vector. Let's call this, I don't know, let's call it $\underline{V}$. OK? And then equal to, let's call this $\underline{W}$. OK?

8.71 So A is this matrix. It's got two rows and two columns. So we call it two-by-two, number of rows by number of columns. The vector is just this single column. So it's got two rows, one column. So it's two-by-one. And I just denote it by a $\underline{V}$ with an underline. The underline just means that it's a vector. OK? And then $\underline{W}$ is the corresponding two-by-one vector over here with an underline.

8.72 How do I actually interpret this? Well, how do I take a matrix times a vector? I take this row and I multiply it by this column and I set it equal to this entry. OK, so that's kind of weird. So think about this. So I take 1, negative 1. I imagine rotating it and multiplying by this. What am I going to get? 1 times x . And then I add negative 1 times y . So I'm going to get x minus y equals 3. I'm getting this equation. Vice versa. If I take 1, 1 and multiply by $\underline{V}$, I'm going to get $1x$ plus $1y$. So x plus y equals 5. Bingo. I'm getting that. OK.

8.73 So why is this useful? Remember we had this method of elimination that said, look, if I get rid of these now—we have this method of elimination that said choose your favorite variable and eliminate it. In my opinion, I would eliminate y because all I have to do is add the equations together. That's simple enough. But if I have this, the matrix-vector form, this gives me what's called my augmented matrix. And the augmented matrix looks like this. You put your coefficient matrix, 1, minus 1, 1, 1, and then a vertical bar 3, 5. You incorporate the data of the coefficients on the left-hand side and the equality on the right-hand side, the 3, 5.

8.74 OK, why do you use this? Why do you use this? Because now, rather than looking at these equations which have x 's and y 's and goodness knows how many variables running around, all you have to do is look at this array of numbers and mess around with these arrays. OK? So for example, what could I do with this corresponding array? I could look at this and say the following. Take this augmented matrix and I'm going to do something to it. OK? So 1, negative 1, 3, 1, 1, 5. Remember, this is representing this system of equations here. It's representing this system.

8.75 What if I did this? Let's call this row two. What if I took row two and I subtract away row one? What is that doing from the system of equations? What am I doing? I'm taking row two and I'm subtracting away this row here. I'm eliminating the variable x . That's all I'm doing. OK, what would happen? Well, x minus x is zero. OK, so how does it change row two? x minus x is zero, y minus minus y is 2. So this row is going to be, what? One minus 1, zero. One minus minus 1, 2. And then 5 minus 3 is 2. And I didn't change row one.

8.76 What did I literally do? I just subtracted the equations to eliminate one of my variables, which is what we were doing earlier. So what did I get? I now have x minus y equals 3, and x minus x is 0, y minus $2y$ is $2y$, and 5 minus 3 is 2. This augmented matrix is just

encoding this one, which is all we did by taking this equation minus this equation. That was it. So we haven't changed the solution type. I can then do what? I can then take row two. I can take one-half times row two.

- 8.77** What am I really doing? I'm just saying multiply both sides by one-half. I'm not changing anything. Balancing the equation. What do I get? I end up getting 1, minus 1, 3, 0, 1, 1. OK, what did I just do there? I just divided both sides by 2. So now I have y is equal to 1. Now, I could take y equals 1 and plug it in here and solve for x . Or I could take equation one and add equation two. So if I did that, if I took row one plus row two, what do I end up getting? One plus zero is 1, negative 1 plus 1 is zero, 3 plus 1 is 4, zero, 1, 1. And now pull this back. What did I just do?
- 8.78** If I took row one, equation one, plus equation two, x plus nothing is x , negative y plus y is 0, 3 plus 1 is 4, and this one's telling me y equals 1, we just found the unique solution to our system of equations. We found the unique solution to our system of equations, which in the augmented matrix form, this one is telling me x equals 4, and this one is telling me y equals 1. So the unique solution to this system is x is 4, y is 1. Four minus 1 is 3, 4 plus 1 is 5. It satisfies both. But now we have a systematic method via matrices and augmented matrices to actually, systematically, find this.
- 8.79** So let's go back to our best-fit line example. If I give you three data points, if I give you three data points, so let's say I give you the origin, I give you, say, 1, 1, and I give you the point, say, I don't know, 2, 0. So those are my three data points. And I want to come up with a line which approximates this as close as I can. You can kind of imagine, it's probably going to be that horizontal line y equals one-half, realistically. OK, that's probably what it's going to be. But if we don't know what it is, we could say y equals mx plus b , and we can start computing our vector of errors associated to this.
- 8.80** So what are our error vectors? So if I look, y plus the error is equal to mx plus b , well, if I plug in the point 0, 0, what am I getting? I'm getting error 1 is equal to b . I'm getting error 2 is equal to what? That would be this point. So that would be m plus b minus 1. And error three is equal to what? So that's when x is 2. So that's $2m$, y is zero, so plus b . That's what I get. So I now have a vector of errors, error 1, error 2, error 3, which now looks like error 1 is b . This one is m plus b minus 1. And this is $2m$ plus b , as my corresponding vector of errors.
- 8.81** Now, what we want to be able to do is, we want to look at this vector and we want to say, OK, I want to choose m and b such that this thing is as small as possible in a very precise way. So you look at the length of the vector, you make it as small as possible. That requires you to optimize and do all of that. But the point of all of this is understanding, how do I come up with a line which fits my data as closely as possible? This is precisely what's kind of known in the literature as supervised learning.
- 8.82** Goes back to the example that we gave at the very beginning of this course, where we were saying things like, if you have a recommendation engine—we were talking about

this in terms of functions—you have a recommendation engine which takes your purchase history, your user history, into account. It then outputs recommendations, of course, corresponding to that. Right? The same principle applies when you're talking about linear regression, where I have this data set and I want to approximate it with a particular line.

8.83 What I'm trying to do is, I'm trying to learn that data as closely as possible. I'm trying to come up with a line which approximates this data as closely as possible, so that I can then make recommendations to you, which is close to your user history. That's what you're trying to do. And that's kind of the foundations of what's called supervised learning. That's completely revolutionizing the world. But it all stems from linear algebra, which has its basis in the fundamentals of algebra.